



BLUEBIRD: a phone-brained blended-wing survey UAV for conservation and agriculture

Design, analysis and simulation of a low-cost, repairable autonomous aircraft

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Abstract

BLUEBIRD is a 1.8 m blended-wing-body survey aircraft for wildlife and agricultural surveys. One student can build it with a desktop 3D printer, CNC-cut foam and free software. An ordinary Android phone is its companion computer, while ArduPlane owns flight control and every failsafe. Landing sites are chosen before takeoff from open terrain, land-cover and map data and compiled into the autopilot’s mission, so any failsafe ends at a vetted site without the phone.

We present the design as a sequence of decisions, each with the analysis that decided it.

- **Stall.** Transitional RANS CFD (SU2, Langtry–Menter) showed that the outer-wing section stalls about 25 % below the prediction of the fast XFOIL-trained model. This cut the aircraft’s maximum lift coefficient to 0.90 and puts the stall speed near 9.3 m/s at 2.52 kg.
- **Performance.** Trimmed cruise L/D is 11.4 at 14 m/s, which gives about 61 km of still-air range with a 20 % energy reserve.
- **Stability.** Reflex and 3° of washout trim the tailless wing at an 8 % static margin. Taller fins canted outboard raise Dutch-roll damping from 0.088 to 0.103 without destabilising the spiral mode.
- **Approach.** A split-elevon crow airbrake steepens the approach to about 5.2:1.
- **Structure.** The belly landing, not flight, sizes the structure: 84.5 N m of wing-root bending against 31.2 N m for the worst flight case. A deep TPU skid sled holds the design landing load to 50 g. EPP foam outer wings covered in two film layers clear the flutter requirement (44.6 m/s against 41.9 m/s).
- **Safety logic.** The landing-site failsafes pass four end-to-end ArduPilot software-in-the-loop tests.

Nothing has flown yet. We close with the verification status and the test plan for the first airframe.

Contents

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Table 1: BLUEBIRD at a glance. Predicted values at 2.52 kg unless noted; see the sections cited.

Span; wing area	1.8 m; 0.52 m ²	Trimmed L/D at cruise; best	11.4; 12.1 at 12 m/s
All-up mass	2.52 kg [pending: final CAD]	Range; endurance (20 % reserve)	61 km; 1.2–1.36 h
Cruise; survey height	14 m/s; 100 m AGL	Stall speed	9.3 m/s
Static margin	8 % MAC	RGB ground sample distance	3.9 cm/px at 100 m
Landing design load	50 g at the CG	Avionics BOM (excl. phone)	\$1,059

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1 Introduction

Small uncrewed aircraft have changed how ecologists count animals and map habitat. They fly lower and cheaper than crewed survey aircraft, and in controlled trials drone imagery counted birds more accurately and precisely than observers on the ground [1–4]. Thermal cameras extend this to animals under canopy edges and at dawn [5]. Most of that work uses multirotors, whose endurance of about 20–30 minutes limits a sortie to a few square kilometres. Fixed-wing aircraft cover far more ground per battery. The commercial survey fixed-wings that do this, however, are expensive, closed and hard to repair in the field. The places where wildlife and smallholder-farm surveys matter most are often the places with the least money, spares and runway.

BLUEBIRD (*Blended-wing Long-range Unmanned Explorer for Biodiversity, Imaging and Rural Data*) is our answer to that gap. It is a 1.8 m blended-wing-body (BWB) aircraft that surveys at 100 m above ground at 14 m/s. It is built from a 3D-printed centre body and CNC-cut foam outer wings. Its onboard computer is an ordinary Android phone, which runs the cameras, the geotagging and the landing logic. A proven open-source autopilot, ArduPlane on a Matek H743-WING flight controller [6], flies the aircraft and owns every failsafe. The guiding rule of the architecture is short:

The flight controller flies; the phone thinks.

If the phone crashes, overheats or is removed, the aircraft still flies to the nearest pre-vetted landing site and lands on its own.

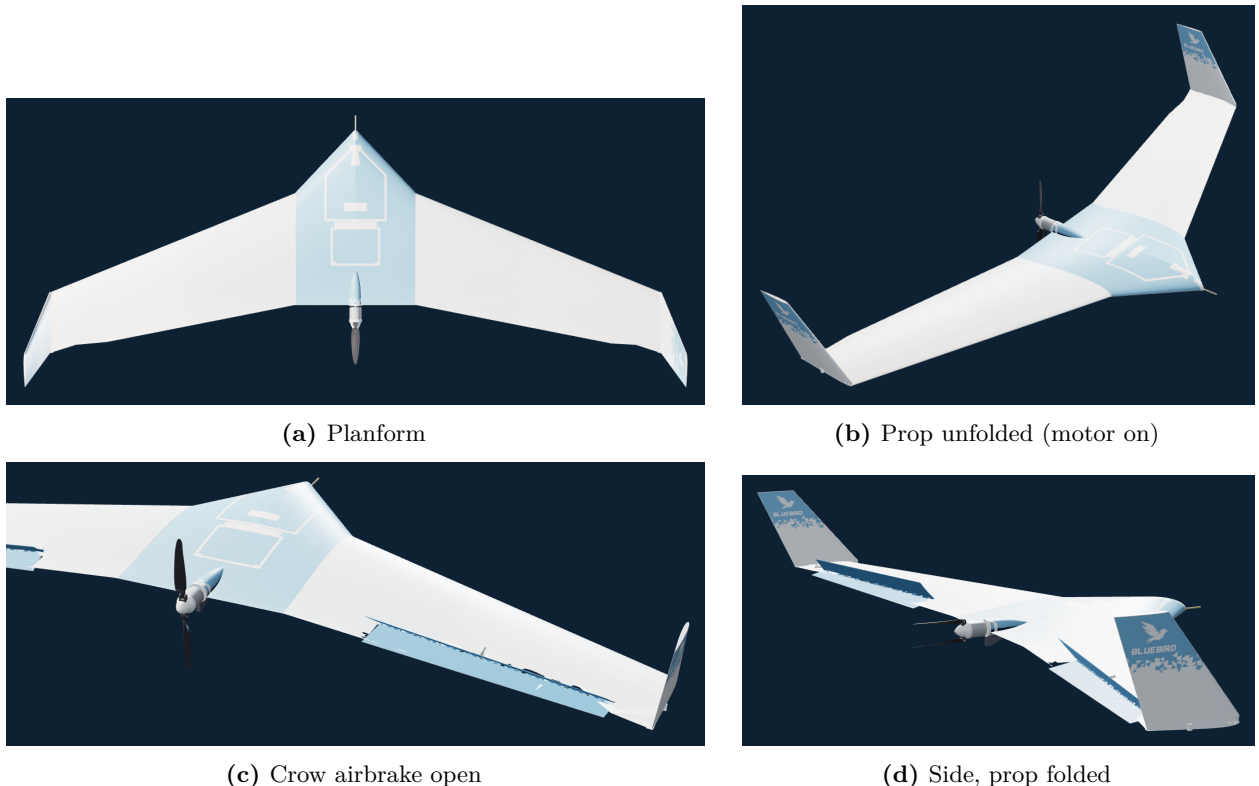


Figure 1: BLUEBIRD rendered from the design model in the Rev C livery. The moving parts sit on their CAD hinge axes: the prop blades fold on their pins, and each elevon is split into an upper and a lower half that open 51° up and 12° down as the crow airbrake.

1.1 Contributions

This paper documents the design, analysis and simulation of BLUEBIRD before its first build. The main contributions are:

1. **A phone-as-companion-computer architecture** with a strict split of authority. The phone improves decisions such as landing-site choice, divert and survey quality, but it is never needed to land safely (Section 9).
2. **Landing sites planned before takeoff.** A ground planner scores candidate sites from open terrain, land-cover and map data, corrects the required landing run for density altitude, and compiles each site into an ArduPilot landing sequence. The autopilot then diverts to the nearest one on any failsafe (Section 10).
3. **A split-elevon crow airbrake** on a tailless aircraft. It steepens the approach into short sites without flaps or a tail (Section 6).
4. **A printed and foam airframe designed for field repair**, analysed end to end with free tools:
 - aerodynamics: NeuralFoil, vortex-lattice, VSPAERO and SU2 RANS CFD (Section 4);
 - stability and flight dynamics (Section 6);
 - structures: beam models, CalculiX FEA and landing-impact analysis (Section 8);
 - flight software: ArduPilot software-in-the-loop (SITL) simulation (Section 12).
5. **A single-source-of-truth workflow.** Every dimension lives in one parameter file that the CAD and all the analyses read, so a change propagates everywhere and is re-checked (Figure 2).

1.2 Design process

Figure 2 shows the loop we followed. Each discipline reads the same parameter file and writes a results file. Every design change in this paper came out of that loop, and each one was re-checked by the disciplines downstream of it. Examples include the taller canted fins, the foam outer wing, the shallower lower crow deflection and the deeper skid sled. Where a decision involved real alternatives, we give a trade table with the numbers that decided it.

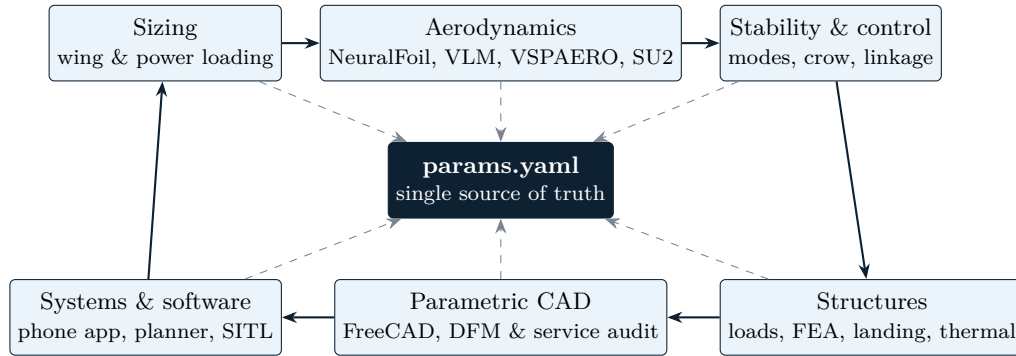


Figure 2: Design loop. Every analysis reads the shared parameter file and writes a results file. Dashed arrows are the design changes each discipline fed back. The loop ran until all checks passed together.

1.3 Status and scope

Nothing in this paper has flown. All results are from analysis and simulation, and we mark clearly where a number still needs a bench test, a flight test or the final CAD export ([pending: ...]). Section 12 lists what has been verified, how, and the test plan for the first build.

2 Mission and requirements

2.1 Concept of operations

A two-person team drives to the survey area with the aircraft in a car boot.

Before takeoff. On a laptop, the ground planner:

1. turns the survey polygon into flight lines;
2. finds and scores landing sites from open data;
3. checks that every point on the route can reach a site with the energy reserve;
4. uploads the mission to the autopilot.

The phone is strapped under the hatch and connected by USB, and it runs a preflight checklist. One of its checks is that the phone is in airplane mode.

Launch and survey. The aircraft launches from a wheeled dolly on a rough strip, or by hand or bungee. It climbs to 100 m above ground and flies the lines at 14 m/s, taking overlapping RGB and thermal images.

Landing. It lands on its belly at a planned site, steepening the approach with the crow airbrake. Before takeoff, the people named for each planned site are notified. The operator has to press Send; nothing goes out automatically.

2.2 Requirements

Table 2 lists the requirements that drove the design. Some have a hard number. Others are goals that we maximised, and those are marked as goals. The last column says where each is addressed.

Two requirements shaped the design more than any others.

R9 (safe landing without the phone). Every safety function had to live in the autopilot. This is why landing sites are compiled into the autopilot’s mission, and not held only in the phone app.

R14 (buildable without composite layup). This ruled out glass-fibre skins. It led to printed PETG-CF structure in the centre body and EPP foam with iron-on film on the outer wings.

3 Configuration and sizing

The mission sets the drivers: a 14 m/s survey cruise at 100 m above ground, a 20 % energy reserve for reaching a vetted landing site, a belly landing on unprepared ground, dolly or hand launch, and a payload stack (a 150–240 g phone over a 600 g battery) that must sit at the centre of gravity (CG). Every printed part must also fit a 256 mm printer bed.

3.1 Configuration trade

The layout was chosen at the concept stage by a qualitative trade between a conventional tailed aircraft, a plain flying wing and a blended wing body (BWB). We did not analyse the two rejected layouts numerically. Table 3 lists the criteria and the repository facts that decided the trade.

The BWB wins the first two rows, which are hard requirements. Its cost is the narrow CG window of a tailless aircraft. A 90 g change in phone mass moves the CG by 4.9 mm, and the full

Table 2: Top-level requirements and where they are addressed.

ID	Requirement	Value	Section
R1	Survey altitude above ground (inside the 14 CFR 107 ceiling of 400 ft)	100 m AGL	9
R2	Cruise speed for the photo cadence and wind penetration	14 m/s	5
R3	RGB ground sample distance at 100 m	≤ 4 cm/px	9
R4	Thermal: a deer-sized animal resolvable on a counting pass	$\approx 7 \times 2$ px at 50–60 m	9
R5	Energy reserve kept for reaching a landing site	20 %	5, 10
R6	Endurance and range (goal)	maximise	5
R7	Launch without a runway: dolly on a rough strip, or hand/bungee	crosswind ≤ 3 m/s, bumps ≤ 15 mm	6
R8	Belly landing on an unprepared site	run ≥ 60 m, slope ≤ 4 %	6, 10
R9	Safe landing without the phone (autopilot-owned fail-safes)	always	12
R10	Field service: battery swap, phone swap, open hatch	30 s, 30 s and 5 s	8
R11	Fits a range of phones, not one model	$170 \times 82 \times 12$ mm, 150–240 g	9
R12	Hot and high: East African plateau sites	1500–2000 m, up to 35 °C	5
R13	Regulatory: FAA Part 107 and Remote ID, FCC- authorised radios, phone in airplane mode in flight	compliant	11
R14	Built by one student with a desktop printer; no composite layup	—	8
R15	Designed with free tools only, so the design can be shared openly	—	Appendix B

battery trim travel (165–199 mm) moves it by only 8.4 mm, so the window is met by a per-phone battery notch that the companion app enforces at preflight.

Pusher or tractor. A tractor would sit ahead of the nadir cameras and the phone bay, wash the belly, and meet the ground first in a belly landing. The centre pusher at $x = 575$ mm, behind the 520 mm trailing edge, leaves the nose in clean air, and its blades fold aft when the ESC brake stops the motor. We carry its three costs explicitly. It works in the centre-body wake (a $0.002 C_{D0}$ add-on and a 0.93 static thrust factor). Its running disc reaches 134 mm below the thrust line, so the aircraft cannot take off from its belly and uses a dolly (55 mm seated clearance) or a hand or bungee launch. A blade stopped pointing down touches the ground at 2.3° tail-down, so landings must stay flat.

3.2 Planform

Table 4 and Figure 3a give the planform. The centre body has a straight trailing edge, so its leading edge sweeps at 46.5° from the 520 mm root to the 330 mm wing root at $y = 180$ mm. The outer wing tapers to a 180 mm tip with 22° of leading-edge sweep and -3° of washout; Section 4.4 tests these three choices.

3.3 Sizing: wing loading and power loading

The first sizing (`sizing/sizing.py`) took the area from the planform and assumed $L/D = 8.5$, $C_{L,\max} = 0.9$ and a propulsive efficiency of 0.5. It gave 2.25 kg, 43.3 g/dm², an 8.9 m/s stall and

Table 3: Configuration trade. Ratings are qualitative; the numbers are for BLUEBIRD.

Criterion	Conventional tail	Plain flying wing	BWB, tip fins (chosen)
Phone over battery at the CG	fuselage pod	MH45 on a 330 mm root is 32 mm deep: pod needed	57 mm deep at the pack station, inside the lifting surface
Belly landing	boom and tailplane exposed	flat, but no protected bay	flat 520 mm centre body on a replaceable TPU sled
CG range	wide (tail volume)	narrow	narrow: SM 6–10 % is a 12.5 mm window
Trim	no reflex needed	reflex, sweep, washout	reflex, sweep, washout; −2.9° elevon at cruise
Parts	most; long boom	fewest	bed-sized centre segments, two removable foam wings

Table 4: Planform and reference values (`params.yaml`, `aero/results.yaml`).

Span b	1.800 m	Reference area S	0.5203 m ²
Root chord (centreline)	520 mm	Mean aerodynamic chord $\bar{c}$	312.3 mm
Centre-body half width	180 mm	Aspect ratio $A = b^2/S$	6.23
Wing root / tip chord	330 / 180 mm	Quarter-chord sweep (outer)	19.4°
LE sweep, outer / centre	22 / 46.5°	Washout (tip, nose down)	−3°
Dihedral	1°	Tip fins: height, cant out	200 mm, 10°
Elevon chord / span	22 % / 25–90 %	Sections, centre / outer	MH60×1.15 / MH45

54 km of range. The analysed aircraft weighs 2.432 kg (detailed CAD) or 2.52 kg (foam wings): 46.7 and 48.4 g/dm², or 45.9 and 47.5 N/m².

Level flight links the wing loading W/S to the shaft power loading P/W . With the trimmed polar $C_D = C_{D0} + kC_L^2$ of Section 4.7,

$$\frac{P}{W} = \left[\frac{q C_{D0}}{W/S} + \frac{k (W/S)}{q} \right] V + \dot{h}, \quad q = \frac{1}{2} \rho V^2, \quad \text{and} \quad \frac{W}{S} \leq \frac{1}{2} \rho V_s^2 C_{L,\max}. \quad (1)$$

Figure 3b plots Equation (1) for $C_{D0} = 0.0220$, $k = 0.0786$ and $C_{L,\max} = 0.90$. At 14 m/s cruise power is least at $W/S = q\sqrt{C_{D0}/k} = 63.5 \text{ N/m}^2$, but that loading would raise the stall to 10.7 m/s. The design sits where the power curve is flat: it needs only 4 % more cruise power than the optimum and keeps the stall near 9.3 m/s. The wing loading is therefore set by stall, launch and landing speeds, not by cruise power. The belly slide grows with V_s^2 (12 m at 9.1 m/s on grass, $\mu = 0.35$), and the dolly lift-off is at 11.2 m/s.

The launch sets the power loading. The dolly and hand launch need at least 18 N of installed static thrust. The chosen motor and propeller give 23.0 N ($T/W = 0.93$ at 2.52 kg) and 9.3 W/N of shaft power at 14 m/s, against the 1.2 W/N cruise needs. Cruise therefore runs the motor at about 10 % of its launch power, where it is least efficient (Section 5.1).

3.4 Mass basis and design evolution

The mass basis changed twice. The first budget in `params.yaml` was 2.250 kg. The detailed CAD, which `aero/results.yaml` uses, weighed 2.432 kg, but its printed outer wings lacked their structural lattice; restoring it and the other fixes would have made the printed aircraft 2.72 kg. The outer wings were therefore changed to EPP foam cores with two layers of iron-on film. The

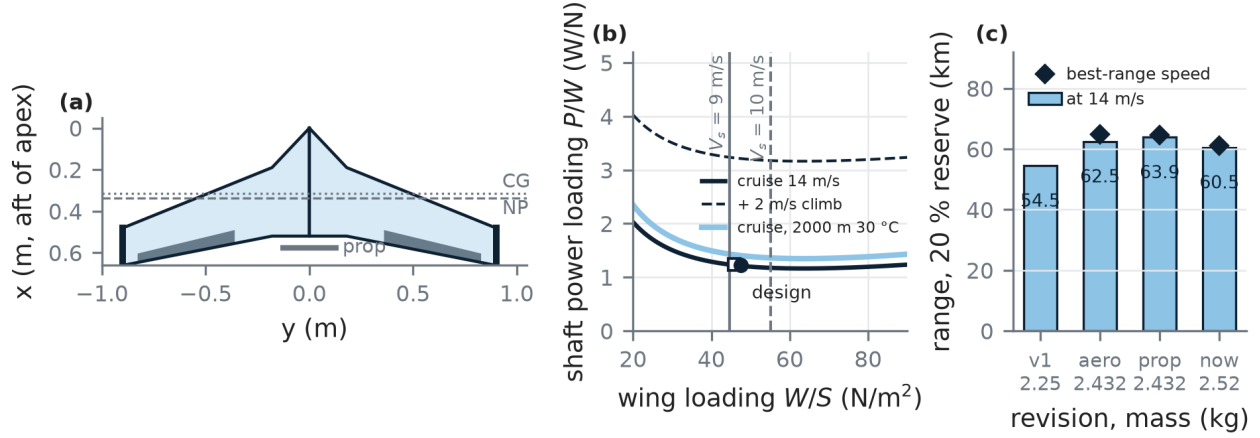


Figure 3: Sizing. (a) Planform from `aero/geometry.py`, with the elevons (dark), the AeroBuildup neutral point and the CG at 8% static margin. (b) Constraint diagram, Equation (1); open square 2.432 kg, filled circle 2.52 kg. The available shaft power at 14 m/s, 9.3 W/N, is off the plot. (c) Still-air range at 14 m/s (bars) and at the best-range speed (diamonds) by revision: `sizing.py`, `aero/results.yaml`, `docs/prop.md` and this paper.

structures estimate for that aircraft is 2.51 kg at 353 g per wing; the wing CAD weighs 357.1 g per side, giving 2.52 kg. We re-ran the trim and performance models at 2.52 kg and report both masses. The largest items are the battery (600 g), the phone (200 g) and the motor (107 g). The foam-wing CG has not yet been re-extracted from the CAD: [\[pending: CG and static margin with the foam wings\]](#).

Figure 3c shows how the range estimate moved. It rose when the analysed L/D replaced the assumed 8.5. It then fell as the mass grew from 2.25–2.52 kg and the avionics estimate from 4.0–9.3 W. An early figure of about 80 km is obsolete.

4 Aerodynamics

4.1 Reynolds numbers and airfoil selection

At the 10 m/s stall-check speed the chord Reynolds number $Re = \rho V c / \mu$ is 353 000 on the centreline, 224 000 at the wing root and 122 000 at the tip. Over the envelope (9.3–22 m/s) it spans about 1.1×10^5 to 7.8×10^5 . In this regime laminar separation bubbles control drag and maximum lift [7].

The reflex requirement. A tailless aircraft trims on its own pitching moment. About the CG,

$$C_{m,cg} = C_{m0} + C_L \frac{x_{cg} - x_{np}}{\bar{c}} + C_{m\delta} \delta_e = C_{m0} - SM C_L + C_{m\delta} \delta_e = 0, \quad (2)$$

so with neutral elevons it trims at $C_L = C_{m0}/SM$. At $SM = 0.08$ and the cruise $C_L = 0.40$ the aircraft needs $C_{m0} \approx +0.03$. A cambered section such as the Clark Y ($c_{m0} \approx -0.09$) cannot get there with any practical washout, so the choice was restricted to reflexed sections with c_{m0} near zero, and sweep and washout supply the rest [8].

Candidates. To document the choice we ran NeuralFoil, the section model of the aircraft analysis, on seven sections at the root and tip Reynolds numbers (Figure 4). The Hepperle MH45 and MH60 [9] combine a small c_{m0} (-0.009 , -0.011) with low drag at $c_l = 0.4$ (0.0092) and $c_{l,max} = 1.19$

at the tip. The strongly reflexed MH49 and E186 trim more easily (c_{m0} of +0.029 and +0.008) but lose 15–17 % of $c_{l,\max}$ and have 18–24 % more drag. The MH62 has 3 % less drag but is thinner (9.3 %) and has 5 % less $c_{l,\max}$. The S5010 is the closest rival: it has 5 % more tip $c_{l,\max}$ and a less negative c_{m0} , but 6 % more drag at the cruise lift coefficient. We kept the MH45 because cruise drag drives range and the SU2 campaign was run on it. If flight tests show that stall margin is the binding constraint, the S5010 is the next section to check.

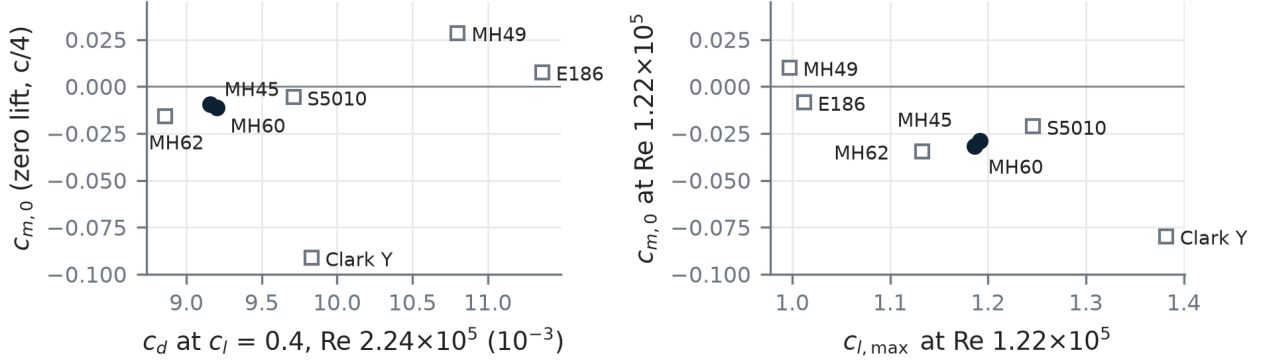


Figure 4: Post-hoc airfoil trade (NeuralFoil). Left: zero-lift moment against drag at the cruise c_l , wing-root Reynolds number. Right: c_{m0} against $c_{l,\max}$ at the tip Reynolds number. Filled: MH45 and MH60, used on BLUEBIRD.

Centre-body thickness. The MH60 is 10.1 % thick, which is 49.3 mm of depth at the battery station ($x = 205$ mm) of the 520 mm root. Scaling it by 1.15 to 11.6 % gives 56.7 mm, enough for the 12 mm phone envelope over the 25 mm pack with room for the hatch, tray and skins. NeuralFoil puts the cost at 6 % more centre-body section drag, about 0.4 % of the aircraft’s cruise drag, with a slightly higher $c_{l,\max}$ (1.26 against 1.21).

4.2 Section polars and the SU2 RANS cross-check

All section data in the aircraft model come from NeuralFoil [10], a neural surrogate trained on XFOIL [11] polars, called through AeroSandbox 4.2.10 at each section’s Reynolds number and Mach 0.03. It inherits XFOIL’s e^N transition model and its tendency to over-predict $c_{l,\max}$ near $Re = 10^5$. XFOIL itself is not in the project toolchain, so the independent check is two-dimensional RANS.

Set-up. We ran incompressible RANS in SU2 8.5.0 [12] on the three sections that set the stall estimate: MH60 at $Re = 353\,165$, MH45 at 224 124 and MH45 at 122 249. Turbulence used SST (2003 variant) [13, 14], and transition used the four-equation Langtry–Menter $\gamma-Re_\theta$ model [15] (“LM”) at a free-stream turbulence intensity of 0.1 %. A fully turbulent SST run (“FT”) bracketed it. Convection was flux-difference splitting with MUSCL reconstruction, and time stepping was implicit Euler with FGMRES and ILU(0), with the CFL number ramped from 2 to a cap of 10–25. Each α restarted from the previous one. The coefficients are the mean of the last 300 iterations. A point counts as steady if c_l moved less than 0.01 peak-to-peak over that window.

Mesh. The gmsh mesh has its far field at $50c$, a quad boundary layer (first cell $1.5 \times 10^{-5}c$, y^+ of about 0.1–0.25) and refined triangles over the suction side and a $3c$ wake box. The medium mesh

(400 surface points, 45 layers at growth 1.12, 98 200 cells) matches the fine one (188 500 cells) at MH45, 122 249, 4° to 0.6 % in c_l , 7 % in c_d and 0.006 in c_m . The coarse mesh could not resolve bubble reattachment and converged to a fully separated flow, so it was rejected.

Convergence. All LM points from 0–6°, which set the cruise numbers, are steady. Near stall the LM solutions are not unique. At the same α , chains started differently settle on different branches (MH60 at 11°: $c_l = 1.05$ or 1.32), depending on whether the bubble bursts or reattaches in pseudo-time. This is a known limit of steady RANS with transition at Re between 1×10^5 and 4×10^5 , so we quote $c_{l,\max}$ as a range. The campaign cost about 43 process-hours.

Table 5: SU2 against NeuralFoil (NF); c_d at $c_l = 0.35$; LM $c_{l,\max}$ is the range over solution branches (aero/cfd/results.yaml).

Case	$c_{l,\max}$			c_d at $c_l = 0.35$			c_{m0}
	NF	LM	FT	NF	LM	FT	NF / LM
MH60, 353 165	1.21	1.06–1.44	1.37	0.0075	0.0083	0.0135	−0.004 / −0.011
MH45, 224 124	1.22	1.02–1.10	1.33	0.0091	0.0111	0.0147	−0.010 / −0.020
MH45, 122 249	1.19	0.87–0.92	1.18	0.0129	0.0169	0.0161	−0.029 / −0.031

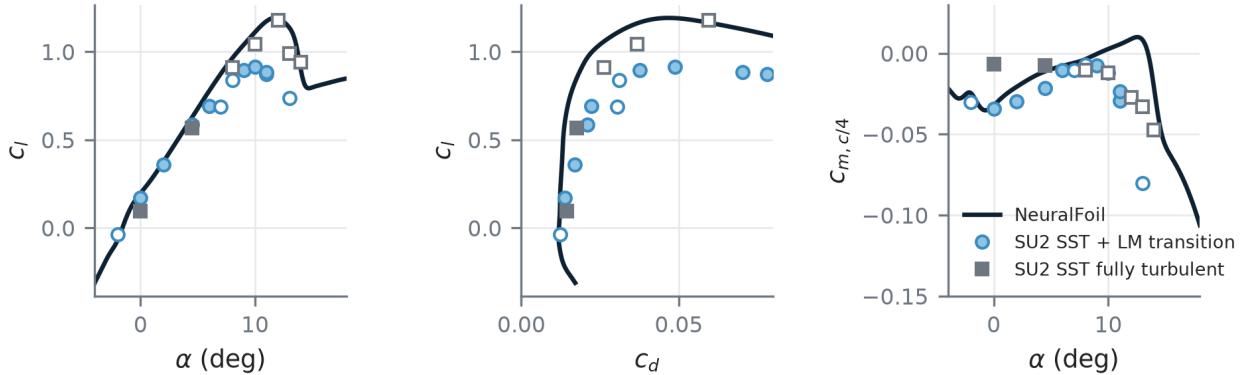


Figure 5: MH45 at the critical tip Reynolds number (122 249): NeuralFoil against SU2 with Langtry–Menter transition and fully turbulent SST. Hollow markers were still moving at the iteration cap; shedding points are omitted.

What agreed and what did not. In the linear range (Table 5 and Figure 5) LM tracks NeuralFoil to 0.03–0.04 in c_l , and its c_m agrees to 0.002 at the two higher Reynolds numbers. At 122 000, LM is 0.010 more nose-down at cruise because of a mid-chord bubble. NeuralFoil’s drag is 10–31 % below LM’s, and the gap grows as Re falls; fully turbulent SST over-predicts drag by 25–80 %, as expected. The important disagreement is stall. Every transitional MH45 solution stalls earlier and lower than NeuralFoil, because the leading-edge bubble bursts at 9–11°. At the tip LM gives $c_{l,\max}$ of 0.87–0.92 against 1.19. Fully turbulent SST matches NeuralFoil there only because it cannot form the bubble, so it is not a lower bound.

What changed in the design. SU2 changed three things. (i) The three-dimensional $C_{L,\max}$ knock-down in `aero/analyze.py` went from 0.90 to 0.80. This is equivalent to a tip $c_{l,\max}$ of 1.05, the midpoint of LM and FT and 12 % below NeuralFoil, and it lowered $C_{L,\max}$ from 1.012 to 0.90.

(ii) A 0.002 laminar-bubble drag add-on entered the build-up (Table 7). (iii) The minimum airspeed rose from 11 to 12 m/s, and later to 12.5 m/s once the trimmed stall was computed. The MH60 never limits stall (every model gives $c_{l,\max} \geq 1.06$). One gap remains: the MH60 was meshed at its native 10.1 %, not the 11.6 % flown.

4.2.1 Flow-field visualisation

The polars show *that* the tip section stalls early; the flow fields show *why*. We read the SU2 restart files directly (pressure, velocity, and the turbulence and intermittency variables at every node), matched them to the medium mesh (Figure 6) and computed the surface quantities ourselves: $C_p = 2p$ and $C_f = 2\mu u_t(y_1)/y_1$ from the first node off the wall, with $\rho = U_\infty = 1$ and $y_1 = 1.5 \times 10^{-5}c$ inside the viscous sublayer. Integrated round the airfoil, they reproduce SU2’s own c_l and c_d to better than 0.1 % for every solution shown. Reversed flow is where the surface-parallel velocity u_s is negative. On the upper surface we mark laminar separation S and reattachment R (zero crossings of C_f) and transition T, the first station where the Langtry–Menter intermittency γ exceeds 0.5 in the boundary or shear layer. The medium mesh places S and R within about $0.02c$ of the fine mesh (Figure 11).

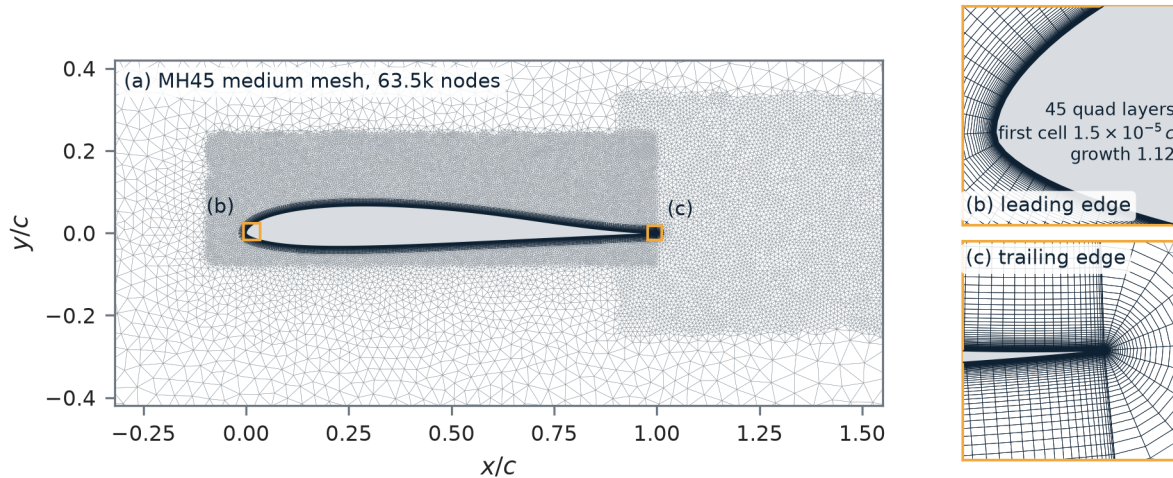


Figure 6: Medium mesh used for every production run, shown for the MH45 (63 522 nodes). (a) Near field, with the refined suction-side and wake boxes (far field at $50c$). (b) Leading edge and (c) trailing edge: 45 quad layers (first cell $1.5 \times 10^{-5}c$, growth 1.12) wrap the sharp trailing edge in a fan, so no cell is degenerate.

The cruise bubble. At the 4.5° reference cruise angle the MH45 tip section ($Re = 122\,249$) carries a laminar separation bubble over mid-chord (Figures 7 and 8). The laminar layer leaves the wall at $x/c = 0.22$, where the pressure starts to rise behind the suction peak. The separated shear layer stays laminar over a nearly constant-pressure plateau while the reversed flow beneath it thickens to 0.8 % of the chord. At $x/c = 0.45$ the shear layer transitions. Turbulent mixing brings high-momentum fluid down, the pressure recovers steeply, and the layer rolls up into a vortex whose reversed flow reaches $0.17U_\infty$ at $x/c = 0.48$. This vortex is the closed bubble at $x/c \approx 0.45$ – 0.55 noted in the mesh study. A very thin layer of slow reversed flow then persists to $x/c = 0.82$, where C_f turns positive. The bubble costs more pressure drag than laminar flow saves in friction: integrated over the surface, the transitional solution has $c_{d,p} = 0.0149$ and $c_{d,f} = 0.0058$, against 0.0064 and 0.0109 fully turbulent, so its total c_d is higher (0.0207 against 0.0173). The plateau also

shifts suction aft, which is the 0.010 extra nose-down c_m at this Reynolds number. At higher Re the bubble shrinks: it spans $x/c = 0.25$ – 0.54 at the MH45 root ($Re = 224\,124$), and 0.34 – 0.64 , at most 0.4% of the chord thick, on the MH60 centre body ($Re = 353\,165$, Figure 10).

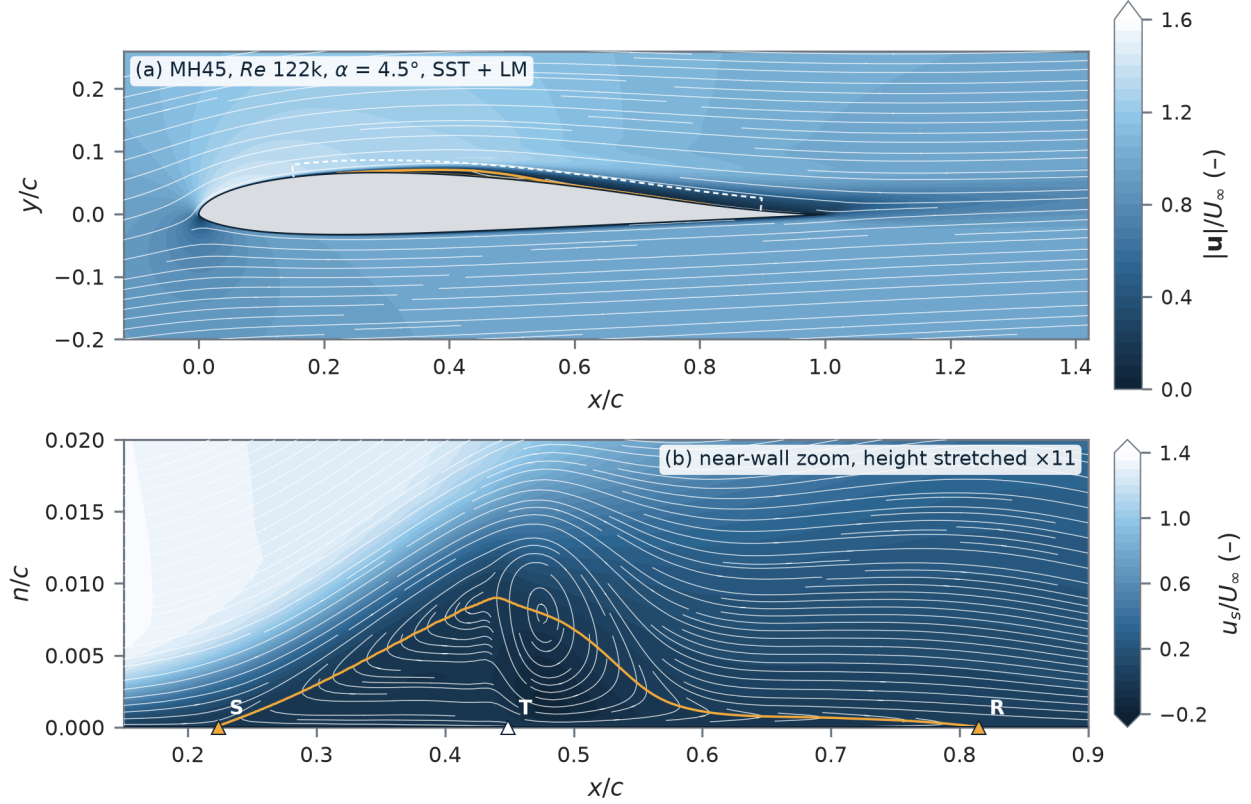


Figure 7: MH45 tip section at $Re = 122\,249$, $\alpha = 4.5^\circ$, SST + LM (steady). (a) Speed with streamlines. The amber line is $u_s = 0$, the edge of reversed flow, and the dashed band is the region drawn in (b). (b) Surface-parallel velocity in surface coordinates (x/c of the foot point on the upper surface, wall-normal height n/c , stretched $\times 11$): laminar separation S at $x/c = 0.22$, transition T at $x/c = 0.45$, the recirculating vortex at $x/c = 0.48$ and reattachment R at $x/c = 0.82$.

Why the bubble moves forward and bursts. Raising α sharpens the suction peak ($C_{p,\min}$ goes from -1.21 at 4.5° to -3.25 at 10°) and moves the adverse pressure gradient towards the nose, so the laminar layer separates earlier: at $x/c = 0.48$ at 0° , 0.22 at 4.5° and 0.025 at 10° (Figure 8(c)). Transition follows it forward (from 0.77 to 0.10), and the mid-chord bubble becomes a short leading-edge bubble, $x/c = 0.025$ – 0.17 at 10° (Figure 9). Behind it, the reattached turbulent layer must carry the whole pressure recovery to the trailing edge, starting from the momentum deficit the bubble left. It separates earlier as α rises: from $x/c = 0.49$ at 10° , where c_l peaks at 0.915 , and from 0.26 on the 11° hi-chain solution ($c_l = 0.872$), whose bubble reattaches at 0.23 and leaves only a sliver of attached flow. By 12 – 13° the shear layer no longer reattaches. Reversed flow runs from $x/c \approx 0.02$ to the trailing edge, the suction peak collapses ($C_{p,\min} = -1.44$ in the last 13° iterate) and c_l falls to about 0.67 – 0.74 , with vortex shedding at 12° . This is bubble bursting, and it gives an abrupt leading-edge stall. Whether the bubble just reattaches or bursts at 10 – 11° depends on the solver’s pseudo-time path, which is why the chains disagree and why $c_{l,\max}$ is a range, 0.87 – 0.92 .

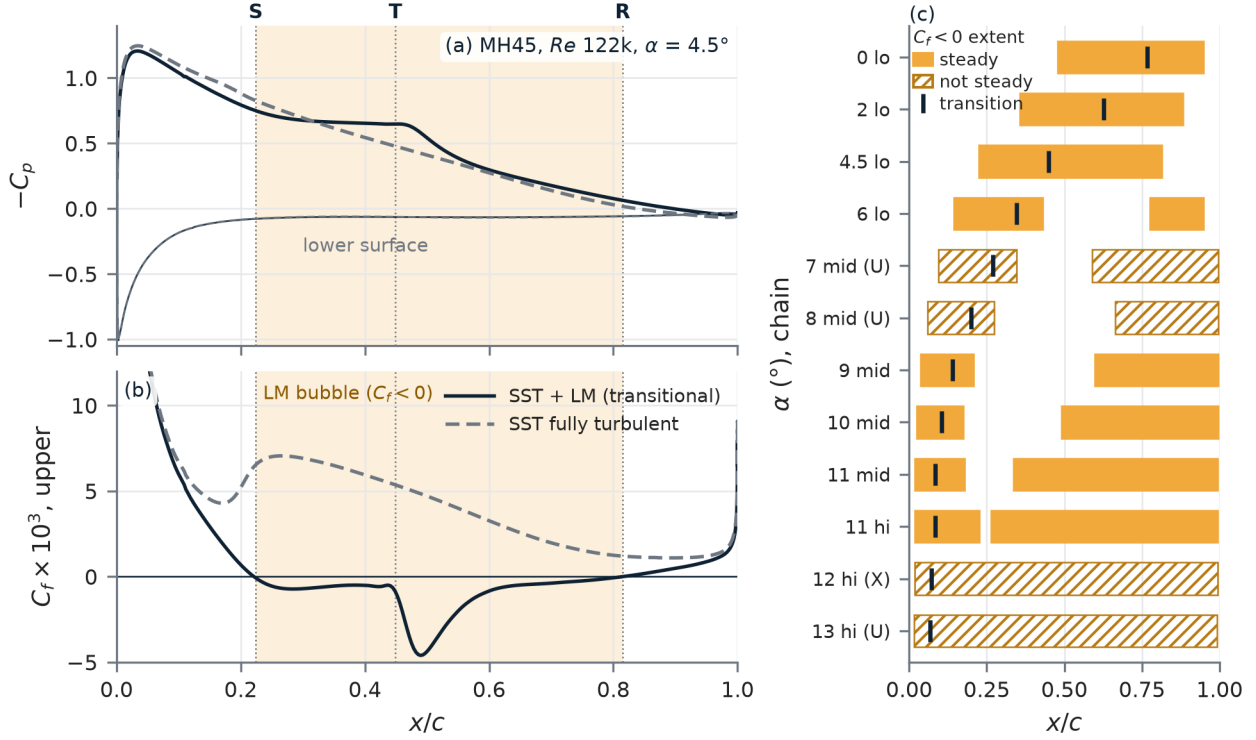


Figure 8: (a) Pressure and (b) upper-surface skin friction on the MH45 at $Re = 122\,249$, $\alpha = 4.5^\circ$: transitional SST + LM against fully turbulent SST (thick lines upper surface, thin lines lower). The shaded band is the LM reversed-flow region between S and R; T is transition. C_f is drawn with a three-point $[1\ 2\ 1]/4$ filter that removes a 1–2% node-to-node ripple. (c) Extent of $C_f < 0$ on the upper surface for every LM solution of the three chains (lo, mid, hi), with transition marked; hatched bars were not steady (U: still drifting at the iteration cap, X: shedding).

Why fully turbulent RANS misses it. The fully turbulent run has no intermittency to hold its eddy viscosity down. At 4.5° its eddy viscosity stays below the molecular value only up to $x/c \approx 0.19$ and then grows quickly, just ahead of the point where the transitional solution’s laminar layer separates (0.22). The dip in its C_f near $x/c = 0.17$ (Figure 8(b)) is this delayed growth: at 0.1% free-stream turbulence, even the “fully turbulent” SST run is not turbulent from the nose. A turbulent profile is fuller and carries more momentum near the wall, so it rides through the same adverse gradient, and C_f stays positive over the whole upper surface. With no bubble to burst, SST stalls gradually from the trailing edge: at 12° its separation has only crept forward to $x/c = 0.40$, and c_l is still 1.18. That matches NeuralFoil’s 1.19, but for the wrong reason. The fully turbulent model over-predicts friction drag and cannot represent the mechanism that limits $c_{l,\max}$ at $Re \approx 1 \times 10^5$. Neither it nor NeuralFoil, which is trained on XFOIL [11], is a safe bound here.

What this means for the design. The outer wing will stall by bubble bursting: abruptly, from the leading edge, and at a section $c_{l,\max}$ 23–27% below the NeuralFoil value the stall model used. This is why the CFD check cut the $C_{L,\max}$ knock-down to 0.80 and raised the minimum airspeed to 12 m/s rather than trusting the polar. At cruise the same bubble dominates the tip section’s pressure drag and trims its c_m slightly nose-down, so the tip should not be modelled with a fully turbulent or bubble-free polar. All of this depends on where a laminar layer separates and transitions, which in turn depends on free-stream turbulence and surface finish. Printed skin and

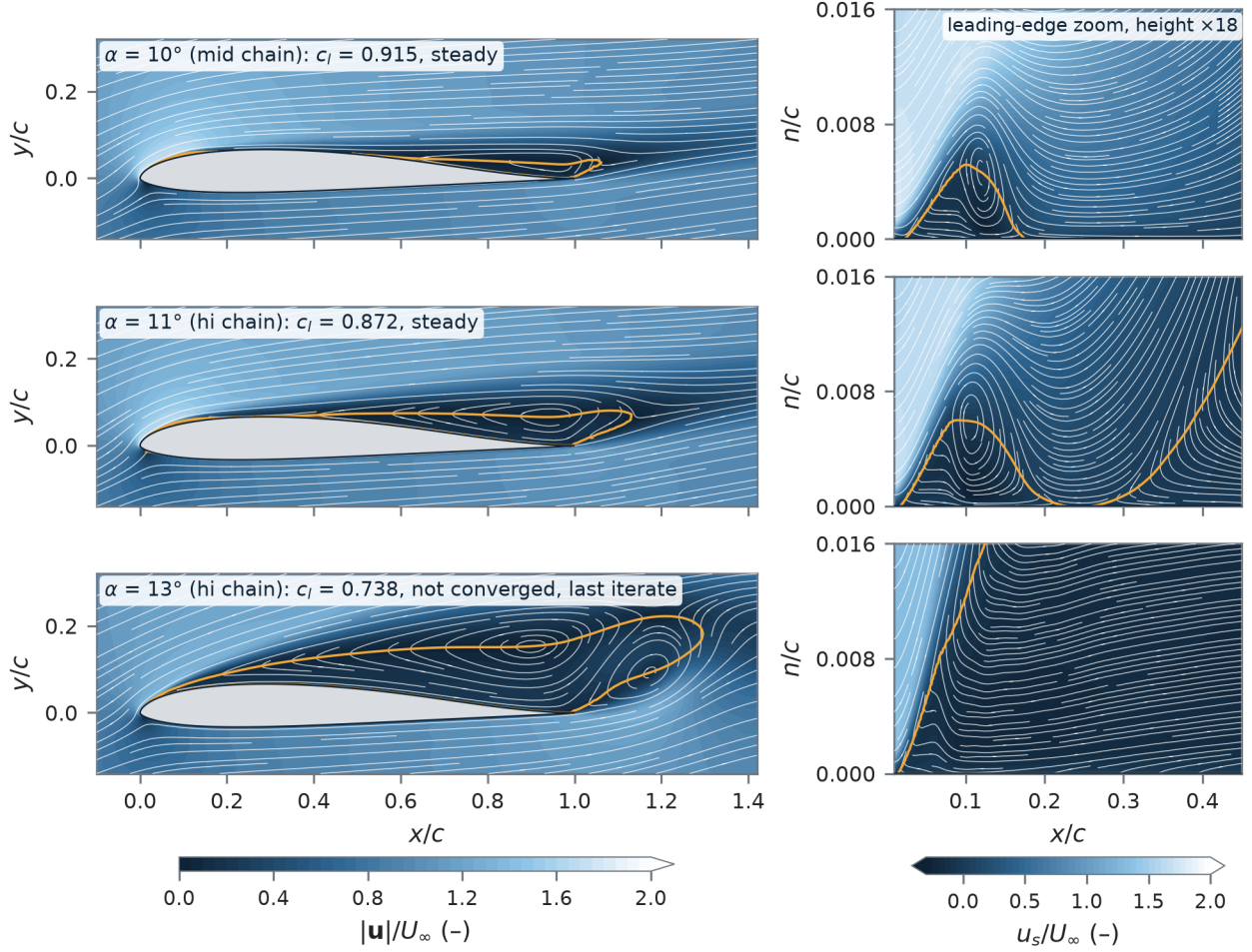


Figure 9: MH45 at $Re = 122\,249$ near and past stall, SST + LM. Left: speed with streamlines and the $u_s = 0$ line (amber). Right: the front 45 % of the upper surface in surface coordinates (height $\times 18$), coloured by u_s/U_∞ . At 10° (mid chain, steady) a short leading-edge bubble sits at $x/c = 0.025$ – 0.17 and the trailing edge separates from 0.49 . At 11° (hi chain, steady) the bubble reattaches at 0.23 and the flow separates again at 0.26 . At 13° (hi chain, last iterate of a run still drifting at the iteration cap) the bubble has burst and the flow is separated from $x/c = 0.017$ to the trailing edge.

film seams may trip the layer earlier or later than these smooth-wall solutions. The pictures also show single steady-RANS states of a flow that is unsteady near stall. The stall behaviour must therefore be confirmed with a slow-flight test at altitude.

4.3 Three-dimensional analysis

The working model is AeroSandbox’s AeroBuildup [16]. It is a strip build-up that uses NeuralFoil section polars at the local Reynolds number, adds lifting-line induced drag and sweep corrections, and includes the fins. It is fast enough to trim at every speed, and it sets the neutral point, the CG target and the performance. It does not model stall, so $C_{L,\max}$ is estimated separately (Section 4.6). Two inviscid vortex-lattice models cross-check it. The first is AeroSandbox’s VLM. The second is VSPAERO 7 on an OpenVSP 3.53 model [17], which `aero/vsp_model.py` builds from `params.yaml` (thickened MH60, MH45 panels, 35° -swept fins, three wake iterations, $Re_{\bar{c}} = 3.0 \times 10^5$).

The lift slopes agree to 6 %: 0.081 per degree from AeroBuildup and 0.076 from VSPAERO. The

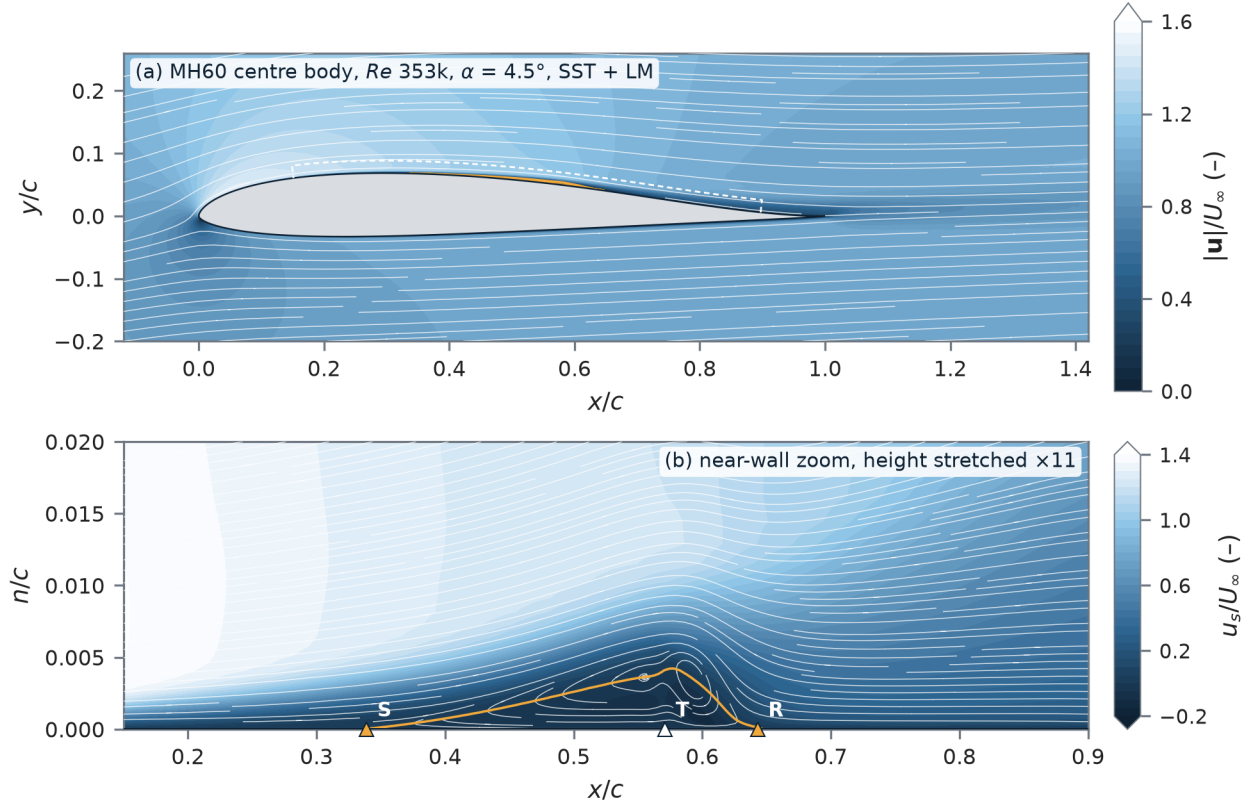


Figure 10: MH60 centre body at $Re = 353\,165$, $\alpha = 4.5^\circ$, SST + LM (steady), in the same format and at the same scales as Figure 7. The bubble is shorter and thinner than on the MH45 tip: S at $x/c = 0.34$, T at 0.57, R at 0.64, and at most 0.4% of the chord thick.

neutral points are 338.0 (AeroBuildup), 340.3 (VSPAERO) and 350.9 mm (VLM) aft of the apex, a spread of 4% of $\bar{c}$. AeroBuildup is the most forward, so placing the CG from it is conservative: at the 313 mm CG, VSPAERO gives 8.8% static margin against the design 8%. Both models give a positive zero-lift moment (+0.014 to +0.016) although both sections have negative c_{m0} : sweep and washout supply the difference, as Equation (2) requires. VSPAERO's span efficiency is 0.87–0.90. The lateral and rate derivatives are in the stability section. One result feeds back here: the fins grew from 160 to 200 mm, with 10° of outboard cant, for Dutch-roll damping. That costs 0.6 W at cruise and is included in every number below.

4.4 Planform trade: sweep, taper and washout

The sweep, taper and washout were set at the layout stage from flying-wing practice. For this paper we tested them one at a time with the trim model at 2.52 kg. For each variant we re-located the neutral point, put the CG at 8% static margin and trimmed at 14 and 11 m/s. A VLM spanload at $C_L = 0.75$, divided by the NeuralFoil $c_{l,max}$ at each station's Reynolds number, shows where the wing stalls first (Table 6 and Figure 12a).

Washout. With no washout the first section to stall is at $\eta = 0.63$, inside the elevon span, and cruise needs -7.5° of elevon. At -3° the first stall moves to $\eta = 0.40$, the inboard end of the elevon, so the outer elevon keeps roll authority as the stall develops. The cruise trim falls to -2.9°

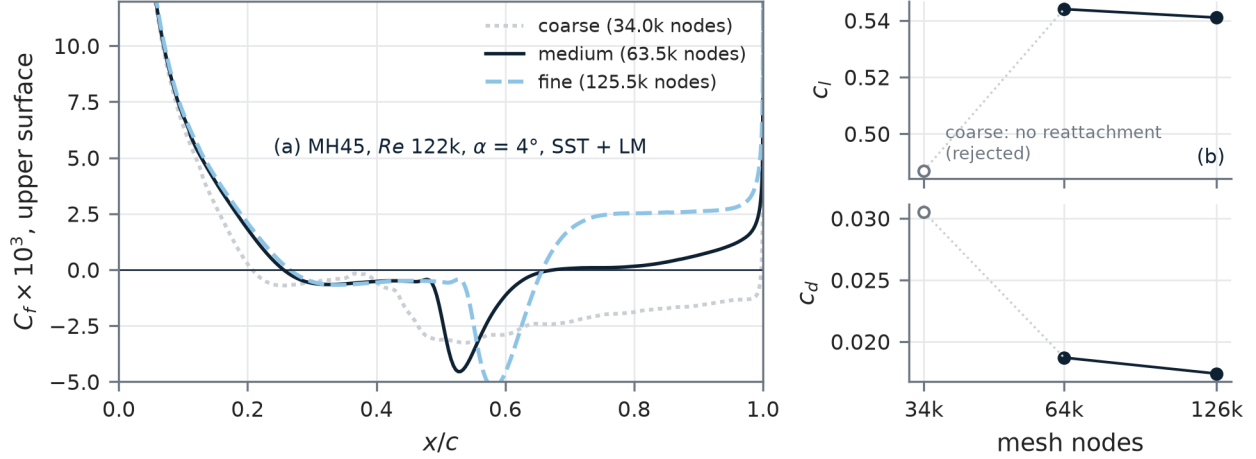


Figure 11: Mesh sensitivity on the MH45 at $Re = 122\,249$, $\alpha = 4^\circ$, SST + LM. (a) Upper-surface C_f . The medium and fine meshes agree on laminar separation ($x/c = 0.256$ and 0.267) and reattachment (0.675 and 0.656). Transition moves from 0.49 to 0.53 , and the fine mesh recovers more turbulent skin friction behind the bubble. Their 7% difference in c_d is pressure drag (0.0124 on the medium mesh against 0.0107 on the fine one). The coarse mesh never reattaches. (b) c_l and c_d , mean of the last 300 iterations; the coarse solution was rejected.

Table 6: Planform trade at 2.52kg , CG at 8% SM. $\eta_s = 2y/b$ of the first station to reach its $c_{l,\max}$ (figscripts/aero_trim.yaml).

Variant	S (m ²)	x_{np} (mm)	δ_e at 14	L/D at 14	L/D at 11	η_s	Re_{tip} at 10
Design (22° , 180 mm , -3°)	0.520	338.0	-2.9	11.44	11.91	0.40	122 000
Washout 0°	0.520	329.7	-7.5	11.12	11.13	0.63	122 000
Washout -1.5°	0.520	333.3	-5.3	11.31	11.56	0.56	122 000
Washout -4.5°	0.520	341.9	-0.4	11.50	12.10	0.40	122 000
LE sweep 15°	0.520	302.9	-5.0	11.32	11.52	0.20	122 000
LE sweep 30°	0.520	383.6	-1.4	11.49	12.06	0.40	122 000
Tip chord 140 mm	0.492	330.0	-4.0	11.82	12.00	0.56	95 000
Tip chord 220 mm	0.549	345.6	-2.0	11.09	11.75	0.40	149 000

and L/D rises by 2.9% . Going on to -4.5° would save another 2.4° of elevon for 0.5% more L/D , which is within the model’s uncertainty. We kept -3° , which the stability and loads work already use. Washout is cut into the foam cores, so it is cheap to change after flight test.

Sweep. Sweep moves the neutral point strongly: -35 mm at 15° and 46 mm at 30° . The payload stack fixes the achievable CG, and the battery trim moves it by only 8.4 mm . At 22° the required CG (313 mm) is where the detailed CAD puts it (312.5 mm). More sweep would cut trim drag slightly (0.4% L/D at 30°) but would need a new internal layout.

Taper. A 140 mm tip gains 3.3% in L/D from the smaller area. It also raises the wing loading by 6% , moves the first stall into the elevon span ($\eta = 0.56$) and drops the tip Reynolds number to $95\,000$, below the $122\,000$ at which SU2 already shows the bubble cutting $c_{l,\max}$. A 220 mm tip costs 3.1% in L/D . The 180 mm tip is the smallest that keeps the tip above $Re = 1.2 \times 10^5$ at 10 m/s and the first stall inboard of the elevon.

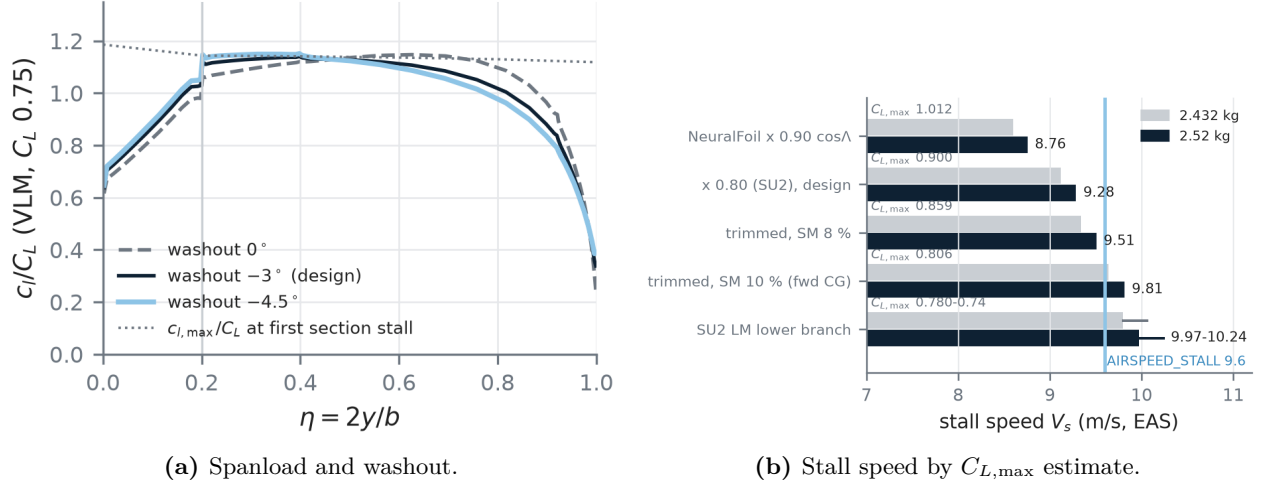


Figure 12: Stall. (a) VLM section c_l over C_L at $C_L = 0.75$ (fins off, elevons neutral). The dotted line is the NeuralFoil section $c_{l,max}$ scaled to the C_L at which the first section reaches it; the step at $\eta = 0.2$ is the centre-body junction. (b) Stall speed at both masses for the estimates of Section 4.6, against the current AIRSPEED_STALL.

4.5 Drag build-up

AeroBuildup sees only clean, smooth wings and fins. Table 7 lists what is added, referenced to S . The clean model gives $C_D = 0.010$ at zero lift, so the add-ons are more than half the zero-lift drag and about 40 % of the cruise drag. They are the least certain part of the model: each 0.001 of C_{D0} costs about 2.5 % of range at 14 m/s.

Table 7: Parasite-drag add-ons on S (aero/analyze.py CD0_EXTRA).

Item	ΔC_{D0}	Basis
Printed-skin roughness and seams	0.0030	about 25 % over smooth skin friction
Belly skid and camera lip	0.0030	sled and lip in the free stream
Camera windows	0.0015	two nadir openings
Control gaps and hinges	0.0015	split elevons, unsealed
Pusher interference and spinner	0.0020	prop in the centre-body wake; thrust deduction
Pitot, antennas, miscellaneous	0.0010	
Laminar-bubble drag	0.0020	SU2: NeuralFoil c_d 10–30 % low
Total	0.0140	

4.6 Maximum lift and stall speed

We use a critical-section method. The lowest section $c_{l,max}$ on the span (the tip MH45 at $Re = 122\,249$, 1.192) is corrected for sweep and knocked down for three-dimensional and SU2 effects:

$$C_{L,max} = K c_{l,max,crit} \cos \Lambda_{c/4}, \quad V_s = \sqrt{\frac{2W}{\rho S C_{L,max}}}. \quad (3)$$

With $K = 0.80$ and $\Lambda_{c/4} = 19.4^\circ$ this gives $C_{L,max} = 0.90$, and $V_s = 9.12$ m/s at 2.432 kg or 9.28 m/s at 2.52 kg. The VLM spanload check above is independent: with NeuralFoil section limits and no knock-down, the first section stalls at $C_L = 1.07$, close to the 1.012 that $K = 0.90$ gave before SU2.

Trim lowers this further. At the 13.9° stall angle the elevons must deflect trailing-edge up to hold $C_{m, cg} = 0$, and that removes lift. The trimmed $C_{L, max}$ is 0.859 at 8% static margin and 0.806 at the forward CG (10%). Figure 12b collects the five estimates. At 2.52 kg the trimmed stall at the forward CG is 9.81 m/s, above the 9.6 m/s that the ArduPilot parameters [6] set for 2.43 kg. With the foam-wing mass, AIRSPEED_STALL should become 9.8 m/s and AIRSPEED_MIN ($1.3V_s$) 12.8 m/s. The worst case, the lower SU2 branch, is 10.0–10.2 m/s. A slow-flight test at altitude must settle it.

4.7 Trim, drag polar and lift-to-drag ratio

At each airspeed the trim solves $L = W$ and $C_{m, cg} = 0$ for α and the symmetric elevon angle by alternating root-finding, with the add-ons included. Our re-run at 2.432 kg reproduces `aero/results.yaml` to three significant figures. At 2.52 kg and the 14 m/s cruise (Figure 13) it gives $\alpha = 5.1^\circ$, $\delta_e = -2.9^\circ$, $C_L = 0.396$, $C_D = 0.0346$ and $L/D = 11.4$, with drag 2.1% above the lighter aircraft. The best L/D is 12.1 at 12 m/s ($C_L = 0.54$). A parabola through the trimmed points,

$$C_D = 0.0220 + 0.0786 C_L^2, \quad (4)$$

matches them to about 1% near cruise. Its k corresponds to an effective Oswald factor $e = 1/(\pi Ak) = 0.65$, far below the inviscid span efficiency of 0.87–0.99. The difference is lift-dependent profile drag and the up-elevon trim penalty, both of which grow at low speed.

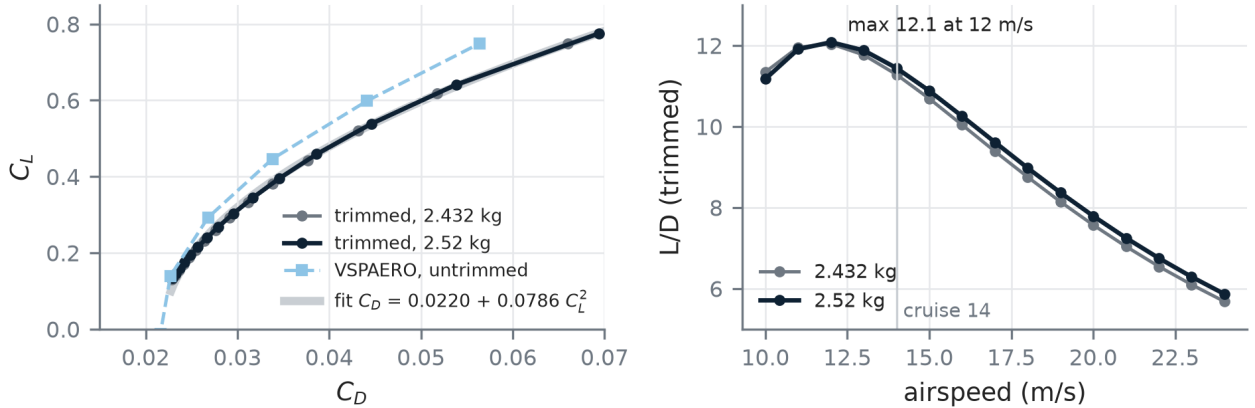


Figure 13: Left: trimmed drag polars (AeroBuildup plus add-ons) at both masses, the fit of Equation (4), and the untrimmed VSPAERO sweep with its own parasite estimate. Right: trimmed L/D .

5 Propulsion and performance

5.1 Propeller and motor

Selection on measured data. The candidates were all 18 aeronaut CAM Carbon folding propellers of 10–13 in diameter and 5–9 in pitch in the UIUC database [18, 19], crossed with four T-Motor motors (72 combinations). Each motor was reduced to a model fitted to the maker’s supply-side test rows [20]:

$$Q = K_t(I - I_0(n)), \quad V_m = \frac{n}{K_v} + IR, \quad I_0(n) = I_{0, ref} \sqrt{n/n_{ref}}, \quad K_t = \frac{60}{2\pi K_v}. \quad (5)$$

This model is coupled to the measured $C_T(J, n)$ and $C_P(J, n)$ and a $35\text{ m}\Omega$ pack, and solved for the rpm at which thrust equals the trimmed drag. For the AT2814 900KV the fit gives $K_v = 806\text{ rpm/V}$, $R = 59\text{ m}\Omega$ and $I_0 = 3.2\text{ A}$ at 9000 rpm (the datasheet says 1.2 A), with 2.0% rms power error over 46 rows. The large no-load current is where the cruise losses are: the motor and ESC are only 68% efficient at 14 m/s.

Table 8: Propeller-motor matching (docs/prop.md; 2.432 kg, 7.7 W avionics). Range at the best-range speed; $\eta_{14} = TV/P_{\text{cells}}$ at 14 m/s; launch is installed static thrust at 16.4 V and $\leq 40\text{ A}$.

Prop	Motor	Range (km)	η_{14}	Launch (N)	Verdict
11×8	AT2814 900KV	65.5	0.510	21.4	+1.4% range; throttle cap
11×7	AT2814 900KV	64.6	0.501	22.8	chosen
10×8	AT2814 900KV	64.3	0.501	18.1	0.1 N launch margin
11×7	AT2820 880KV	54.4	0.420	22.2	31 g heavier; poor part-load
11×7	AT2312 1150KV	66.7	0.507	15.2	fails launch; 4 mm shaft
12×9	AT2814 900KV	68.6	0.533	21.8	dolly clearance 42 mm (min. 45)

Decision. The 11×7 on the AT2814 900KV (Table 8) is within 1.4% of the best propeller that fits the dolly, which is inside the model’s uncertainty, and it keeps the most launch thrust. Larger discs gain about 6% in range because lower rpm raises the motor’s part-load efficiency, but they break the 45 mm dolly clearance. The 12×9 is the documented upgrade if the dolly seat is raised. The light AT2312 is good in cruise but cannot reach 18 N within its 25 A limit.

BEMT check. A blade-element momentum model with Prandtl tip and hub losses [21] was run on the CAD blade (22 stations, NeuralFoil polars at local Re and Mach). With the nominal 7-in pitch it reads 7% low: 14.2 N and 207 W against 15.2 N and 223 W measured at 8000 rpm. It also reaches zero thrust at $J = 0.74$ instead of 0.82. Adding 2° of pitch puts it on the measured C_T for $J \geq 0.5$ (Figure 14, left). The retail blade has more effective pitch than its label, which is why the selection used measured data.

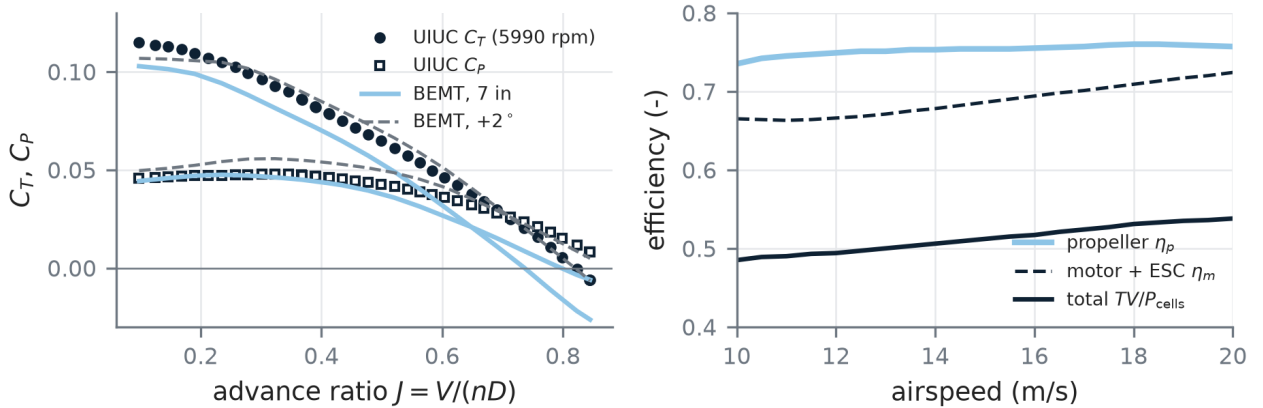


Figure 14: Left: UIUC C_T and C_P of the aeronaut CAM Carbon 11×7 at 5990 rpm against BEMT on the CAD blade. Right: efficiency along the trimmed cruise curve at 2.52 kg (measured prop, fitted AT2814 900KV, 14.8 V pack).

Operating points at 2.52 kg. At 14 m/s the propeller turns at 4920 rpm ($J = 0.61$, $\eta_p = 0.75$). The total $\eta = TV/P_{\text{cells}}$ is 0.51, which confirms the 0.50 assumed in `aero/results.yaml`, and the pack supplies 4.0 A. At full throttle on a fresh 16.4 V pack the installed static thrust is 23.0 N at 39.9 A, just inside the 40 A ESC. The full-throttle rpm (about 9300 in flight, 10 200 static) is beyond the highest UIUC sweep (5990 rpm) and static point (7540 rpm), so the model holds the coefficients at the last measured values there.

5.2 Battery

The pack is 4S2P Molicel P45B 21700 Li-ion [22]: eight 4.5 A h, 70 g cells, 14.4 V nominal. The mission energy is

$$E_{\text{mission}} = n_s n_p C V_{\text{cell}} f_u (1 - f_r) = 4 \cdot 2 \cdot 4.5 \text{ A h} \cdot 3.6 \text{ V} \cdot 0.80 \cdot 0.80 = 82.9 \text{ W h}, \quad (6)$$

where $f_u = 0.80$ of the rated 129.6 W h counts as usable and $f_r = 0.20$ is the landing-site reserve. At 600 g (560 g of cells) the pack stores 216 W h/kg.

Li-ion or LiPo. The ratio of peak to cruise power decides. Cruise draws 4.0 A, or 2.0 A per cell (0.45 C). Even the launch, capped at 40 A, is 20 A per cell (4.4 C), below the cell's 45 A (10 C) rating. The high discharge rate that is a LiPo's main advantage is not needed, so specific energy decides. We did not model a specific LiPo pack; any LiPo alternative must exceed 216 W h/kg at pack level, at the same usable fraction, to match the range. The Li-ion costs are voltage sag (1.4 V at 40 A through 35 m Ω , handled by a sag-compensated failsafe) and heat soak. On a 35 °C day the pack reaches 47 °C at launch under a light skin, against a 60 °C discharge limit.

5.3 Power required and available; climb

Figure 15 (left) shows the power curves at 2.52 kg. The required shaft power is DV from the trim table. The electrical power adds the propulsion losses and 9.3 W of avionics. The available power is full-throttle thrust times airspeed at 14.8 V. The climb follows from the excess power,

$$\sin \gamma = \frac{T_{\text{avail}} - D}{W}, \quad \dot{h} = V \sin \gamma, \quad (7)$$

with D taken at $L = W$, which is conservative for steep climbs. The best climb rate is 8.3 m/s at 15.5 m/s on a mid-charge pack and 7.2 m/s on an empty one (14.0 V). ArduPilot limits the commanded climb to 4 m/s (TECS_CLMB_MAX), so even an empty pack at 2000 m and 30 °C keeps a margin of about 1.5. Full throttle in flight draws 35 A.

5.4 Endurance and range

A battery aircraft does not get lighter in flight, so endurance and range follow directly from energy [23]:

$$t = \frac{E_{\text{mission}}}{P_{\text{cells}}(V) + P_{\text{av}}}, \quad R = Vt; \quad \text{for } P_{\text{av}} = 0: \quad R = \frac{E_{\text{mission}}}{W} \eta_{\text{tot}} \frac{L}{D}. \quad (8)$$

The electric Breguet form on the right gives 69.9 km at 14 m/s and 2.52 kg ($\eta_{\text{tot}} = 0.506$, $L/D = 11.4$). With the 9.3 W of avionics the range is 60.5 km, so the avionics cost 13 % of the range. Table 9 compares the published numbers with our re-computation.

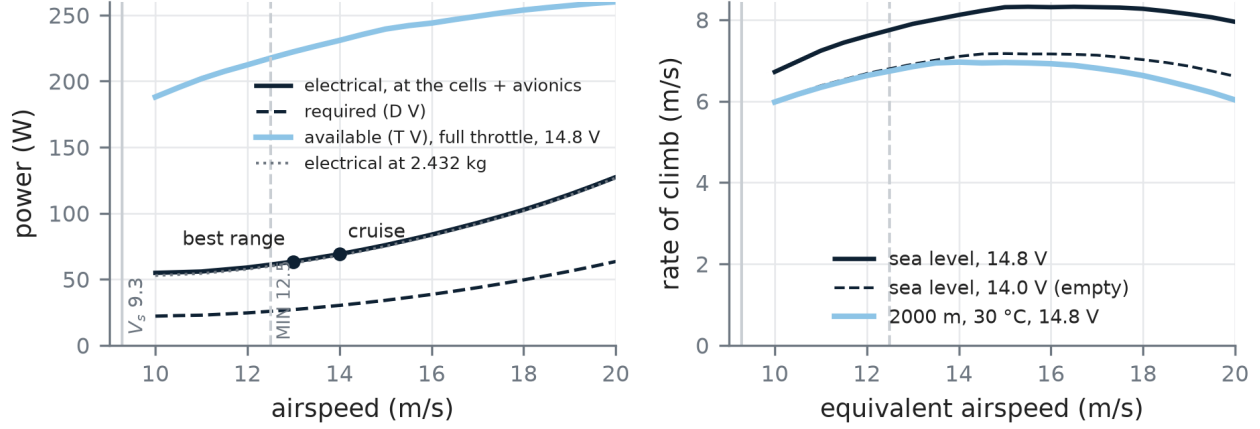


Figure 15: Left: shaft power required, electrical power (cells plus avionics) and full-throttle power available at 2.52 kg; dotted, electrical power at 2.432 kg. Right: rate of climb at sea level (mid and empty pack) and at 2000 m, 30 °C.

Table 9: Sea-level performance, still air, 20% reserve. A: `aero/results.yaml` (fixed $\eta = 0.50$, 7.7 W avionics). B, C: this paper (measured prop, fitted motor, 9.3 W).

	A: 2.432 kg	B: 2.432 kg	C: 2.52 kg
Stall, $C_{L,max} = 0.90$ / trimmed at SM 8% (m/s)	9.12 / 9.34	9.12 / 9.34	9.28 / 9.50
Cruise 14 m/s: C_L ; L/D	0.382; 11.3	0.382; 11.3	0.396; 11.4
Cruise: shaft / total electrical power (W)	29.6 / 66.9	29.6 / 68.1	30.3 / 69.1
Cruise: endurance (h) / range (km)	1.24 / 62.5	1.22 / 61.4	1.20 / 60.5
Best range: speed (m/s); range (km)	12; 64.9	13; 62.3	13; 61.2
Endurance at AIRSPEED_MIN 12.5 m/s (h)	–	1.38	1.36
Best climb at 14.8 / 14.0 V (m/s)	–	8.65 / 7.45	8.32 / 7.17

The 2.52 kg mass (2.1 % more drag) and the higher avionics estimate (7.7 to 9.3 W) together cost 3% of cruise range. The measured propeller data move the best-range speed from 12 to 13 m/s, because the total efficiency rises with speed (0.49 at 12, 0.51 at 14 m/s). The honest summary is a still-air range of about 61 km and an endurance of 1.2–1.36 h after the reserve. The largest uncertainties are the drag add-ons and the motor’s part-load efficiency, and both need measurement: [\[pending: thrust-stand efficiency and in-flight power at 14 m/s\]](#).

5.5 Hot-and-high operation

The East African survey sites that motivate BLUEBIRD lie at about 1500–2000 m in 25–30 °C air, and the design case adds a 35 °C day. The ground planner’s density model (`density_ratio` in `landing.py`) takes the elevation as the ISA pressure altitude [24] with the actual temperature:

$$\sigma = \frac{\rho}{\rho_0} = \frac{p(h)}{RT\rho_0}, \quad p(h) = 101\,325 \text{ Pa} \left(1 - 2.255\,77 \times 10^{-5} h\right)^{5.25588}, \quad (9)$$

with $R = 287.05 \text{ J/(kg K)}$ and $\rho_0 = 1.225 \text{ kg/m}^3$. ArduPilot flies equivalent airspeed, so the coefficients at each EAS are unchanged and

$$V_{TAS} = V_{EAS}/\sqrt{\sigma}, \quad P_{req} = P_{req,0}/\sqrt{\sigma}, \quad s_{land} \propto V_{TAS}^2 \propto 1/\sigma. \quad (10)$$

The planner scales its 60 m minimum landing run by $1/\sigma$ above a 300 m density altitude. We re-solved the propeller and motor at each site density (Table 10 and Figure 16, 2.52 kg).

Table 10: Hot-and-high performance at 2.52 kg. Cruise at 14 m/s EAS; speeds in m/s TAS; launch is installed static thrust on a fresh pack.

Site	σ	DA (m)	V_s	Cruise	P (W)	t (h)	R (km)	Launch (N)
ISA sea level	1.000	0	9.28	14.00	69.1	1.20	60.5	23.0
Sea level, 35 °C	0.935	690	9.60	14.48	71.0	1.17	60.9	22.0
1500 m, 25 °C	0.807	2180	10.33	15.59	75.7	1.10	61.5	19.9
1500 m, 30 °C	0.793	2350	10.42	15.72	76.2	1.09	61.6	19.7
2000 m, 25 °C	0.758	2790	10.66	16.08	77.7	1.07	61.8	19.1
2000 m, 30 °C	0.746	2950	10.75	16.21	78.3	1.06	61.9	18.8
2000 m, 35 °C	0.734	3110	10.83	16.35	78.8	1.05	61.9	18.6

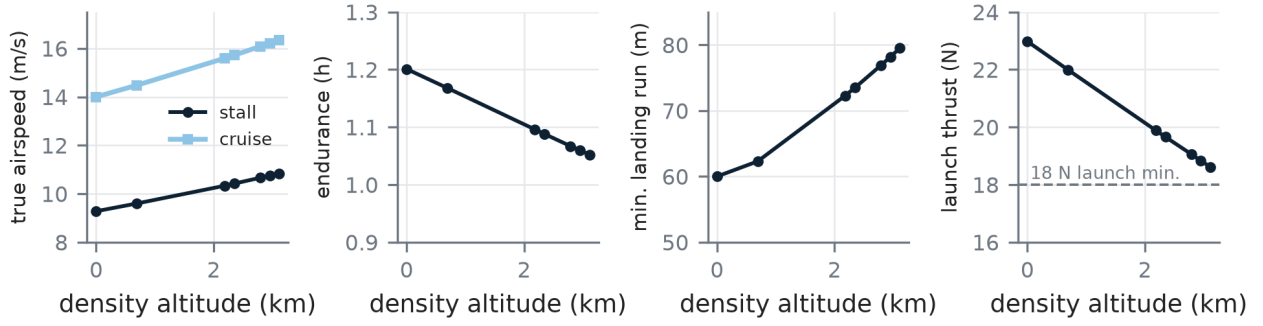


Figure 16: Hot-and-high trends at 2.52 kg against density altitude (the sites of Table 10): true stall and cruise speeds, endurance at 14 m/s EAS, the planner’s minimum landing run, and installed launch thrust against the 18 N minimum.

At the worst case (2000 m, 35 °C, $\sigma = 0.73$) the density altitude is 3.1 km:

- **Stall and landing.** The indicated stall is unchanged, but the true stall rises from 9.3 to 10.8 m/s. The minimum landing run grows from 60 to 80 m (72 m at 1500 m, 25 °C), so fewer patches qualify.
- **Endurance and range.** Cruise power rises by $1/\sqrt{\sigma}$, from 69 to 79 W, and endurance falls 12 % to 1.05 h. The true speed rises by the same factor, so still-air range stays near 62 km.
- **Reynolds number.** Reynolds numbers fall by 17 %, to about 94 000 at the tip at stall. An AeroBuildup trim at the 2000 m, 30 °C Reynolds numbers gives 2.6 % more drag than the sea-level coefficients in the table, so endurance and range there are about 2.5 % lower. NeuralFoil’s tip $c_{l,max}$ drops 2 % at 95 000, but SU2 has not been run there: [pending: SU2 MH45 polar at $Re \approx 9.5 \times 10^4$].
- **Launch.** Static thrust falls from 23.0 to 18.6 N, just above the 18 N set for sea level, while the true lift-off speed rises by $1/\sqrt{\sigma}$. Scaling the simulated dolly roll ($s \propto mV_{TAS}^2/T$) takes 13.0 m at 2.43 kg to about 23 m at 2.52 kg, 2000 m and 30 °C: [pending: dolly simulation at site density]. A bungee launch is the robust option at high sites.
- **Climb.** The best climb falls to 6.9 m/s (5.9 on an empty pack), still above the 4 m/s autopilot limit.

These results change operating rules, not the airframe. At high sites the planner needs the forecast temperature (its `--temp` input), and the ArduPilot airspeeds stay in EAS. The phone is a separate limit: above about 38 °C it cannot run the full payload, so a 35 °C day is near its edge.

Table 11: Design decisions in Sections 6 and 7, the alternatives, and the number that decided each.

Decision	Alternatives	Deciding number
Reflexed sections, -3° washout	no washout; conventional camber	cruise trim -2.8° against -5.9° / -10.5° ; L/D 11.3 against 11.1 / 10.3
Static margin 6–10 %, design 8 %	5 %; 11 %	short period meets the phugoid at 5 %; trimmed stall 9.6 m/s at 10 %
Fin 200 mm, 10° cant	160 mm; 200 mm vertical	Dutch-roll ζ 0.088 $\rightarrow$ 0.103 with the RSS-low spiral kept ≥ 8 s
Split-elevon crow	flaps, spoilers, parachute	no extra surface; ΔC_D 0.05–0.11; $ \Delta C_m \leq 0.06$
Crow lower halves 12°	51° , symmetric	at 51° the lower trailing edge hung 15 mm below the skid
Parallelogram 1:1 linkage	offset four-bar	split error 0.0° against 1.2°
Dolly seat at the 8° lift-off attitude	rotate on a rear pin	roll 13 m (33 m at 2°); no prop strike at 7.8°
4-inch wheels, 3.5 Hz torsion bars	tyres only; 4.5 Hz	66 of 66 runs clean over 5–20 mm bumps
Deep TPU skid sled	wheels; thin skid	abuse-case load at the CG 94 g $\rightarrow$ 43 g, no bottoming

Everything in this part of the paper is predicted, not measured. Five measurements would close the largest gaps: thrust-stand efficiency at 40–80 W; a glide test for L/D and the drag add-ons; a forward-CG stall test for AIRSPEED_STALL; the foam-wing CG; and a launch at the first high site.

6 Stability, control and flight dynamics

A tailless aircraft has to trim, damp and control itself with one set of trailing-edge surfaces. On BLUEBIRD those surfaces are also the landing airbrake and a drag rudder. This section derives the margins, the linear modes and the control power, and shows the number that forced each design choice (Table 11). Everything is predicted; nothing has flown. The tools are AeroSandbox’s AeroBuildup with NeuralFoil sections and its vortex-lattice method (VLM) [10, 16], cross-checked with VSPAERO [17]. The equations of motion follow Etkin and Reid [25], and the modes are judged against MIL-F-8785C Class I [26].

6.1 Model chain and mass properties

All 131 mass items came from the detailed CAD through headless FreeCAD and match its mass report to the gram (2431.7 g, $x_{cg} = 312.5$ mm). About the CG, $I_{xx} = 0.224$, $I_{yy} = 0.054$, $I_{zz} = 0.277$ and $I_{xz} = +0.0018$ kg m²; I_{xx} is 46 % above the early budget because most of the outer-wing mass is outboard. The later foam-core outer wing brings the aircraft to about 2.51 kg [pending: modes re-run with the foam-wing mass properties]. AeroBuildup gives the trim, C_{L_α} , C_{m_α} and the polar; the VLM gives the rate and lateral derivatives; strip terms add profile-drag yaw damping, the centre-body Munk moment and the crow drag. We found and fixed two AeroSandbox problems. Its VLM applied p, q, r about the nose apex, which gave $C_{m_q} = -2.69$ instead of -1.67 about the CG. And a separate fin surface did not converge (C_{n_β} 0.0245 to 0.0189 with refinement), while a winglet-form fin converged to within 1 %.

Table 12: Trim with and without reflex and washout (AeroBuildup, design CG).

Variant	C_{m_0}	$\delta_{\text{trim}}, 10 / 14 \text{ m/s}$	$L/D, 14 \text{ m/s}$	Margin, 10 m/s
MH60/MH45, -3° washout (design)	+0.014	$-7.2^\circ / -2.8^\circ$	11.3	17.8°
MH60/MH45, no washout	-0.008	$-10.5^\circ / -5.9^\circ$	11.1	14.5°
NACA 2410, -3° washout	-0.032	$-16.9^\circ / -10.5^\circ$	10.3	8.1°

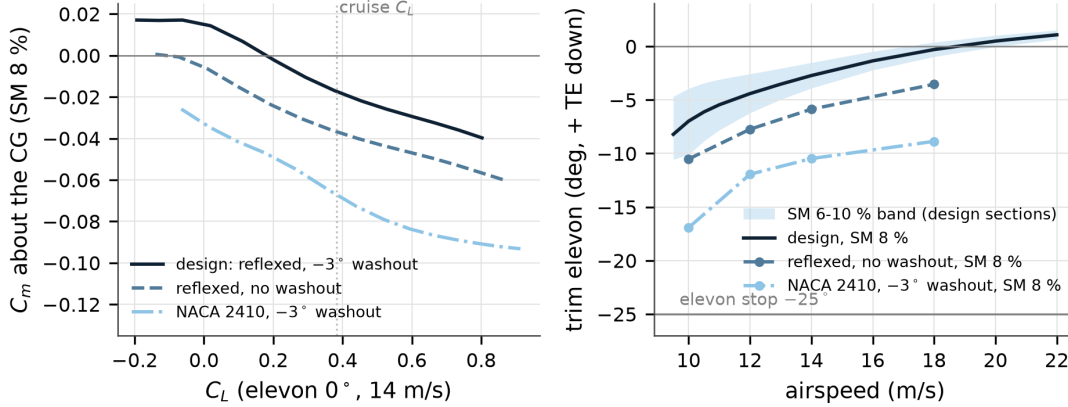


Figure 17: Left: moment about the CG at zero elevon (AeroBuildup, re-run for this paper). Right: trim elevon; the band is the design at SM 6–10 % (aero/controls.yaml), the markers the variants at SM 8 %.

6.2 Longitudinal static stability and trim

The static margin is

$$\text{SM} = \frac{x_{\text{np}} - x_{\text{cg}}}{\bar{c}} = -\frac{C_{m_\alpha}}{C_{L_\alpha}}, \quad (11)$$

with $\bar{c} = 312.3 \text{ mm}$. AeroBuildup puts the neutral point at 337.5 mm (VSPAERO 340.3 mm, VLM 350.9 mm), which gives 8.0 % at the design CG (8.8 % by VSPAERO). The margin softens with angle of attack: C_{m_α} is -0.346 at cruise and -0.197 at the approach, local margins of 7.6 % and 4.9 %. The mode analysis uses these conservative values.

How a tailless aircraft trims. Without a tail, the moment about the CG must be nose-up at zero lift. Linearising in C_L ,

$$C_{m,\text{cg}} = C_{m_0} - \text{SM} C_L + C_{m_\delta} \delta = 0 \quad \Rightarrow \quad \delta_{\text{trim}} = \frac{\text{SM} C_L - C_{m_0}}{C_{m_\delta}}, \quad (12)$$

with $C_{m_\delta} = -0.0076$ per degree. Two features make C_{m_0} positive. The reflexed MH60 and MH45 sections have section c_{m_0} between -0.004 and -0.03 , close to zero. And the -3° washout on a wing swept 22° unloads the tips, which lie behind the CG, so that at zero net lift the tips push down and the root up: a nose-up couple. We measured each by re-running AeroBuildup with the washout removed, and with a conventional NACA 2410 in place of both reflexed sections, at a fixed CG (Table 12, Figure 17). Washout adds $\Delta C_{m_0} = +0.021$ and reflex $+0.046$. Without either, trim takes most of an elevon travel that roll, the crow and the flare also need, and trim drag costs 1.4 % (washout) or 8.3 % (reflex) of cruise L/D .

Table 13: The static-margin trade.

	SM 6 % (aft)	SM 8 % (design)	SM 10 % (forward)
Trim elevon at 10 m/s	-3.9°	-7.0°	-10.1°
Trimmed $C_{L,\max}$ / stall speed	0.92 / 9.0 m/s	0.86 / 9.3 m/s	0.81 / 9.6 m/s
Short period on approach, ω_n (rad/s) / ζ	7.1 / 0.79	8.8 / 0.67	10.3 / 0.59
Dolly lift-off / roll (seat + 1°)	–	11.2 m/s / 13.0 m	12.9 m/s / 18.4 m

Static-margin target. The battery tray trims the CG anywhere in 6–10 % (x_{cg} 318.8 mm to 306.3 mm), and Table 13 shows the ends of the band. Aft, the short period softens; at 5 % on the approach it falls to 6.2 rad/s and couples with the phugoid. Forward, trimming at the stall angle costs lift. The trimmed $C_{L,\max}$ falls to 0.81, the stall speed rises to 9.6 m/s, and the flare needs -12.4° of elevon at $\alpha = 13.6^\circ$, close to the stall angle of 13.9° . We design at 8 %. Two later changes move the CG aft. The taller fins of Section 6.3 move it about 1.4 mm. The foam outer wings (357.1 g per side at $x = 475.5$ mm, against 346.3 g at 468.4 mm printed) move it about 3.4 mm. The battery tray has only about 1.2 mm of forward CG trim left, so the margin falls to about 7 %. That is still inside the band [pending: mass and CG roll-up with the final export].

Keeping the CG in the band with any phone. The bay accepts phones from 143 to 235 g; 28 are profiled in `phones.yaml`. Each profile gives a battery notch (18 detents 2 mm apart; one notch is about 9 g of phone, 0.5 mm of CG or 0.16 % SM). Phones lighter than about 160 g take steel-shot ballast in the insert’s front pocket, and of the 28 only one (143 g) needs it, 6.2 g. The app reads the phone model and shows “SET BATTERY TO NOTCH N”, which must be acknowledged. A two-minute check on a two-tower knife-edge stand then confirms the CG at 313(2) mm (SM $8 \pm 0.64\%$). In flight the app estimates the CG from the mean elevon trim in `SERVO_OUTPUT_RAW` (0.19° per mm at 14 m/s, from the trim tables of Figure 17). It warns above 6 mm before a baseline is learned, which catches a missing phone or a wrong notch, and above 3 mm (about 1 % SM) after one stand-verified flight. The phone only warns; the autopilot flies.

6.3 Lateral-directional stability and fin sizing

Directional stability needs $C_{n_\beta} > 0$ and a stable dihedral effect needs $C_{l_\beta} < 0$. A blended wing has little side area, so C_{n_β} comes almost entirely from the two tip fins:

$$\Delta C_{n_\beta, \text{fins}} = 2 a_v \frac{S_v l_v}{S b}, \quad a_v = 1.3 \frac{2\pi A_v}{2 + \sqrt{A_v^2 + 4}}, \quad (13)$$

where the fraction is Helmbold’s lift slope, 1.3 credits the wing as an end plate, and $A_v = h^2/S_v$. For the original 160 mm fins ($S_v = 0.0223 \text{ m}^2$, $A_v = 1.15$, $l_v = 0.266 \text{ m}$) this gives $a_v = 2.18/\text{rad}$ and $\Delta C_{n_\beta} = 0.0276$, or 0.026 with the centre body. The converged VLM gives 0.0229, and with the body term we use $C_{n_\beta} = 0.0214$. The VLM also gives $C_{l_\beta} = -0.103$. The wing alone contributes only -0.046 (and an isolated fin -0.008). The rest is the fins’ side load carrying over onto the tip loading, a winglet dihedral effect that AeroBuildup misses: its C_{l_β} is 6 times too small.

The Dutch-roll and spiral trade. For small roots, the spiral mode is stable when

$$E = C_{l_\beta} C_{n_r} - C_{n_\beta} C_{l_r} > 0. \quad (14)$$

Table 14: Fin sensitivity: cruise 14 m/s | clean approach 11.5 m/s (C_{n_β} , C_{l_β} at cruise). Level 1 needs Dutch-roll $\zeta \geq 0.08$ and $T_2 \geq 20$ s (cruise) or 12 s (approach); Level 2 needs $T_2 \geq 8$ s.

Fin	C_{n_β}	C_{l_β}	DR ζ	DR ζ RSS-low	Spiral T_2 (s)	T_2 RSS-low (s)
120 mm	0.012	-0.087	0.071 0.071	0.051 0.047	68 31	16.3 10.9
160 mm (original)	0.021	-0.102	0.088 0.085	0.067 0.060	30 19	11.8 8.3
200 mm vertical	0.033	-0.116	0.106 0.099	0.083 0.073	22 14.5	9.7 6.8
240 mm	0.046	-0.130	0.122 0.114	0.097 0.086	19 12.1	8.6 6.0
160 mm, 15° cant	0.019	-0.121	0.084 0.081	0.062 0.056	160 78	15.0 11.4
200 mm, 10° cant	0.031	-0.136	0.103 0.097	0.079 0.069	39 27	10.9 8.2

With the 160 mm fins at cruise, $E = (-0.1026)(-0.01657) - (0.02142)(0.1015) = -4.7 \times 10^{-4}$, a mildly divergent spiral with a time to double of $T_2 = 30$ s. A taller fin raises C_{n_β} and $|C_{n_r}|$, which damps the Dutch roll. But $C_{n_\beta}C_{l_r}$ grows faster than $C_{l_\beta}C_{n_r}$, so the spiral diverges sooner. Canting the fin tips outboard acts as strong local dihedral: it adds $|C_{l_\beta}|$, removes a little C_{n_β} , and moves the spiral root toward stability (Table 14, Figure 18d).

How the fin evolved. The 160 mm fin sat on the Level 1 Dutch-roll boundary ($\zeta = 0.088$, RSS-low 0.060). The rule we applied was to take the tallest fin, in 10 mm steps, that keeps the clean-approach spiral at $T_2 \geq 14.4$ s (Level 1 plus 20%). The crossing is at 202 mm, so the fin became 200 mm. As a vertical fin it dropped the RSS-low approach spiral to 6.8 s, below the Level 2 floor, so we added 10° of outboard cant. That restored 8.2 s, stretched the nominal spiral from 19 to 27 s, and kept most of the damping gain. It costs 9.3 g, 0.58 W at cruise (-0.9 % endurance), 1.56 times the fin-root bending moment and 35 mm of span per side. A fin tall enough to lift even the RSS-low approach damping above 0.08, about 220 mm, would lose more spiral margin than cant can recover, so we stop at 200 mm until flight data exist.

6.4 Dynamic modes

The equations of motion are linearised numerically about each trim, $\dot{\mathbf{x}} = A\mathbf{x}$, with $\mathbf{x}_{\text{lon}} = [V, \alpha, q, \theta]$ and $\mathbf{x}_{\text{lat}} = [\beta, p, r, \phi]$ in stability axes [25, 27]. For each eigenvalue λ ,

$$\omega_n = |\lambda|, \quad \zeta = -\frac{\text{Re } \lambda}{|\lambda|}, \quad \tau_{\text{roll}} = -\frac{1}{\lambda_{\text{roll}}}, \quad T_2 = \frac{\ln 2}{\lambda_{\text{spiral}}}, \quad \text{CAP} = \frac{\omega_{n,\text{sp}}^2}{n/\alpha}. \quad (15)$$

Thrust is fixed-throttle ($dT/dV = -0.34$ N s/m at cruise). The $\dot{\alpha}$ terms are dropped, since there is no tail. The uncertainty band is the root-sum-square of one-at-a-time changes of $\pm 25\%$ in C_{n_β} , $\pm 30\%$ in C_{n_r} , $\pm 20\%$ in C_{l_β} , $\pm 10\%$ in C_{l_p} , $\pm 50\%$ in C_{n_p} and $\pm 10\%$ in inertia. Table 15 adds a column for the current configuration: 200 mm canted fins, a 12.5 m/s approach and the 51°/12° crow of Section 6.6. We evaluated it for this paper with `aero/dynamics.py`'s own `evaluate()`.

Longitudinal. The short period is well damped (ζ 0.59–0.67). Its control-anticipation parameter (11–14) exceeds the piloted-aircraft limit of 3.6, which guards a human against abrupt response; Froude-scaled from an 11 m span the limit becomes 22. The phugoid has $\zeta \approx 0.1$ and TECS closes that loop. The static margin moves the short period and little else (Figure 18a–b).

Lateral. The roll mode is fast (τ 0.06–0.08 s). The Dutch roll has a period near 1.7 s with $|\phi/\beta| \approx 1$, so it is not a wing rock. Its RSS-low damping (0.07–0.08) is why it is measured first

Table 15: Rigid-body modes, with RSS-low values in brackets. Speeds in m/s, ω_n in rad/s. Level 1 limits: Category B (cruise) / C (approach).

Fin Case	160 mm cruise 14	200 mm, 10° cruise 14	160 mm appr. 11.5	200 mm, 10° appr. 12.5	200 mm, 10° crow 12.5	Level 1
Short period ω_n, ζ	13.0, 0.59	as 160 mm	8.8, 0.67	10.1, 0.65	11.0, 0.60	ζ 0.3–2 / 0.35–1.3
Phugoid T (s), ζ	7.5, 0.12	as 160 mm	6.7, 0.09	7.1, 0.10	6.8, 0.18	$\zeta \geq 0.04$
Roll τ (s)	0.064	0.062	0.078	0.069	0.069	≤ 1.4 / 1.0
Dutch roll ω_n	3.50	3.94	3.30	3.71	4.05	≥ 0.4 / 1.0
Dutch roll ζ	0.088 (0.067)	0.103 (0.079)	0.085 (0.060)	0.098 (0.073)	0.142 (0.104)	≥ 0.08
Spiral	T_2 30 s (12)	T_2 39 s (11)	T_2 19 s (8.3)	T_2 31 s (9.1)	$T_{1/2}$ 4.2 s	$T_2 \geq 20$ / 12 s

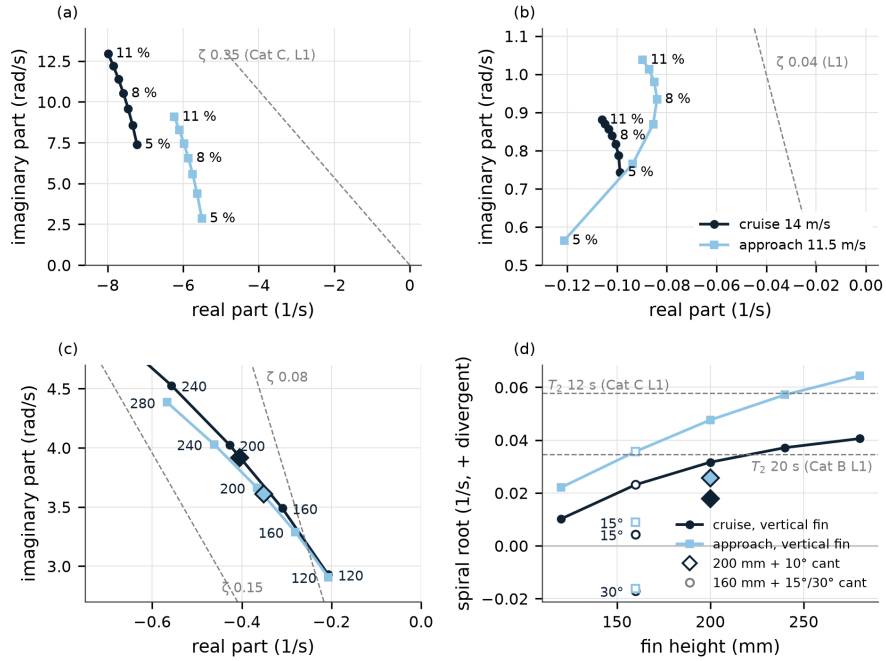


Figure 18: Root-locus sensitivities from `aero/dynamics.yaml`: (a) short period and (b) phugoid against static margin; (c) Dutch roll and (d) spiral against fin height. Navy: cruise 14 m/s; sky: approach 11.5 m/s. Diamonds: the adopted 200 mm, 10° fin. Open markers in (d): 160 mm fins canted 15° and 30°. Dashed: MIL-F-8785C Level 1 boundaries.

in flight. ArduPilot’s roll-rate loop helps: 50 % extra C_{l_p} lifts the 160 mm airframe from 0.088 to 0.108. A drag-rudder yaw damper (0.113) is held in reserve, since split drag grows as $\sin^2 \delta$.

The crow as a yaw damper. Open crow drag turns with sideslip and opposes yaw rate. C_{n_r} doubles ($-0.025 \rightarrow -0.047$), the Dutch roll reaches $\zeta = 0.142$ (0.130–0.173 across the drag models), and the spiral converges. The 12° lower halves halved the crow drag and cut this from 0.181, but the approach is still the best-damped phase.

6.5 Control surface sizing

Each elevon spans 25–90 % of the outer wing at 22 % chord (0.0251 m² per side, 43.9 % of the reference area in its span) and is split into two halves, each with its own ES08MD II servo (2.4 kg cm at 6 V). VLM control derivatives are multiplied by $\eta = 0.8$ for viscous loss and the unsealed gap, and by DATCOM’s plain-flap factor K' above 10° ($K' = 0.67$ at 25°) [28].

Roll. The steady roll rate and the roll time constant are

$$\frac{pb}{2V} = \frac{C_{l_{\delta a}} \eta \delta K'(\delta)}{-C_{l_p}}, \quad \tau_{\text{roll}} = \frac{I_{xx}}{-L_p}, \quad L_p = \frac{1}{2} \rho V^2 S b C_{l_p} \frac{b}{2V}. \quad (16)$$

At 10 m/s, $C_{l_{\delta a}} \eta = 0.00443$ per degree and $C_{l_p} = -0.437$ give $pb/2V = 0.170$ (108°/s, 30° of bank in 0.36 s), 2.4 times the 0.07 guideline and well inside the Category C Level 1 limit of 1.3 s; with the crow open it is 0.64 s. Roll does not size the elevon. Adverse yaw is the weak point ($C_n = -0.021$ at full roll, including the roll-rate term), so rudder mixing is kept for flight test.

Pitch. $C_{m_{\delta}} = -0.0075$ per degree gives $|\Delta C_m| = 0.126$ at $\pm 25^\circ$. The forward-CG flare leaves 12.6°, and full nose-down elevon still gives $C_m = -0.08$ to -0.21 at the stall angle. The elevon is therefore sized by forward-CG trim and by the crow drag, which scales with the flapped area.

6.6 Split-elevon crow airbrake

Why an airbrake. Each landing is a loiter down to 40 m above the site and a straight final to a run that may be only 60 m long (Section 10.2). A clean glide at $L/D = 12$ needs 480 m of clear final; at 5:1 the final is 200 m and 11.3° steep, which clears the tree lines around a field. Separate trailing-edge flaps would pitch a tailless wing nose-down and have to be trimmed by the elevons they sit beside. Spoilers add surfaces and servos to a thin outer wing. A parachute gives one steep, unaimed, wind-drifted descent with no go-around. Splitting the elevons instead adds no surface: ArduPlane mixes the halves natively, the same halves give a drag rudder and a yaw damper, and the crow gives $\Delta C_D = 0.05$ –0.11 with a pitch change of only -0.057 to $+0.024$. The costs are four servos instead of two and the 15 m/s limit derived below.

Drag model. Each open half is a split flap with the Torenbeek section-drag increment [29], summed over the halves and scaled by the flapped fraction:

$$\Delta c_{d,\text{half}} = 1.7 \left(\frac{c_f}{c} \right)^{1.38} \sin^2 \delta \cos \Lambda_h, \quad \Delta C_D = \frac{S_{\text{flapped}}}{S} \sum_{\text{halves}} \Delta c_{d,\text{half}}(\delta_i), \quad (17)$$

with $c_f/c = 0.22$, $\Lambda_h = 13.6^\circ$ and $S_{\text{flapped}}/S = 0.439$. This is the same order as split-flap tunnel data [30]. The low model is 0.75 times nominal. The high model is a bluff wedge with $C_D = 1$ on the frontal area [31]. Both halves at 51° give $\Delta C_D = 0.108$, and each half twists about 3° under load about its single horn (a 50 g bench load should twist it 4.4°).

Table 16: Crow settings (power-off glide ratio for the low / nominal / high drag models).

W1/W2	Upper / lower opening	ΔC_D	L/D	Outcome
40/40	18–24° each	–	7–9 at 11.5 m/s	cannot hold a 4:1 path
85/85	51°/51° (48° after twist)	0.098	4.7 / 3.9 / 3.0 at 11.5 m/s	lower halves strike first
85/20	51°/12°	0.053	6.1 / 5.2 / 3.8 at 12.5 m/s	adopted

How the crow setting evolved. The first parameter file opened the halves only 18–24° (L/D 7–9), which cannot fly a 4:1 path. The opening was raised to 51° per half. The landing-shock analysis then found that at 51° the lower halves hang 15 mm below the skid plane: they would be the first contact on every crow landing, back-driving the servos and loading the horns. Two fixes were compared. A Lua script that retracts the lower halves at pre-flare keeps the steep path but puts flight code on the landing path. Limiting the lower halves to 12° (DSPOILER_CROW_W2 20) is a parameter change, and it keeps the trailing edge at least 10 mm clear at 4° of touchdown pitch. Touchdown sink is the same (0.88 against 0.91 m/s), so the parameter won. It halves the crow drag and flattens the path to about 5.2:1 at 12.5 m/s (Table 16, Figure 19a). The planner’s approach slope accordingly went from 4.0 to 4.5 and then 5.0 [pending: crow glide test at 50/75/100 % flap].

Pitch coupling. The lower split flap adds section lift and the upper one removes it. Re-trimmed, the symmetric crow gives $\Delta C_m = -0.016$ (–0.057 to +0.025). With the upper halves at 51°, the trailing-edge-up authority left is –17° against –17.4° needed in the worst case, which leaves no margin there; FLAP_SLEWRATE 25 opens the crow over 4 s to give the pitch loop time, and the first crow flights log trim against flap. The 51°/12° split moves the expected change toward nose-up, the direction with margin.

Hinge moments and the crow speed limit. For a closed elevon, Glauert’s thin-airfoil flap theory [32] gives an upper bound

$$H = C_h q \int c_e^2 dy, \quad C_h = c_{h_0} + c_{h_\alpha} \alpha + c_{h_\delta} \delta, \quad (18)$$

with $c_{h_0} = +0.082$ (reflexed camber), $c_{h_\alpha} = -0.526/\text{rad}$, $c_{h_\delta} = -0.931/\text{rad}$ and $\int c_e^2 dy = 1.364 \times 10^{-3} \text{ m}^3$. At –25°, 22 m/s and $n = -3.5$, the pair carries 0.216 N m. A design share of 0.6 per half gives 1.32 kg cm, a servo margin of 1.82. An opened half has $c_n = \min(1.7 \sin \delta, 1.3)$ at mid half-chord, times $(u_e/V)^2$ and a 1.25 buffet factor. The linkage is 1:1, so servo torque equals hinge moment and grows as V^2 (Figure 19b). Holding a margin of 1.5 on stall torque at 6 V gives

$$V_{\text{crow}} = V_{\text{ref}} \sqrt{\frac{T_{\text{stall}}/1.5}{T(V_{\text{ref}})}} = 15.0 \text{ m/s} \quad (19)$$

for an upper half opened to 60° by crow plus roll (14.2 m/s at 4.8 V). The lower half at 12° is lightly loaded, so it does not set the limit. ArduPlane has no crow speed limit, so this one is operational: crow opens only on the landing leg at 12.5 m/s, and no manual flap channel is mapped in cruise modes.

6.7 The parallelogram linkage

The two halves of an elevon must follow one command, or they split and add drag. In the chosen parallelogram the servo arm and horn are both 17 mm and the 45 mm pushrod is parallel to the

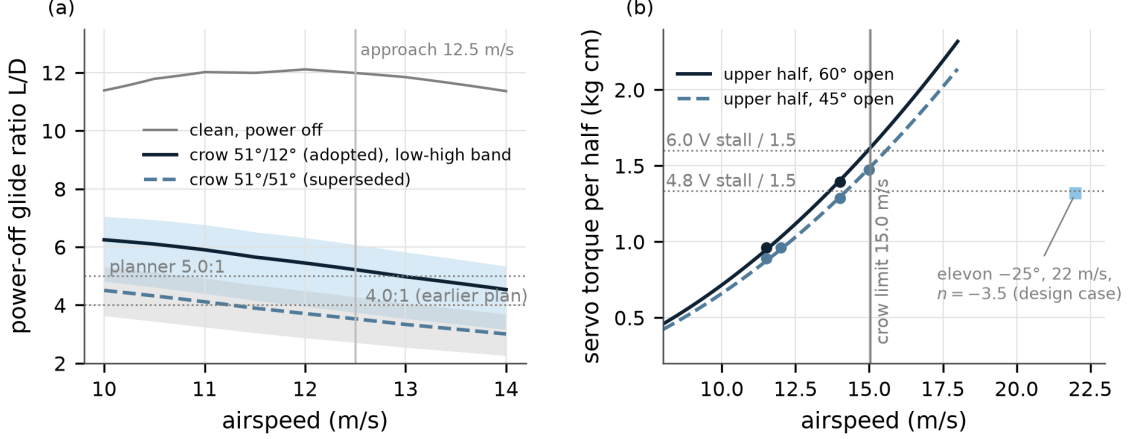


Figure 19: (a) Power-off glide ratio for the symmetric and the adopted 51°/12° crow, with the low-high drag band (`aero/controls.py` split-flap model applied per half). (b) Servo torque per half (points from `aero/controls.yaml`, lines $\propto V^2$) against stall torque divided by 1.5.

shaft-to-hinge line, so the surface angle equals the servo angle exactly and the halves track with 0.0° of split. An offset four-bar would have given 1.38 times the torque, but 1.2° of split, and it could not follow the 50° closing the mixer commands at full pitch plus full roll. The minimum transmission angle is 35°; the rod buckling factor is 10 and the horn stress factor 7. Horn and arm radii must match to ± 0.2 mm (0.5 mm gives 0.8° of split).

6.8 ArduPilot mixing and parameters

ArduPlane’s differential-spoiler mixer [6] drives the four halves. In its ± 4500 output scale,

$$u_{\text{upper,lower}} = G_{\text{mix}} (u_{\text{pitch}} \pm u_{\text{roll}}) \pm 0.45 W_{1,2} f, \quad \delta = 60^\circ \frac{u}{4500}, \quad (20)$$

with f the flap percentage and the result clamped at ± 4500 . The 1:1 linkage maps full servo throw (1000–2000 μs) to $\pm 60^\circ$, so `MIXING_GAIN` 0.42 makes full pitch or full roll 25° and both together 50°, and `DSPOILER_CROW_W1/W2` = 85/20 open the halves 51° and 12°. The other settings in `sim/sparrow.parm` follow from the analysis above: `DSPOILER_RUD_RATE` 35 (a 21° drag-rudder split, the servo limit at 22 m/s); `FLAP_SLEWRATE` 25 and `LAND_FLAP_PERCNT` 100 (crow over 4 s, only on landing); `SERVO_AUTO_TRIM` 0 with the halves parked 1° apart so they never fight; `KFF_RDDRMIX` and `YAW2SRV_DAMP` 0; `AIRSPEED_STALL` 9.6 (the trimmed stall at the forward CG) and `AIRSPEED_MIN`, `TECS_LAND_ARSPD` and `LAND_PF_ARSPD` all 12.5 m/s ($1.3V_s$); and `TECS_LAND_PMAX` 4° to keep the prop off the ground.

7 Launch and recovery

BLUEBIRD has no landing gear. It launches from a passive wheeled dolly, or by hand or bungee, and it lands on its belly on a TPU skid sled. The dolly is the primary launch (Table 17). It loads the structure least, it lets the autopilot fly the takeoff with nobody near the propeller, and it is the method we could simulate. Hand launch is kept for the first AUTOTUNE flights and for sites with no strip.

Table 17: Launch methods.

	Dolly (primary)	Hand	Bungee
Longitudinal load	1.5 g limit via the rear cups	about 3 g (95 N grip)	4 g limit (hook 132 N ultimate)
Site and kit	719 g dolly (\$117); 20 m walked strip, bumps ≤ 15 mm, crosswind ≤ 3 m/s	none	bungee, stake, a clear line
Motor start	2 s warning, operator clear	on 15 m/s ² forward acceleration, 0.2 s delay	as hand
Verification	2-D and roll simulation, 3-D clearance	not simulated	not simulated

7.1 Dolly launch dynamics

Passive release by lift. The aircraft sits on three ball-tipped posts in open 20° cones, one moulded into the TPU sled at (180, 0) mm and a pair at (365, ± 160) mm. There is no latch on the aircraft. The seat holds it at 8°, its lift-off attitude, so it leaves the dolly when lift equals weight:

$$V_{lo} = \sqrt{\frac{2W}{\rho S C_L(\alpha_{seat})}}, \quad C_L = \frac{23.9 \text{ N}}{\frac{1}{2}(1.213 \text{ kg/m}^3)(11.2 \text{ m/s})^2(0.520 \text{ m}^2)} = 0.60. \quad (21)$$

Once airborne, two belly feelers trip a sear at 15 mm of lift. A spring slider then frees the front mast and the two rear towers, each tower folds 72° outboard once its own ball is unloaded, and the towers drop two brake spikes. The rear cups are a pair, not one centre-line post, because the pusher disc passes over the dolly after lift-off; the simulation put the prop into a rigid centre-line post.

Why the seat angle. The first concept rotated the aircraft on one rear pin from a near-level deck. That needed up to -18° of elevon at 11 m/s and put the prop into the ground at 7.8° of pitch. Seating at the lift-off attitude removes the rotation. Seats of 0, 2, 4, 6 and 8° need no lift-off within 7 s, then 33, 21, 17 and 13 m of roll.

Simulation. `sim/dolly_sim.py` integrates the aircraft (x, z, θ) and the sprung dolly with penalty contacts at the balls, AeroBuildup tables with ground effect, a UIUC-based thrust model (21.9 N static) and ArduPlane 4.7’s takeoff and pitch-rate logic. At the design point (SM 8%, rolling resistance 0.10, still air) the nose unloads at 10.2 m/s and the aircraft lifts off at 11.2 m/s after 13.0 m (Figure 20). The release fires 0.066 s later, the prop clears the folding towers by 54 mm (3-D check), and all 27 envelope cases pass (grass 0.05–0.15, SM 6–10%, ± 3 m/s of wind), lifting off at 10.7–13.25 m/s.

Torsion-bar suspension at 3.5 Hz. Rev. 4 of the dolly rode on 3-inch wheels with no springs. In the bump sweeps (both wheels, met at 3–11 m/s), a 5 mm bump threw the aircraft off early and an 8 mm bump gave a stalled lift-off. Bigger 4-inch wheels alone still struck the prop at 15 mm. Springs at 4.5 Hz with 15 mm of travel failed at 20 mm, and 5-inch wheels re-contacted at 10 mm. Rev. 5 uses 4-inch foam wheels on 60 mm trailing arms, each sprung by a music-wire torsion bar,

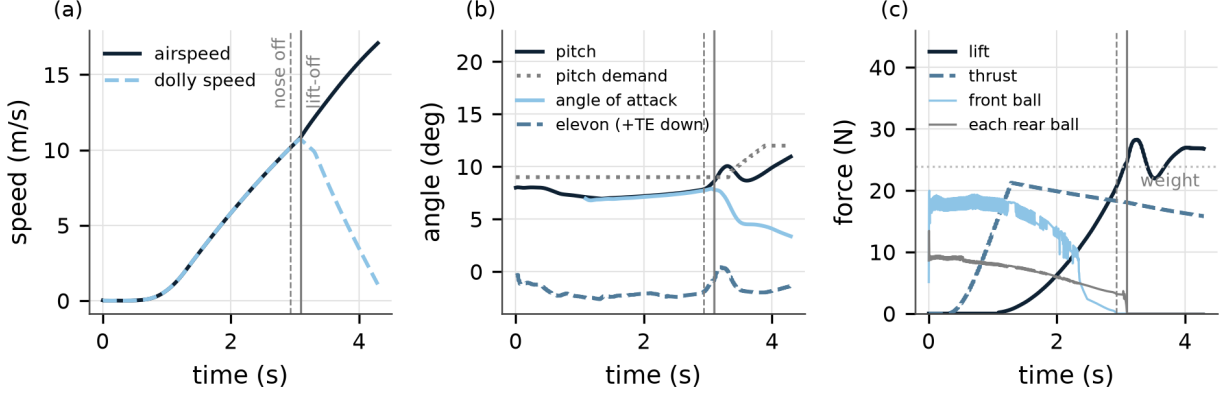


Figure 20: Dolly takeoff at the design point (sim/dolly_sim.py re-run; throttle ramp from 0.3s). Dashed line: nose off the front ball; solid line: lift-off at 11.2 m/s after 13.0 m.

with every corner tuned to its static load:

$$k_w = m_c (2\pi f)^2, \quad \delta_s = \frac{g}{(2\pi f)^2} = 20.3 \text{ mm}, \quad k_w = \frac{G \pi d^4}{32 L r_{\text{arm}}^2}. \quad (22)$$

At 3.5 Hz this gives 503 N/m front and 203 N/m rear, from bars of 2.03 mm by 73 mm and 1.57 mm by 66 mm, with 18 mm of travel to a TPU stop. All 66 bump runs from 5 to 20 mm were then clean. Stiffer rear springs (5–6.5 Hz) released the aircraft early at the forward CG and were rejected. In the same revision the front cone moved into the TPU sled, because the landing analysis banned rigid parts near the skid face (Section 7.2); nose-low lift-offs then passed too. Rev. 5 added 77 g and \$25.

Crosswind limit and strip rule. A roll model adds crosswind, motor torque and ArduPlane’s roll loop, using the C_{l_β} and C_{l_p} of Section 6.3. At a 3 m/s crosswind the aircraft leaves banked 3.9° downwind and reaches 7.9° . The wingtip keeps 156 mm of its 187 mm clearance, and the prop passes 48 mm from the towers (15 mm required). At 5 m/s the upwind tower folds before lift-off, and the margins fall to 34 mm and 117 mm. The limit is therefore a 3 m/s crosswind component. The strip rule comes from the bump sweeps. A bump met at 9.0–9.35 m/s, about 4 m before lift-off, can bounce the aircraft back onto the released dolly and put 75 N on a rear cup rated 60 N ultimate. Hence the rule: nothing over 15 mm in the first 20 m, and nothing over 12 mm from 8 to 13 m.

TKOFF_GND_PITCH. On the dolly this ground pitch sets how hard the pitch loop lifts the nose off the front ball. Settings from seat -1° to seat $+3^\circ$ all pass. But the sprung dolly pitches $0.7\text{--}0.9^\circ$ nose-down under the thrust, so at the forward CG seat $+1^\circ$ never lifts the nose and the aircraft leaves nose-low at 12.9 m/s after 18.4 m; seat $+2^\circ$ gives 11.6 m/s and 14.3 m. The dolly overlay therefore sets 10 (seat $+2^\circ$, the parameter’s maximum), with TKOFF_ROTATE_SPD 12 and a 6 s bog-down timeout.

Hand and bungee launch. The motor arms on 15 m/s^2 of forward acceleration and 4 m/s of GPS speed and starts 0.2 s later; static thrust is 0.92 of the weight, and the pusher keeps the prop behind the thrower. Hands grip only the printed centre body or root rib, since a finger on the EPP wing drives the foam to about 50 % strain. The bungee hook became a pocket in the keel pad after

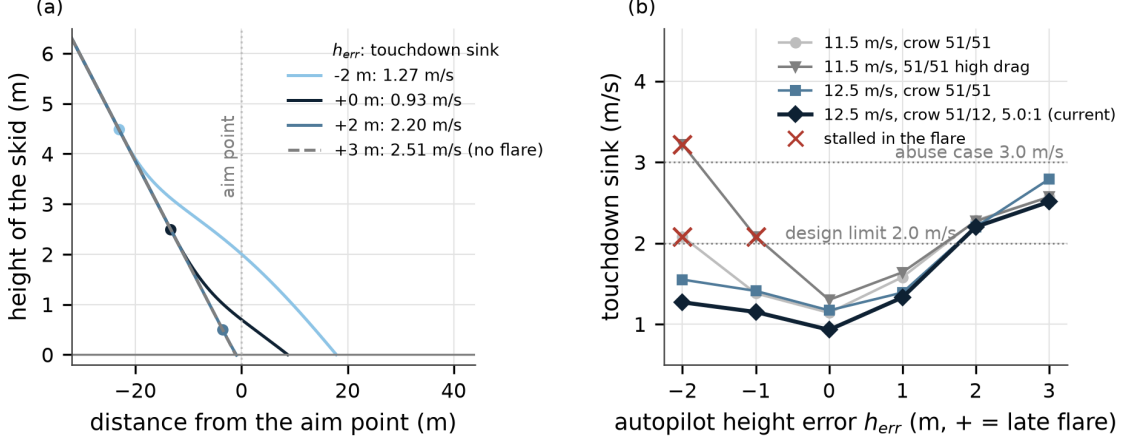


Figure 21: (a) Flare trajectories for the current configuration (12.5 m/s, 5:1, crow 51°/12°); dots mark the flare start. (b) Touchdown sink against height error: the `systems/out/flare.yaml` cases and the current configuration (re-run of `systems/flare.py`). Crosses: flares that stalled.

the landing analysis found the first hook block 2 mm below the skid face. SITL cannot test either launch.

7.2 Approach, flare and touchdown

Flare model. On the final, TECS holds 12.5 m/s (1.3 V_s) on the 5:1 path, about 2.5 m/s of sink. ArduPlane flares when the estimated height falls below 2.5 m or below 1 s of sink. It then blends the sink demand to 0.6 m/s by 1 m, with the throttle closed and the pitch held between 2° and 4°. `systems/flare.py` flies this logic on the aircraft’s own aerodynamics, as a point mass with a lagged pitch:

$$\begin{aligned} \theta_{cmd} &= \gamma_{dem} + \alpha_{1g}(V) + K \frac{\dot{h} - \dot{h}_{dem}}{V}, & \dot{\theta} &= \frac{\theta_{cmd} - \theta}{\tau}, \\ C_D &= 0.0218 + 0.0783 C_L^2 \phi + \Delta C_{D,crow}, & \phi &= \frac{16(h/b)^2}{1 + 16(h/b)^2}, \end{aligned} \quad (23)$$

with $K = 1.2$, $\tau = 0.35$ s, $C_L = 3.15(\alpha + 2.2^\circ)$ capped at 0.90, and McCormick’s ground-effect factor ϕ [33]. The key input is the height error h_{err} (estimate minus truth), which is ± 1 –2 m with a barometer and site heights from a 30-m surface model.

Touchdown sink. The nominal touchdown is at 0.93 m/s and 3.9° of pitch (Figure 21). A height error of ± 1 m gives 1.15–1.33 m/s, a flare 2 m late gives 2.20 m/s nose-down, and no flare gives 2.51 m/s. The earlier setting (11.5 m/s, 5° pitch cap, full symmetric crow) stalled when it flared early, hitting at up to 3.2 m/s. The 12.5 m/s approach and 4° cap removed that. The flare is weak by design: the 4° cap keeps the tail and folded prop off the ground. The design limit sink is therefore 2.0 m/s, which covers ± 1.5 m of height error and matches the CS-23.473 floor of 2.13 m/s [34]. A missed flare is the 3.0 m/s abuse case. A downward TF-Luna lidar cuts the height error to ± 0.1 m and should keep routine touchdowns at 0.9–1.2 m/s.

ArduPlane SITL on the stock airframe (`systems/sitl_touchdown.py`) checks the logic, not our aerodynamics. It gives 0.1–0.8 m/s with a true height, and 1.92 m/s at -9.9° with the barometer 2 m high: the unflared, nose-down touchdown that `flare.py` predicts.

Table 18: Landing run measured from the start of the planner’s run (current configuration).

Flare	Float past aim (m)	V_{td} (m/s)	Slide, μ 0.35 / 0.45 (m)	Total on soft turf (m)
Nominal	8.7	11.2	18.3 / 14.2	37.0
2 m early	17.9	10.1	14.9 / 11.6	42.8
2 m late	−0.9	12.6	23.1 / 18.0	32.2
None	−1.1	12.6	23.1 / 18.0	32.0

The skid sled. Landing gear was not pursued: wheels need a smooth run, which crop and grass sites lack, and would add mass and drag. The skid must absorb the touchdown instead. As first drawn, it was a hollow TPU section only 12 mm deep at the front, with two rigid parts at or below its face (117 g at the CG in the 3 m/s abuse case). Thinner walls with the hard points raised still bottomed, at 94 g at the CG and 238 g at the nose. The deep sled drops the wear face 10 mm and thins the walls to 0.9 mm, for 20 mm of stroke at the front and 29 mm aft, and adds a tail skeg. The abuse case falls to 43 g at the CG and 62 g at the nose without bottoming, for 5 g and $\Delta C_{D0} = 0.0002$. The resulting 50 g ultimate load factor is used in Sections 8 and 9.3.

From the landing run to the planner’s minimum run. The planner needs at least 60 m of flat run at sea level, scaled with density altitude (Section 10.3), with the aim point 10 m in. Table 18 fills it with the float past the aim point (Figure 21a) and a belly slide

$$s_{\text{slide}} = \frac{V_{td}^2}{2\mu g}, \quad (24)$$

with $\mu = 0.35$ (soft turf) and 0.45 (firm turf) from the landing-shock model, neglecting residual lift (which lengthens the slide) and crow drag (which shortens it). The worst case, an early flare onto soft turf, uses 43 m, which leaves 17 m for touchdown scatter and wind that are not modelled here. Low crow drag, which lets the aircraft float, and a tailwind would both eat into this margin; the site score favours headwind landings.

8 Structures and materials

The airframe is a printed centre body (PETG-CF frames and keel, foamed LW-ASA skins, a PA-CF motor module, a TPU-95A skid sled), two hot-wire-cut EPP outer wings covered in iron-on film, and a carbon-tube spar that meets in a machined 7075-T6 V-node on the centreline. The main finding is that **the belly landing, not flight, sizes the wing root, the V-node and most centre-body hard points**: the ultimate landing moment at the wing root is about 84 N m, against 31 N m for the worst flight case. Nothing has been built; every number is a prediction from the scripts in **structures/** and **systems/**, and Section 8.11 lists the tests that must confirm them.

Conventions and criteria. x points aft from the centreline leading edge, y to starboard, z up, in mm. Ultimate load is $1.5\times$ limit; the params landing load factors are already ultimate. Carbon tubes are checked with $RF = 600 \text{ MPa}/\sigma$, an allowable set well below the 1000 MPa strength for thin walls, joint mouths and supplier scatter; metals with $SF = \sigma_y/\sigma$; printed parts with

$$SF_{\text{print}} = k \sigma_y/\sigma_{vM} \geq 1 \quad \text{at ultimate load,} \quad (25)$$

where k is a layer-line knock-down (0.5 PETG-CF and PA-CF, 0.6 LW-ASA, 0.8 TPU) and σ_{vM} is taken away from fixed and loaded faces, where point singularities sit. Table 19 is the design history: each row is a decision that an analysis forced.

Table 19: Structural decision log (ultimate loads). Each change was forced by the analysis in the third column.

Decision	Alternatives	Deciding analysis	Outcome
Landing load	8 g rule; as-built skid; thin walls	CG 117 g (as built), 95 g (thin walls) at 3 m/s	Deep sled; 50 g at CG
Spar V-joint	PETG-CF box; 7075 node	Box FEA SF = 0.41 at the joiner mouth	Machined 7075-T6 node
Node to keel	12×4 tab + inserts; 16×8 tab	SF = 0.53; 0.85 at DLF 0.95; bolt 588 N vs 400 N insert	v3 web, M3 8.8 through-bolts
Joiner	10×8; 12×10	Landing at node mouth: 12×10 RF = 0.69	12×8, RF = 1.03
Outer tube	Pultruded; roll-wrapped	Mouth ring bending 136 MPa; hoop allowable 40 vs 300 MPa	Roll-wrapped 14×12
Tube end	$y = 630, 740, 820$	Skin RF = 0.80 past 630; film RF = 0.30 past tube end	820 + 12×10 spigot
Outer wing	Printed; foam + glass; foam + film	Printed + lattice 438 g/side, joiner RF = 0.90; glass needs epoxy	EPP 30 + film
Wing skin	1 film layer; 2–3 layers	1 layer at −20%: flutter ≤ 37.6 m/s vs 41.9	2×Oracover, $Gt \geq 54$ N/mm
Battery lock	As designed	FEA at 400 N: SF = 0.33	Tray v3: SF _{print} = 1.97
Skid screws	Nylon M3	715 N stop: RF = 0.89	Steel M3, RF = 8.1
Motor pod	LW-ASA shell	67 MPa vs 9 MPa; 80 Hz mode on cruise 1P	2 CF tubes, 258 Hz
Top colour	Any	Dark top soaks above 90 °C	$\alpha \leq 0.40 / 0.45$

8.1 Load cases and the V–n diagram

The envelope follows CS-VLA [35] (as ASTM F2245 and pre-Amendment-5 CS-23 [34]) at the detailed-CAD mass of 2.432 kg, with $W/S = 45.8$ N/m², $\bar{c} = 0.312$ m and $C_{L_\alpha} = 4.38$ /rad. With the Pratt alleviation factor [36],

$$n = 1 \pm \frac{K_g \rho_0 U_{de} V C_{L_\alpha}}{2 W/S}, \quad K_g = \frac{0.88 \mu_g}{5.3 + \mu_g}, \quad \mu_g = \frac{2 W/S}{\rho \bar{c} C_{L_\alpha} g}, \quad (26)$$

so $\mu_g = 5.58$ and $K_g = 0.451$. The design gust is $U_{de} = 7.62$ m/s at V_C and 3.81 m/s at V_D , half the manned values, taken as a three-sigma gust below 120 m with flights cancelled in forecast severe turbulence; the manned values are a sensitivity case.

Gusts are capped at the untrimmed $C_{L,\max} = 0.90$, manoeuvres at the trimmed 0.86 , and $C_{L,\min} = -0.50$. With $V_{S1} = 9.34$ m/s, the limit $n = 5.6$ meets the stall line at $V_A = V_{S1}\sqrt{5.6} = 22.1$ m/s, equal to $V_C = 22.0$ m/s (ArduPilot AIRSPEED_MAX); $V_D = 28$ m/s. The UAS gust at V_C gives $+5.43/-3.43$, so the design limit load factors are $+5.6/-3.5$, ultimate $+8.4/-5.25$ (Figure 22). An early params value of $[-2, 5]$ did not cover the gust and was raised; the manned gust would reach $+7.9$. The cases are pull-ups at $C_{L,\max}$ and V_D , gusts at V_C , push-overs at $C_{L,\min}$ and V_D , a rolling pull-up (CS-VLA 349: $\frac{2}{3}n$ with full aileron at V_A , conservatively at zero roll rate), full elevon at V_C , and the belly landing.

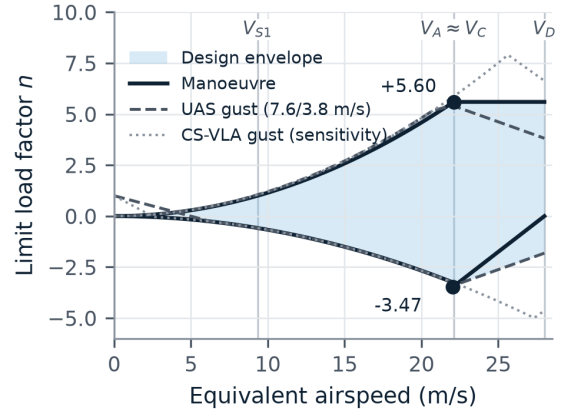


Figure 22: V-n diagram (limit), from the repo's load model.

8.2 Spanwise loads

A vortex-lattice model of both half-wings with camber (`structures/loads.py`) gives the span loads. The elevon is a deflected camber line with the DATCOM plain-flap factor $K'(\delta)$ (0.68 at 25°); unit solutions for α , symmetric and antisymmetric elevon and pitch rate are superposed, manoeuvres trimmed to $C_{m,cg} = 0$ at pull-up rate $q = g(n - 1)/V$, and gusts add $\Delta\alpha$ at fixed elevon. Inertia relief uses the detailed wing masses. Shear V , bending M_b and torsion T are resultants outboard of each streamwise cut about the spar line, which runs from the apex (285, 0) to 45% chord at $y = 630$ (sweep 17.0°). In flight (Table 20 and Figure 23) the rolling pull-up governs bending, 31.2 N m at the apex, and the high- α pull-up governs torsion, 7.97 N m. The belly landing gives -77.8 N m as an equivalent static load (50 g times a root dynamic load factor, DLF, of 0.76) and 84.5 N m as a transient with the in-plane slide: **2.5–2.7 times the flight bending**, of opposite sign.

Table 20: Ultimate section loads V (N) / M_b (N m) / T (N m) about the spar line; the landing transient includes the in-plane slide.

Case	$y = 0$ (node)	$y = 180$ (root)	$y = 340$ (joiner tip)	$y = 630$
A: pull-up, $n = 5.6$	63.6 / 17.3 / 7.97	40.4 / 8.2 / 5.66	23.4 / 3.2 / 4.12	6.5 / 0.0 / 1.44
C: gust up, $n = 5.43$	61.7 / 20.1 / 6.36	42.9 / 10.9 / 4.35	29.1 / 5.2 / 3.04	9.1 / 0.3 / 1.08
G: push-over, $n = -3.5$	-39.8 / -11.9 / -5.35	-26.3 / -6.1 / -4.04	-16.2 / -2.8 / -3.09	-6.1 / -0.2 / -1.15
Rolling pull-up, V_A	76.4 / 31.2 / 4.67	59.6 / 18.9 / 3.06	45.4 / 10.3 / 2.02	16.8 / 1.4 / 0.75
Landing, equiv. static	-165 / -77.8 / 1.28	-135 / -50.0 / 2.20	-99.9 / -30.9 / 2.33	-45.6 / -7.7 / 1.05
Landing, transient	- / 84.5 / 4.66	- / 59.4 / 4.64	- / 42.7 / 4.53	- / 17.9 / 3.51

8.3 Main spar

Layout and stress. A pultruded 12×8 joiner is bonded at least 60 mm into a V-node sleeve and runs to $y = 340$ inside a roll-wrapped 14×12 outer tube that spans $y = 182$ to 820 (overlap 166 mm). A 12×10 tip spigot ($y = 772$ –898) sits 50 mm inside the outer tube and passes through the tip rib; a 4 mm carbon dowel reacts root torsion. With $E = 110$ GPa, the section moduli $Z = \pi(d_o^4 - d_i^4)/32d_o$ are 136.1, 87.8 and 124.0 mm³ for 12×8 , 12×10 and 14×12 , i.e. 81.7, 52.7 and 74.4 N m at 600 MPa. With $\sigma = \sqrt{M_b^2 + M_c^2}/Z$ (including the in-plane moment M_c), flight is

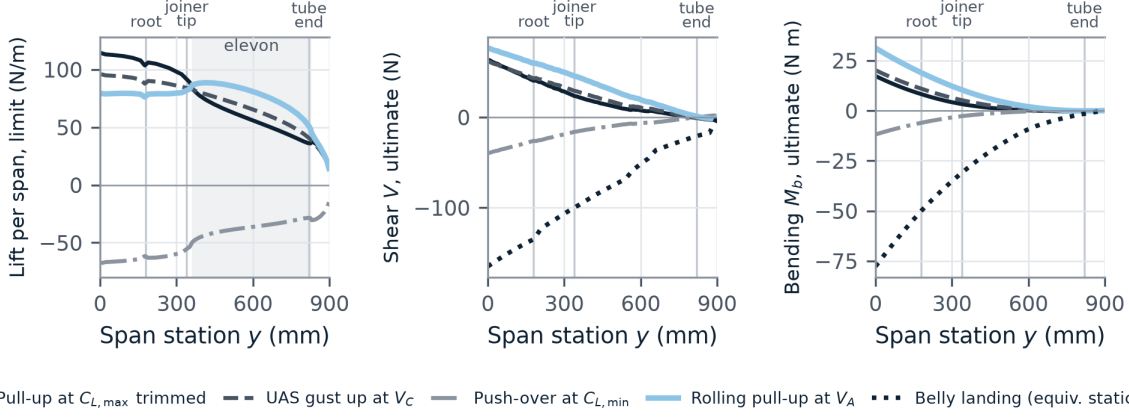


Figure 23: Spanwise lift (limit), shear and bending (ultimate) about the spar for the governing flight cases and the belly landing (equivalent static): landing bending is 2.5 times the worst flight case and of opposite sign. Re-run of the repo’s vortex-lattice load model (`figscripts/struct_vn_loads.py`).

easy (229 MPa at the apex, RF = 2.6) and the landing governs (Table 21 and Figure 25b). At the node mouth ($y = 60$) the foam wing’s transient moment of 78.9 N m gives 580 MPa (RF = 1.03), and the must-not-fail params DLF of 0.95 gives 740 MPa, a factor of 1.35 on strength. Inside the node the aluminium sleeve carries about 60 % of the moment. The node FEA shows a local joiner peak 1.4 times the beam value where the bell-mouth starts, which the strength-to-allowable ratio $1000/600 = 1.67$ covers.

Table 21: Spar stresses at ultimate load (MPa). RF against 600 MPa; SF against 1000 MPa for DLF 0.95. Landing rows for the printed (346 g/side) and foam (353 g/side) wings.

Station	Tube	Flight	Landing transient	Landing, DLF 0.95	Clod stop (30 g)
$y = 0$, apex	12×8	229, RF = 2.6	in sleeve	in sleeve	–
$y = 60$, node mouth, printed	12×8	–	557, RF = 1.08	709, SF 1.41	418, RF = 1.44
$y = 60$, node mouth, foam	12×8	–	580 , RF = 1.03	740, SF 1.35	–
$y = 60$, node mouth, foam	12×10	–	899, RF = 0.67	1147, SF 0.87	–
$y = 180$, outer-tube mouth	12×8	139, RF = 4.3	437, RF = 1.37	530, SF 1.89	RF = 1.8
$y = 337.5$, joiner tip	14×12	83, RF = 7.2	345, RF = 1.74	354, RF = 1.70	RF = 2.4

Joiner evolution. The first CAD pair, 10×8 in 12×10, deflected 74 mm at the gust limit and kept only 0.69 roll effectiveness at V_D ; a 14 mm outer tube cut this to 43 mm. At 50 g landing, a 12×10 joiner reached 864 MPa (RF = 0.69) and 1099 MPa at DLF 0.95, above strength. The 12×8 keeps the outside diameter, so nothing else changes, for 31 g per pair; above about 420 g per wing the next step is 14×10 in 16×14.

Roll-wrapped, not pultruded. The joiner bears on the outer tube at the mouth and at its tip. For a rigid two-point contact with overlap L , $P_2 = M/L$ and $P_1 = |V| + P_2$, and the outer wall sees a ring-bending stress

$$\sigma_\theta = 6 K (P_1/d) R/t^2, \quad K = 0.08, \quad (27)$$

for a cosine bearing of width d (the joiner OD) on a ring of mean radius R and wall t . Under the combined landing moment at the root joint (59.4 N m), $P_1 \approx 525$ N and $\sigma_\theta = 136$ MPa. A pultruded unidirectional tube has a hoop allowable of only 40 MPa (RF = 0.29) and would split;

a roll-wrapped 0/90 tube has about 300 MPa (RF = 2.2). Joiner ends are plugged and the fit is 0.02–0.08 mm diametral, since play adds to the 1.2° root rotation. The price is modulus: a woven tube is nearer 60–70 GPa than the assumed 110, which would raise outer-panel deflection about 40 % and cut roll effectiveness about 0.03; strength is unaffected.

Deflection and tip spigot. Integrating $w'' = M_b/EI$ with the tube stack alone gives 24 mm of tip deflection at the gust limit ($n = 5.43$), 4.4 mm at 1 g; the film adds only $EI \approx 11 \text{ N m}^2$ to the tubes' 90–185 N m^2 . Outboard of the tube, the fin, fin plate and tip rib (about 41 g at a local landing DLF near 1.8) put $V = 42.5 \text{ N}$, $T = 2.19 \text{ N m}$ and $M = 3.25 \text{ N m}$ on the tube mouth. Carried by the film alone, the principal compression exceeds the wrinkling stress on EPP, $\sigma_{\text{wr}} = 0.5 (E_f E_c G_c)^{1/3} = 14.4 \text{ MPa}$ [37] (RF = 0.30). Carbon cap strips fail at their bond, where the shear flow $VQ/I \approx 2.5 \text{ N/mm}$ puts 0.23 MPa into EPP rated for 0.15. The 7.1 g spigot carries V , M and 60 % of T at 35 MPa, leaves the film at RF ≥ 2.5 , and loads the tube mouth to 28 MPa of ring bending (RF = 10.7).

8.4 Torsion and aeroelasticity

`structures/torsion.py` models the wing as beam elements along the swept spar axis (Hermite bending, linear torsion), clamped at the centreline. A closed skin cell has the Bredt–Batho constant $J = 4A^2 / \oint ds/t$ [38]; in the foam wing, core and film act in parallel,

$$GJ = G_c J_c + (Gt)_f \frac{4A^2}{\oint ds}, \quad J_c \approx \frac{1}{3} \int t_c^3 ds, \quad (28)$$

with A the enclosed area, t_c the local core thickness and $(Gt)_f$ the shear stiffness of the film's PET carrier. Streamwise strips carry the swept-wing coupling $\alpha_s = \theta \cos \Lambda - w' \sin \Lambda$. With stiffness $\mathbf{K}$ and aerodynamic stiffness $q\mathbf{A}$, divergence is the lowest q with $(\mathbf{K} - q\mathbf{A})\mathbf{u} = 0$; reversal is where the roll effectiveness $\eta = (L_\delta + \mathbf{r}^T \mathbf{u})/L_\delta$ crosses zero, with $(\mathbf{K} - q\mathbf{A})\mathbf{u} = q\mathbf{B}\delta$ and L_δ the rigid rolling moment; flutter uses the k (V–g) method on six symmetric modes with Theodorsen strip aerodynamics [39], cross-checked by a typical section at 75 % semispan. The code reproduces the Hodges and Pierce typical-section example (2.186 against 2.17) [40].

Table 22: Torsion and aeroelastic results: printed wing (0.6 mm LW-ASA box) and foam baseline (EPP 30, 2×Oracover, tube to 820, spigot). Tip twist is streamwise at 22 m/s, full elevon.

Quantity	Printed LW-ASA	Foam E baseline
GJ at $y = 400$ / outboard (N m^2)	51.2 / 14.5 ($y = 810$)	22.7 / 9.3 ($y = 855$)
First bending / torsion mode (Hz)	9.4 / 50.3	9.0 / 41.0
Divergence	none (eigenvalue 8.8 km/s)	none (eigenvalue 5.8 km/s)
Aileron reversal (m/s); roll effectiveness at V_D	109; 0.83	70 ($2.5 V_D$); 0.85
Flutter, strip / typical section (m/s)	67 / 60	44.6 / 50.4
Flutter, strip, film 20 % soft (m/s)	–	42.0
Flutter requirement (m/s)	$1.15 V_D = 32.2$	$1.3 \times 32.2 = 41.9$
Tip twist, full elevon	−0.98°	−0.74°

Margin argument. Neither wing can diverge: sweep-back makes bending wash the tip out (Table 22). Reversal of the foam wing is at 70 m/s, 2.5 times V_D , and most of its −0.74° tip twist comes from swept-wing bending, so the tube, not the skin, sets roll authority. Flutter binds. The foam wing has less than half the printed wing's GJ , and its strip flutter speed of 44.6 m/s is 1.06

times a requirement that the foam study sets at 1.3 times the usual $1.15V_D$. Because film modulus and ironing tension are uncertain, we also require a pass with the film 20 % soft: 42.0 m/s. The fin clears the NACA TN 4197 criterion [41] at 168 m/s; elevon flutter depends on linkage stiffness and is treated with the control system.

Skin acceptance test. The margin therefore rests on the film shear stiffness Gt (Figure 24b). Only the PET carrier carries shear, 1.45 N/mm per micrometre ($G = 1.45$ GPa). One Oracover layer has $23\mu\text{m}$ (33 N/mm), two give 67 N/mm, and $0.8 \times 67 \approx 54$ N/mm is the soft-film point that still meets 41.9 m/s. A 300 mm filmed EPP bar twisted by T through θ must give, from Equation (28),

$$(Gt)_f = \left(\frac{TL}{\theta} - G_c J_c \right) \frac{\oint ds}{4A^2} \geq 54 \text{ N/mm}. \quad (29)$$

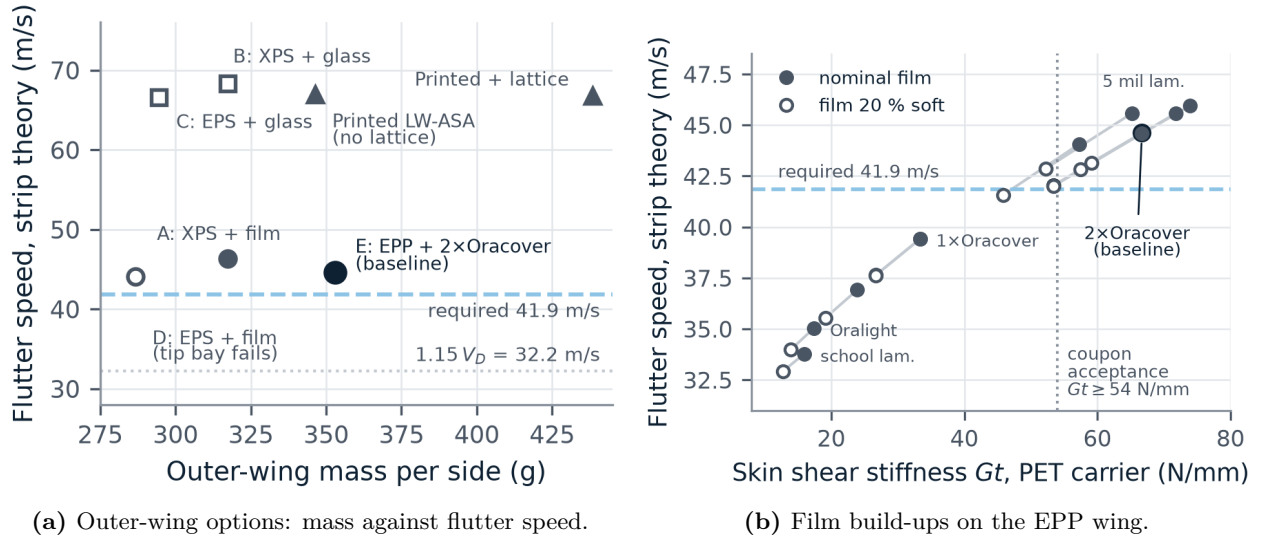


Figure 24: Outer-wing trade (`structures/foam_wing.yaml`). (a) Filled markers are buildable without composites; hollow squares need glass and epoxy; D fails in the tip bay. (b) Each segment joins a nominal film (filled) to the same film 20 % soft (hollow); no single layer of any product passes.

8.5 Foam outer wing

Construction and bonding. Each wing is an EPP core of 30 kg/m^3 , hot-wire cut 0.10 mm under the outer mould line and covered with two full layers of Oracover, transparent then colour (131 g/m^2 , $46\mu\text{m}$ of PET). The film is ironed on flat at $100\text{--}110^\circ\text{C}$ over an EPP primer and *not* shrunk: shrinking starts near 150°C , above the core's 100°C limit, while the adhesive activates at 80°C . Seams are staggered and the skin laps 10 mm onto the root rib so the torsion shear flow enters it. The 14×12 tube lies in a 14.2 mm slot cut from the lower surface and closed with its own foam strip, leaving 1.1 mm of cover at $y = 820$. Bonds into EPP use moisture-curing polyurethane or foam contact adhesive, not epoxy, checked against 0.15 MPa (foam tear 0.20 times 0.75): the tube slot sees 0.008 MPa (RF = 25), the root-rib face 0.043 MPa (RF = 4.7), the tip-rib face 0.024 MPa (RF = 8.2). Carbon-to-carbon and carbon-to-PETG-CF joints use CA. A pulled coupon of two bonded EPP blocks must tear the foam, not the glue line.

Mass. The structural model gives 353 g per side: 180.6 g of kept printed and bought hardware, 85.3 g of core, 48.6 g of film, 19.7 g of tube extension and spigot, and 18.8 g of hinge strip and adhesive. The detailed CAD, with every fastener, weighs 357.1 g per side with its CG at $x = 475.5$ mm [pending: landing and flutter re-run at the 357.1 g CAD mass].

Why foam and film. The printed LW-ASA wing is stiff in torsion (flutter 67 m/s), but with the lattice needed against panel buckling it weighs 438 g per side, drives the joiner to $RF = 0.90$, and takes the aircraft to 2.72 kg. A glass-skinned XPS core (B: 0.7 oz glass at $\pm 45^\circ$ in epoxy) is the best wing, 317 g per side with 68 m/s flutter, but the builder cannot lay up glass and epoxy; it stays as the upgrade for a shop-built core. Film on XPS (A) has more flutter margin, but XPS is limited to 74°C and dimples under the iron. EPP with two film layers (E) needs only a hot-wire shop and a covering iron, is the most crash-tolerant core, and is repaired by pressing out a dent and ironing on a patch (Figure 24a); the flutter margin it gives up is protected by the Gt test. Because a finger pad at 0.32 MPa exceeds EPP’s 0.20 MPa compressive strength, hand launches grip only printed parts.

8.6 Landing impact

Design touchdown and load factors. BLUEBIRD lands on its belly; the pusher propeller, dolly launch and belly cameras fixed a skid at the concept stage, and wheels were not analysed. ArduPilot’s flare, flown on the aircraft’s own aerodynamics and in SITL, touches down near 1 m/s but at up to 2.8 m/s, nose down, when the height estimate is 2.5 m high. The limit sink is therefore 2.0 m/s at 0° – 6° (close to the 2.13 m/s of CS-23.473 [34]), with an abuse case of 3.0 m/s at $\pm 6^\circ$ checked at ultimate. `systems/thermal_shock.py` models the aircraft as a planar rigid body (2.43 kg, $I_{yy} = 0.054 \text{ kg m}^2$) on 60 skid nodes plus hard points, each a skid–soil spring with Hunt–Crossley damping [42] and friction; the skid law comes from a frame FE of the TPU section, soils are anchored to Clegg-hammer readings (0.5, 1.9 and 12 N/mm per mm of skid), and lift at touchdown is $0.95W$. Each station is designed to

$$n_{\text{ult}} = \max[1.5 n_{\text{limit}}(2.0 \text{ m/s}), n_{\text{abuse}}(3.0 \text{ m/s})], \quad (30)$$

rounded up to 5 g. The skid dominates (Figure 25a). The as-built skid (1.2 mm walls, 12 mm deep at the front, a PETG-CF hook block and dolly cup at the wear face) gave 117 g at the CG and 252 g at the nose; thinner walls still bottomed (95 g). The **deep sled** drops the wear face 10 mm to $z = -38$, with 0.9 mm walls, a 30° ramp, 20–29 mm of stroke and every rigid part 8 mm clear. It gives 31 g at the CG at limit and 43 g in abuse, hence **50 g at the CG**, with 65 (nose bay), 55 (cameras), 50 (phone, battery, centre), 60 (servos, skeg) and 70 g (motor), plus 30 g longitudinal, in 13–22.5 ms pulses. These values hold only with the deep sled.

Energy absorption. The abuse sink energy is $\frac{1}{2}mv^2 = \frac{1}{2}(2.43)(3.0)^2 = 10.9 \text{ J}$. The 0.9 mm-wall sled crushes at about 1.0–1.6 N/mm per mm, 310–500 N over its 312 mm flat, so its 20–29 mm stroke absorbs roughly 6–15 J, bracketing the demand. The model crushes 18.7 of the 20 mm front stroke; the mean deceleration $v^2/2s = 24.5 \text{ g}$ against the 43 g peak gives a peak-to-mean ratio of 1.75, near the $\pi/2$ of a half-sine.

Wing transient. `structures/landing.py` base-excites the beam model through the V-node with the vertical and in-plane pulses, the detailed wing masses and 2 % damping. The first modes are 9.3 Hz vertical and 10.9 Hz in-plane, so the pulses are short against the 107 ms bending period

and the root DLF is only 0.49–0.76, below the single-mode bound of 0.95 kept in params. The tip, loaded by the 36 g fin, whips to a local DLF of 1.7–2.1; that whip failed the printed skin outboard of $y = 630$ (RF = 0.80) and pushed the tube to 740, then 820. After the pulse the wing rings back to 68 N m tip-up at about 85 ms.

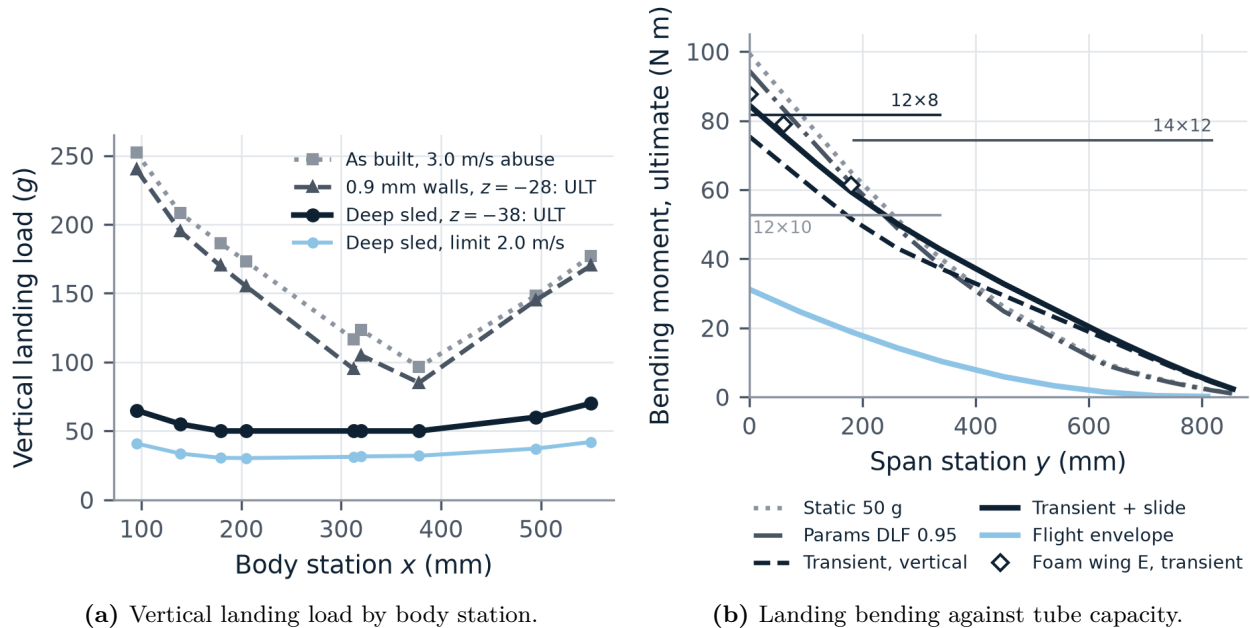


Figure 25: Belly landing. (a) Ultimate design loads for three skids (`systems/out/design_loads.yaml`). (b) Printed-wing landing moments, the flight envelope, foam-wing points and tube capacities at 600 MPa; inside the node ($y < 60$) the sleeve carries about 60% of the moment.

V-node and keel attachment. The wing inertia acts about 180 mm aft of the apex, so the node sees a nose-up pitch moment of 53 N m (transient) or 67 N m (DLF 0.95). The as-designed 12×4 mm tab had to pass all of it to the front bolt (about 1500 MPa by hand calculation), and the bolt pulled 494 N or 588 N against a 400 N heat-set insert. Version 3 has a keel web 16 mm wide, 12 mm deep at the hub and 6 mm at the rear bolt, with R2 fillets, and M3 8.8 through-bolts (RF = 5.9/4.9) under a PETG-CF pad at least 4 mm thick and 20 mm across, which raises pull-through from RF = 0.86/0.72 to 2.9/2.4. Because the ring-down reverses the moment, both bolts are through-bolts. The node gains 23 g (Table 23).

Retention and hard points. Items are retained for *mng*: battery and tray 307 N down and 184 N fore-aft (lock rated 400 N), phone 133 N and 80 N, motor 103 N and 44 N. Under the 715 N stop, nylon skid screws would see 238 N each against 211 N, so they are steel. The pack must bear on the channel roof: a keel FEA with the pack on the rails alone deflected 78 mm. The LW-ASA motor pod would see 67 MPa against 9 MPa under the 25 N m skeg strike, so two $4 \text{ mm} \times 2 \text{ mm}$ carbon tubes carry it (127 MPa, RF = 4.7).

8.7 Finite-element analysis of printed and machined parts

Method and verification. `structures/fea.py` meshes a STEP part with Gmsh [43] into quadratic tetrahedra (1.5–2.5 mm, refined to 0.6–1.2 mm at bores; 134 141 elements for the V-node)

and solves it with CalculiX 2.23 [44] as bundled with FreeCAD [45]. Materials are linear-elastic and isotropic (Poisson ratio 0.35 printed, 0.33 aluminium, 0.45 TPU), so print anisotropy enters only through k . Multi-body parts share one bonded mesh; loads are forces, pressures, spread moments, remote force–moment pairs and body loads, applied as consistent nodal loads. Tube stubs are solid rods of equal EI , $E_{eq} = E[1 - (d_i/d_o)^4]$, with stress scaled back by $1/[1 - (d_i/d_o)^4]$. Against hand solutions for a cantilever and a tube in torsion the worst error is 1.04%; FreeCAD’s own FEM workbench gives the identical cantilever result, and mesh sensitivity is about 6%.

Table 23: FEA at ultimate load: σ_{vM} away from boundaries, SF against yield, SF_{print} for printed parts. Geometry from the current CAD source unless marked “official”.

Part	Case	σ_{vM} (MPa)	SF	SF _{print}	Verdict
PETG-CF spar box, 4 variants	flight (first pass)	118–122	0.41	0.21	fail
7075 node, 12×4 tab	landing, early loads	940	0.53	–	fail
7075 node v2, 16×8 tab	trans. / DLF 0.95	472 / 595	1.07 / 0.85	–	fail
7075 node v3, keel web	trans. / DLF 0.95 / flight	335 / 421 / 117	1.50 / 1.19 / 4.3	–	pass
Root rib (official STEP)	flight / landing	5.3 / 5.3	9.5 / 9.4	4.7 / 4.7	pass
PA-CF motor module	70 g landing / flight	13.9 / 5.4	5.0 / 13	2.5 / 6.5	pass
PA-CF motor module	skeg strike 330 N	21	3.3	1.6	pass
Battery tray, as designed	400 N lock	150	0.33	0.17	fail
Battery tray v3	400 N / 184 N design	27.6 / 12.7	1.81 / 3.9	0.90 / 1.97	pass
Rear dolly cup, 3 mm wall	launch	2.1	24	12	pass
Fin skin / tongue	7.5 m/s gust, V_D	4.4 / 8.5	3.4 / 5.9	2.0 / 3.5	pass

Results. Table 23 lists the governing cases. The PETG-CF spar box failed in every variant, even with carbon caps running at SF ≈ 10 , because the joiner must put its moment into plastic at the mouth; hence the metal node. The landing then moved the critical point from the sleeves to the keel attachment, which three node versions resolved (Figure 26). Other parts failed at load-path details: the tray lock at a 1.5 mm tab, the forward keel at its rails. Motor-module values are linear scalings of the FEA at 65 g and 250 N. The centre-body geometry was built from the CAD source while the official export was regenerated, and the foam wing raises the node-mouth moment about 4%; scaling linearly puts node v3 at SF ≈ 1.44 (transient) and 1.14 (DLF 0.95), above 1.0 but under the 1.2 target.

Re-runs pending on the final export: [pending: V-node v3 with foam-wing loads]; [pending: battery tray v3 and keel with the pack on the channel roof]; [pending: PA-CF motor mount]; [pending: TPU phone insert at 133 N]; and the fin on its split-tongue geometry.

8.8 Materials

The choices follow stiffness per gram, printability on an enclosed CoreXY printer with a hardened nozzle, and temperature (Table 24). **PETG-CF** is the structural filament, stiff and easy to print and bond, but its 75°C HDT is well below what a dark skin reaches in the sun, which drives the colour rules of Section 8.9. **PA-CF** is used only at the motor, whose winding reaches 100°C

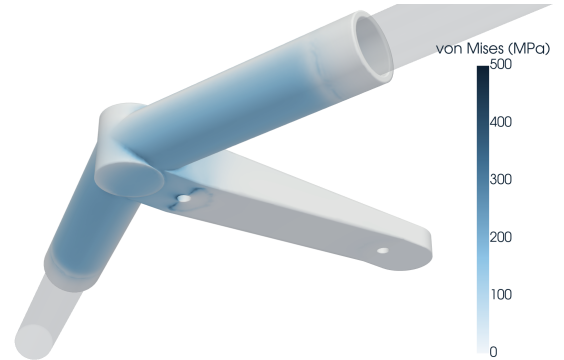


Figure 26: V-node v3 under the ultimate landing transient: von Mises stress to 500 MPa, peak 335 MPa away from the bolt bores (SF = 1.50); joiner stubs translucent.

Table 24: Material data. E , σ_y , density and HDT from params; k and Poisson ratios are our assumptions; foam and film values are typical datasheet values.

Material	E (MPa)	σ_y (MPa)	ρ (g/cm ³)	HDT (°C)	k	Allowable (MPa)
PETG-CF	4500	50	1.30	75	0.5	25
PA-CF	6000	70	1.20	150	0.5	35 (dry)
LW-ASA	1000	15	0.62	95	0.6	9
TPU-95A	26	8	1.21	–	0.8	6.4
EPP 30	4 (G 1.5)	0.20 (comp.)	0.030	100 (service)	–	0.15 (bond)
PET film	4000 (G 1450)	14.4 (wrinkling)	–	75–80 (T_g)	–	–
Carbon tube	110 000	1000	1.55	–	–	600
7075-T6	71 700	503	2.81	–	–	503

after climb at 45 °C ambient; nylon loses about 30 % of its strength wet, so it is printed dry and analysed with dry values. **TPU-95A** is used where compliance is the function (the crushing sled, the preloaded phone clamp), not where stiffness matters: TPU camera posts put a 78 Hz mode on the cruise propeller 1P and were replaced by PETG-CF (841 Hz). **EPP** beat XPS and EPS on crash recovery (it springs back to within about 10 % of an imposed strain) and on its 100 °C limit. **Creep** has not been quantified and is the main open materials question. The design limits it three ways: sustained stresses are small (at 1 g the printed parts carry about 1/8.4 of their ultimate flight stress, some 0.6 MPa in the root rib); temperatures stay below HDT under the colour rules; and no joint above about 300 N relies on a heat-set insert. Preload loss in the TPU clamp and in bolts through PETG-CF after a hot soak must still be measured.

8.9 Thermal: sun heating, adhesives and the thermal study

`systems/thermal_shock.py` solves a 16-node lumped thermal network through a 30 min ground soak in full sun (1000 W/m², calm, bays closed); top skins absorb αG , convect and radiate to a Swinbank sky [46], and the belly sees ground albedo 0.25. **The top-skin colour sets the temperature of the bonded structure:** a black top reaches 93 and 102 °C at 35 and 45 °C ambient, white paint ($\alpha = 0.35$) 59 and 69 °C, nearly linear at 60 K per unit absorptance (Figure 27). The limits are the 75 °C HDT of PETG-CF frames bonded under the centre-body skin, the 80 °C activation of the Oracover adhesive (and the 75–80 °C glass transition of its PET) on the wings, and room-temperature epoxy bonds (glass transition 55–65 °C), which a dark skin soaks at 75–85 °C. We therefore require $\alpha \leq 0.40$ on the foam-wing tops, holding the film near 72 °C at 45 °C ambient, and $\alpha \leq 0.45$ on the centre-body top; the belly sees only reflected sun and is free, so the aircraft must never be parked inverted in the sun. On the ground a reflective shade and late power-on protect both structure and avionics. The same study found the motor-pod mode on the cruise propeller 1P (Table 19).

Sun-coupon rule. Absorptance cannot be read from a colour chart, so each finish is qualified by three matched coupons (sample, white, black) on identical EPP in the same sun. At steady state each balances $\alpha G = h_{\text{eff}}(T - T_{\text{ref}})$ with the same h_{eff} and emissivity, so

$$r = \frac{T_s - T_w}{T_b - T_w} = \frac{\alpha_s - \alpha_w}{\alpha_b - \alpha_w}, \quad (31)$$

independent of sun and wind, and $\alpha_s \approx 0.30 + 0.63r$ for $\alpha_w = 0.30$, $\alpha_b = 0.93$. The limits become $r \leq 0.16$ on the wing tops and $r \leq 0.24$ on the centre-body top, valid when $T_b - T_w \geq 15\text{ K}$. The test has not been run, and the references matter: with $\alpha_w = 0.35$, $r = 0.16$ would mean $\alpha \approx 0.44$.

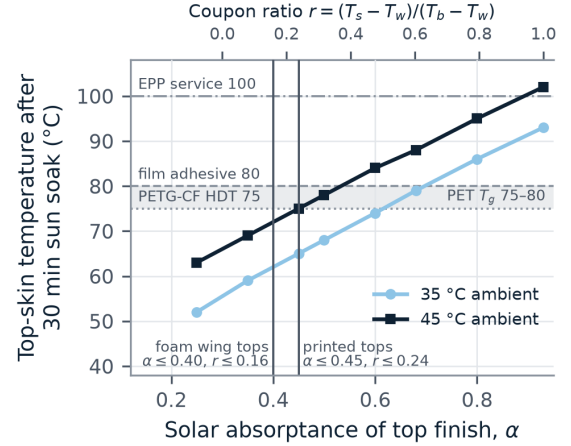


Figure 27: Top-skin temperature after a 30 min sun soak against absorptance, with material limits; the top axis maps α to the coupon ratio r .

8.10 Mass budget, centre of gravity and producibility

The first budget was 2250 g and the aerodynamic design mass is 2525 g. The detailed CAD with printed wings came to 2432 g (centre-body components 1163 g, printed centre body 466 g, hardware 131 g, outer wings 672 g) with the CG on target at $x_{\text{cg}} = 312.5\text{ mm}$ and a static margin of 8.0 %. With the printed lattice restored it would be about 2.72 kg; with foam wings and about 75 g of centre-body fixes it is about 2.51 kg (2.52 kg at the CAD wing mass). The foam wing is heavier and further aft (357.1 g at $x = 475.5\text{ mm}$ against 346.3 g at 468.4 mm), moving the CG roughly 3 mm aft, more than the 1.2 mm of forward trim the battery tray has with the reference phone; the rest goes to nose ballast or a lower static margin (about 7 %, inside the 6–10 % band) [pending: mass and CG roll-up with the foam wings].

Manufacturability and serviceability audits. Geometric audits of the frozen CAD exports (`cad/detail/dfm_check.py`) found seven DFM blockers, all local fixes, among them a wing dowel along y instead of the 17° spar axis (the wing could not slide on), servo trays wider than their pockets, a node insert in a 0.45 mm wall and elevon trailing edges too thin to print. With the fixes the build is about 86 h of printing and 1.35 kg of filament; the foam-wing CAD removes the lattice and dowel blockers and addresses the wing ones. Of 35 line-replaceable units, 10 come out cleanly, 14 only with help and 11 not at all; the worst is the battery, captive behind the coaming lips (breaking the 30 s swap), which trimming the lips over $x = 137\text{--}272$ frees. The avionics cover, XT60 socket, servo-lead joints and soldered harness remain open [pending: DFM and service re-audit on the next export].

8.11 Assumptions and required tests

The analyses assume linear-elastic FEA with one print knock-down, strip-theory flutter of symmetric modes, an LW-ASA modulus of 1 GPa (checked over $\pm 30\%$), 110 GPa for every carbon tube, datasheet foam and film values, and 2 % damping in the landing transient; none has been measured on BLUEBIRD hardware. The tests, in build order, are: (1) coupons for LW-ASA bending, EPP

bond pull-off and the film twist of Equation (29); (2) sun coupons for every top finish; (3) a sled-offcut crush test; (4) a tap test of the first wing (about 9 Hz bending and 41 Hz torsion expected); (5) a drop test with a strain gauge at the wing root; and (6) vibration and landing logs from the first flights.

9 Avionics, onboard computing and payload

One rule shapes the systems design: *the flight controller flies; the phone thinks*. ArduPlane [6] owns stabilisation, navigation, failsafes and landing. The phone does perception, landing-site checks and mission logic. If the phone crashes, overheats or is left out, the aircraft still lands autonomously at the nearest vetted site (Section 10). We judged each choice below against this rule first, then against cost, mass and heat.

9.1 Architecture and the companion computer

Figure 28 shows the avionics. A Matek H743-WING V3 runs ArduPlane 4.7 and carries the power distribution, a 90 A current sensor and 5 V, 6 V (servo) and switchable Vsw regulators. The SiK 915 MHz telemetry radio (about 2 km), the ExpressLRS 2.4 GHz safety-pilot receiver and the GNSS each have a UART. The Dronetag BS Remote ID module takes only 5 V from the FC and has its own GNSS.

The phone connects over USB to a USB 2.0 hub. One hub port feeds a CP2102N USB–UART bridge wired to FC SERIAL2 (MAVLink2 [47], 921.6 kbaud), which leaves the FC’s own USB free for setup; two carry the cameras. The phone sends only high-level messages: divert (MISSION_SET_CURRENT), go-around, mode change, the camera relay and a preflight interlock value. One gate in the app blocks every command unless the FC heartbeat is under 3 s old, the aircraft is armed in an autonomous mode, and it is above a 30 m decision height. In flight the phone is in airplane mode (Section 11.1); its events reach the ground as STATUSTEXT through the FC and the SiK link.

Why a phone. Table 25 compares the phone with a single-board computer (Raspberry Pi or Jetson class); no board was costed in the repo, so that column is qualitative. Energy decides it. The phone runs on its own 17.3 Wh battery (14.7 Wh usable), so its 4.3 W typical load (6.0 W worst) never touches the flight pack. A computer on the pack scales endurance by $P_{cr}/(P_{cr} + P_c)$: at the cruise power $P_{cr} = 66.9$ W, a 4.3 W load costs 6 %. The phone also brings an NPU, a modem for ground sync, a screen and a thermal-status API, and most users own one. Its weaknesses (heat, variation between models, an OS that can kill background work) cannot affect flight safety, because nothing flight-critical runs on it.

9.2 Power

The pack is a 4S2P Molicel P45B (14.4 V, 9 Ah, 129.6 Wh). Of this, 80 % is usable (103.7 Wh), and the mission may spend 80 % of that (82.9 Wh); the rest is the reserve for reaching a landing site. In the worst case every rail runs at or below 60 % of its rating (Table 26). At the pack the avionics draw 9.3 W. The aerodynamic results assumed 7.7 W, giving 66.9 W at 14 m/s and 1.24 h; with 9.3 W these become 68.5 W and 1.21 h [pending: aero re-run with AVIONICS_W = 9.3]. The 42 A peak is 47 % of the pack rating and sags it 1.3 V, so the battery failsafe triggers on consumed charge first.

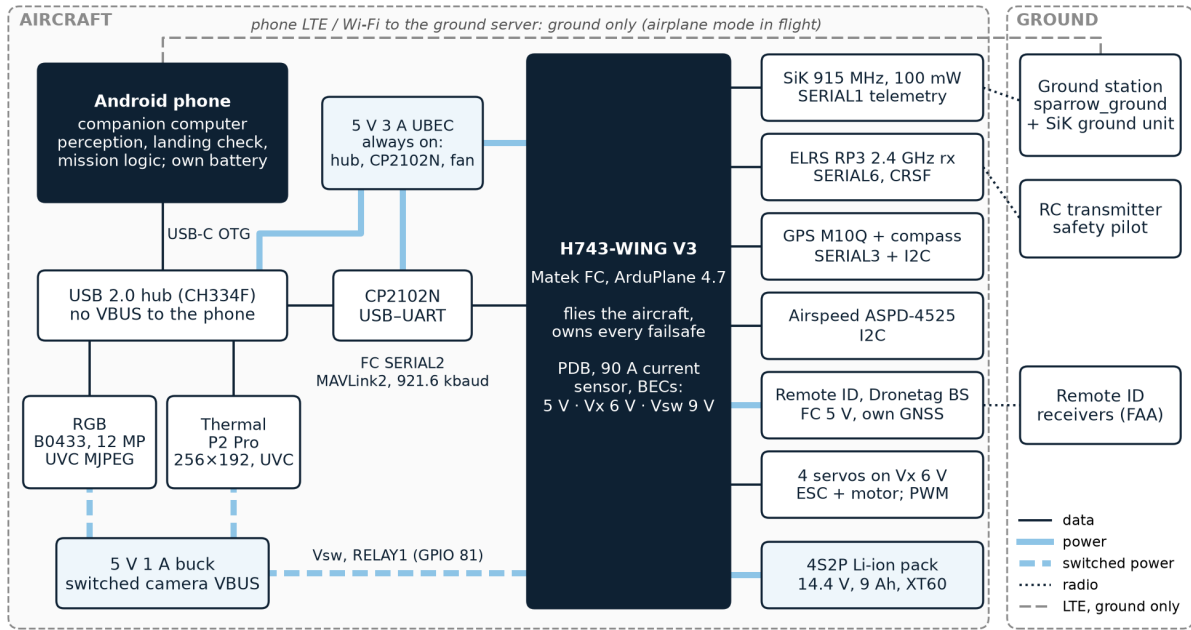


Figure 28: Avionics architecture, drawn from the harness source and UART map. The FC owns every flight-critical link. The phone reaches the FC through the hub and a CP2102N bridge. It can switch the camera power (dashed) but never its own link.

Table 25: Companion-computer trade. The phone column uses the power-budget and phone-bay analyses.

Criterion	Android phone (chosen)	Single-board computer
Energy	Own battery: 3.4 h typical, 2.4 h worst (aircraft 1.24 h)	From the pack: −6 % endurance
Cost	User-supplied (\$0 in the BOM)	Board, storage, cooling, modem
Compute	NPU on Snapdragon-8 or Tensor-class phones	Board-dependent; GPU on Jetson
Cameras	Two UVC streams on one USB 2.0 hub (bench item)	USB 3 or CSI
Heat	Full payload only below $\approx 38^\circ\text{C}$	Sized freely
Fit	28 phone profiles, 143–235 g, one insert each	One fixed board

Table 26: Rail budget (typical and peak current; datasheet or typical values, FC board and servo stall estimated).

Rail (rating)	Loads	Typical / peak	Worst
FC 5 V BEC (2 A)	FC, GNSS, airspeed, ELRS, SiK; Remote ID [pending: current]	0.48 A / 0.72 A	36 %
FC Vx, 6 V (8 A)	4 servos: cruise; gust or crow; all stalled	0.3–0.5 A; 2.0 A; 4.8 A	60 %
Hub UBEC, always on (5 V, 3 A)	hub, CP2102N, phone-bay fan	0.22 A / 0.26 A	9 %
Camera buck, switched (5 V, 1 A)	RGB and thermal camera VBUS	0.38 A / 0.52 A	52 %

Switching the payload. The cameras are the only switched load, after three versions. The trigger was the heat-soak model: after 30 min powered in full sun at 45 °C, the RGB camera reaches 84 °C against about 70 °C. The first fix switched the hub’s regulator through a MOSFET. That breaks the guiding rule: the hub and the CP2102N carry the phone’s MAVLink link, so the phone could switch them off but never back on. The adopted design switches only the camera VBUS, from the FC’s own Vsw output (9 V, ArduPilot RELAY1, default off) through a 5 V buck (1 g, \$6.95). Powered at $T - 2$ min, the soaked RGB camera launches at 61 °C (52 °C under a shade). The phone switches the cameras on at “Cameras on” or arming and off 60 s after disarm. The FC does not report the relay state, so the phone re-sends it on every change, every new FC power session and every 10 s while armed. A DO_SET_RELAY in the mission (Section 10.5) covers a launch without the phone.

We also rejected charging the phone from the hub: it would add 8.3 W to the pack (11 % of endurance) for nothing, since the phone’s own battery lasts 2.4 h in the worst case against 1.5 h at best-range speed.

9.3 Phone bay: heat, vibration and landing shock

The phone rides screen-down under a flip-forward hatch in a TPU insert printed for each model (its CG trim is in Section 6). Lumped models asked whether the phone overheats, whether prop vibration reaches it, and whether it survives belly landings.

Heat. The phone dissipates 3–5 W, with peaks of 8 W, from ground LTE (1.1–1.7 W [48]), two UVC streams (1–2 W, unverified), NPU inference (0.5–2 W) and base load. We assume, since it varies by maker, that Android reports “moderate” at a back temperature of 43–46 °C and “critical” at 50–55 °C. The as-built bay had a flush-hole inlet and a 3–14.5 mm dead-air cavity. Our SU2 section solution gives $C_{p,\text{in}} = -0.97$ at the inlet and $C_{p,\text{ex}} = -0.51$ at the exits: the inlet is in *stronger* suction. With ram recovery η_r the inlet total pressure is $p_\infty + q[C_{p,\text{in}} + \eta_r(1 - C_{p,\text{in}})]$, so air flows only if

$$\eta_r > \frac{C_{p,\text{ex}} - C_{p,\text{in}}}{1 - C_{p,\text{in}}} = 0.23. \quad (32)$$

The fix is a NACA submerged inlet [49] in the hatch (a 7.4° ramp with throat $w/d = 4$, inside Mossman and Randall’s recommended bands), a 3 mm duct over the phone’s back, and three 4 mm × 60 mm exit slots over a rear plenum that every phone from 144 to 170 mm long ends under. The flow network (inlet recovery, losses, exit discharge coefficient 0.62) raises the flow from 0.48 to 1.42 L/s. The duct is a laminar channel heated on one side, $Nu = 5.385$ [50, 51] plus an entry term, with the air marching as $\dot{m}c_p dT_a/dx = h b (T_{\text{ph}} - T_a)$; the film coefficient rises from 12 to 37 W/(m² K). As checks, the duct Reynolds number is about 1600 ($D_h \approx 4.4$ mm), so the flow is laminar, and the air warms only 3 K carrying 5 W: the film coefficient, not the air’s heat capacity, is the bottleneck.

In cruise at 35 °C with 5 W and full sun, the duct lowers the phone from 57 °C to 47 °C with a dark hatch, and from 53 to 45 °C with a white one. The ground is harder: at a 2 W idle a dark closed hatch reaches 60 °C in 20 min, while a white hatch, a 0.75 W fan in the insert and a shade card hold 39 °C. Above about 38 °C ambient no passive fix suffices, so the app sheds load before Android throttles. On the worst of thermal status, headroom and battery temperature it steps down inference, the RGB save rate, the RGB stream, and finally everything but the landing check, which never drops below 2 Hz. A hot phone blocks preflight.

Table 27: Phone-mount trade: peak phone acceleration in g at 2 m/s sink (mount travel), and vibration transmissibility in the prop bands.

	Stiff clamp, 830 Hz (chosen)	Lip insert, 200 Hz	Isolator, 45 Hz, 3 mm stop
Firm turf	31.7 (0.01 mm)	36.2 (0.23 mm)	148.9 (bottoms)
Hard dirt	59.3 (0.02 mm)	87.7 (0.54 mm)	239.2 (bottoms)
Transmissibility, 1P and 2P	≤ 1.2	up to 6.3	0.03–0.4

Table 28: Camera trade at 100 m above ground (approximate catalogue cost and mass).

Camera	Pixels	GSD	Swath	Cost, mass	Verdict
ELP AR0234 (3.6 mm)	1920×1200	8.3 cm	160 m	\$85, 40 g	Rev 1; global shutter
Arducam B0433 (4.0 mm)	3840×3032	3.9 cm	149 m	\$85, 30 g	Chosen ; rolling shutter
IMX568 global shutter	2472×2064	5.7 cm	140 m	\$300–500, 50–80 g	Rejected: cost, USB 3
InfiRay P2 Pro	256×192	41 cm	106 m	\$250, 10 g	Kept
FLIR Boson 640 (8.7 mm)	640×512	15 cm	—	\$3560, 71 g	Funded upgrade

Vibration and shock: a stiff mount, not an isolator. The prop excites once per revolution (1P) at 83–167 Hz and at blade passage (2P) at 167–333 Hz over 5000–10 000 rpm; cruise is at the low edge. A mount of natural frequency f_n and damping ratio ζ transmits base motion by

$$T(r) = \sqrt{\frac{1 + (2\zeta r)^2}{(1 - r^2)^2 + (2\zeta r)^2}}, \quad r = f/f_n. \quad (33)$$

The as-built lip insert ($f_n \approx 200$ Hz) sits in the blade-pass band and amplifies up to $6.3\times$. A soft 45 Hz isolator would cut vibration to 0.03–0.4, but landing pulses of 11–36 ms put their energy at 15–50 Hz, right where the isolator is, and with 3–5 mm of sway space it bottoms out (Table 27; two-mass model on a Hunt–Crossley soil contact [42]). Halving the phone’s peak on hard dirt would need $f_n \leq 20$ Hz and 19 mm of travel. We chose a stiff, preloaded TPU clamp at 830 Hz ($T \leq 1.2$ in both bands), so the phone follows the airframe. Its worst case, 59 g for 12 ms on hard dirt at 2 m/s sink, is 25 times below the 1500 g board-level drop qualification [52].

9.4 Cameras

Both cameras look straight down through belly windows. An external review flagged image quality as the weak point of the first revision: 8.3 cm pixels from a 2.3 MP global-shutter camera, and about four thermal pixels on a deer at 100 m. The second revision changed the RGB camera and kept the thermal one (Table 28).

Survey geometry. For a nadir camera at height h with pixel pitch p , focal length f , $N_x \times N_y$ pixels and horizontal field of view θ_H ,

$$\text{GSD} = \frac{h p}{f}, \quad W = 2h \tan \frac{\theta_H}{2} = N_x \text{ GSD}, \quad L = W \frac{N_y}{N_x}, \quad (34)$$

$$s = (1 - o_s) W, \quad d = (1 - o_f) L, \quad \Delta t = d/V, \quad (35)$$

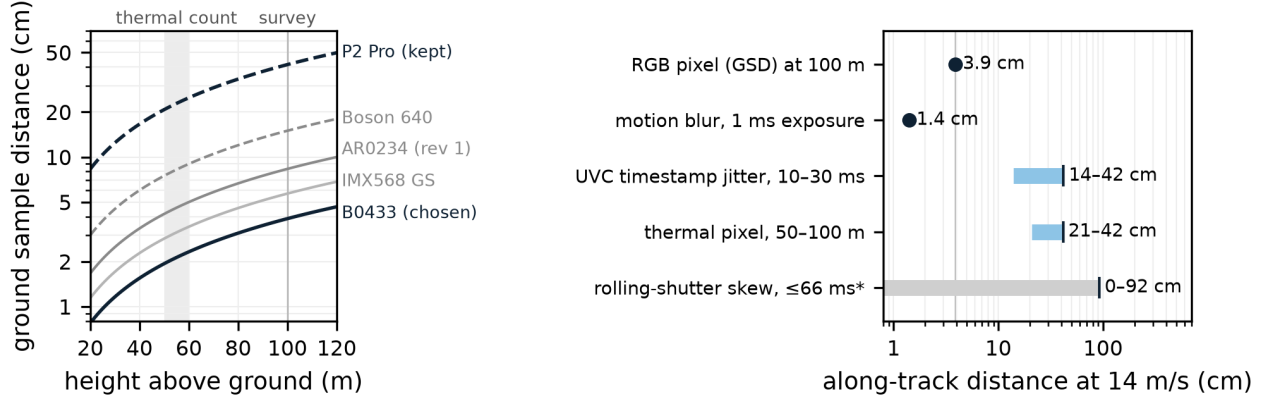


Figure 29: Left: ground sample distance against height (Equation (34)). Right: along-track errors at 14 m/s and 100 m, against the 3.9 cm RGB pixel (grey line). *Rolling-shutter skew varies across the frame and is removed in photogrammetry once the readout time is measured.

with footprint $W \times L$, line spacing s , trigger distance d and photo interval Δt . For the B0433 ($p = 1.55 \mu\text{m}$, $f = 4.0 \text{ mm}$, $\theta_H = 73.3^\circ$) at 100 m, $\text{GSD} = 3.88 \text{ cm}$, $W = 148.8 \text{ m}$ and $L = 117.5 \text{ m}$. With $o_s = 0.70$ and $o_f = 0.75$, the lines are 44.6 m apart and a photo is due every 29.4 m, or 2.1 s at 14 m/s. The line coverage rate sV is 225 ha/h, against 242 ha/h for the rev-1 camera, so twice the linear resolution costs 7% of coverage.

Smear, skew and timing. Three along-track errors matter (Figure 29, right). *Blur*, Vt_e : the app caps exposure at 1 ms, giving 1.4 cm or 0.36 pixel. *Rolling-shutter skew*, Vt_r : the readout takes at most one 15 fps frame period (66 ms), up to 0.92 m (24 pixels) from top to bottom of the frame. This is geometric, not blur, and photogrammetry removes it once t_r is known [53, 54], so we will measure it on the bench. *Timestamp jitter*, $V\delta t$: UVC frames arrive with 10–30 ms of jitter, or 14–42 cm (4–11 pixels). This is why we did not fit RTK GNSS: centimetre positions cannot help when the photo time is uncertain by ten pixels. Survey-grade maps use ground control points instead, and RTK would pay off only with a hardware-triggered camera. Vibration adds blur too. TPU cradle posts put the camera’s lateral mode at 78 Hz, on the cruise 1P, giving 10.5 pixels of blur with the rev-1 camera; the B0433’s 0.39 mrad pixel is half the rev-1’s 0.83 mrad, so that would double. The stiff PETG-CF posts (841 Hz, 0.01 pixel) are therefore required.

Thermal pixels on target. The P2 Pro spans an assumed 56° with 256 pixels, so its pixel is $2h \tan 28^\circ / 256$: 41 cm at 100 m and 21–25 cm at 50–60 m. A deer covers about 4 pixels at 100 m and about 7×2 at 50–60 m, so thermal counting missions fly lower, at the price of the RGB swath (74–89 m instead of 149 m). A Boson 640 would give 15 cm at 100 m but alone costs 3.4 times the whole avionics BOM. It is the funded upgrade, and it is US-made (Section 11).

9.5 On-board image pipeline

Decoding and re-encoding every 12 MP frame would heat a phone already at its thermal limit, so the app saves the camera’s own MJPEG frames (passthrough). Many UVC cameras omit the Huffman tables and rely on the standard ones [55, 56]; Android copes, but PIL, OpenCV and OpenDroneMap do not, so the app inserts the standard tables (unit-tested). Each frame carries the position and attitude interpolated to its capture time in EXIF and XMP tags, with camera latencies (40 ms RGB, 60 ms thermal) assumed until measured. Thermal frames are stored as raw

Table 29: Avionics, propulsion and battery bill of materials, grouped by function.

Group	Cost (\$)	Mass (g)
Flight control and navigation (FC, GNSS, airspeed)	189.00	54.0
Radios and Remote ID (SiK pair, ELRS receiver, Dronetag BS)	214.00	33.0
Payload and phone interface (cameras, hub, bridge, regulators, fan)	381.80	81.9
Actuation (4 servos and extensions)	56.00	88.0
Propulsion (motor, ESC, folding prop)	105.00	163.0
Battery (8 cells, pack kit)	74.00	608.0
Wiring, fuse, mounts, connectors	39.00	33.0
Total without the phone	1 058.80	1 060.9

16-bit counts. Detection never touches full frames: the app decodes at the largest power-of-two reduction that keeps the width at or above 640 px, here quarter scale (960×758). The landing check runs on thermal frames at about 5 Hz, 30–150 m above the site. A frame fails if a blob more than 6°C above the patch median, at least 0.1 m^2 and 2 pixels, touches the projected 20 m-wide patch; three failures in five conclusive frames reject the site. The RGB obstacle detector is still a stub.

Storage and bandwidth. At the planned 2.1 s interval a flight yields about 2400 frames of roughly 3 MB (7–8 GB), so the phone needs 16 GB free; the app’s default 1 s save interval would double this and should follow the planner. At 15 fps, 3 MB frames are about 360 Mbit/s, three quarters of USB 2.0’s 480 Mbit/s before the thermal stream and overhead, so both cameras on one hub is a bench item, with 3840×2160 as the fallback.

9.6 Bill of materials

The 29 BOM lines total \$1 058.80 and 1060.9 g without the user-supplied phone (Table 29). The thermal camera is the largest single cost (24%), and the battery is 57% of the mass. The printed airframe, foam wings, spars and the RC transmitter are not included.

9.7 How the systems design evolved

Table 30 lists the main changes and what forced each one. Most came from coupling: a thermal model changed the power harness, and a regulation changed the telemetry path and the app.

10 Ground planning software

The ground package, `sparrow_ground` (Python), turns a survey polygon and a home point into a complete ArduPilot mission: survey lines, landing sites from free global data, a reachability check, the compiled mission, and draft notices to people near the sites.

Why landing sites are planned before takeoff. Choosing a landing site only in flight, with the phone’s cameras, would make landing depend on the phone and break R9. Choosing only before takeoff cannot react to a person or animal on the site. We do both. The planner vets up to eight sites and compiles each into the FC mission as a `DO_LAND_START` landing sequence; with `RTL_AUTOLAND = 2`, any failsafe sends ArduPlane to the *nearest* one with no phone involved. The

Table 30: Systems changes and the analysis or review that forced them.

Item	Change	Forced by
RGB camera	10.9 cm draft → AR0234 8.3 cm → B0433 3.9 cm	External image-quality review
RGB software	re-encode → passthrough + Huffman tables	Phone heat; ODM compatibility
Camera cradle	TPU posts (78 Hz) → PETG-CF (841 Hz)	10.5 pixels of blur at cruise 1P
In-flight link	LTE → airplane mode, STATUSTEXT over SiK	47 CFR 22.925
Payload power	always on → hub → camera VBUS switched	Sun soak; phone must keep its link
Phone mount	200 Hz lip → 830 Hz clamp	6.3× resonance; isolator bottoms
Phone cooling	flush hole → NACA inlet, duct, fan	Inlet suction; 57 °C in cruise
Remote ID	battery beacon → FC-powered module	40 °C rating, 73 °C soak
Landing run	fixed 60 m → density-altitude scaled	Hot-and-high sites (R12)
Approach	4:1 → 5:1 in the planner	Crow lower halves cut to 12°

phone adds judgement: it re-ranks the sites every 2s on reach, wind and battery (changing only for a 15 % better score), checks the patch on final with the thermal camera, and diverts if the check fails. SITL tests confirm both paths (Table 34).

10.1 Survey planning

The polygon becomes a lawnmower pattern along the long side of its minimum rotated rectangle (fewest turns), with spacing and trigger distance from Equation (35), 40 m turn extensions, and the start at the corner nearest home. Each line is flown level at its highest surface-model point plus the survey height, relative to home, so the FC needs no terrain database. Over lower ground the aircraft is therefore more than 100 m up: in Section 10.6 the worst point is 115 m, inside 400 ft (122 m). Over more than about 22 m of relief along a line the planner would have to split the line or step its altitude, which it does not yet do.

10.2 Landing-site finder

The finder works on a 10 m grid. It uses the Copernicus GLO-30 surface model [57], ESA World-Cover [58] and OpenStreetMap [59] (buildings, roads, power lines, water). Cells that pass the per-cell tests of Table 31 are eroded by one cell, so every run lies in a corridor at least 30 m wide. Runs are measured along 12 headings, up to 200 m, and each (cell, heading) candidate must then pass the approach tests. GLO-30 includes only part of the canopy and roofs, so obstacle heights are added on top of it, which is conservative. Candidates are scored as

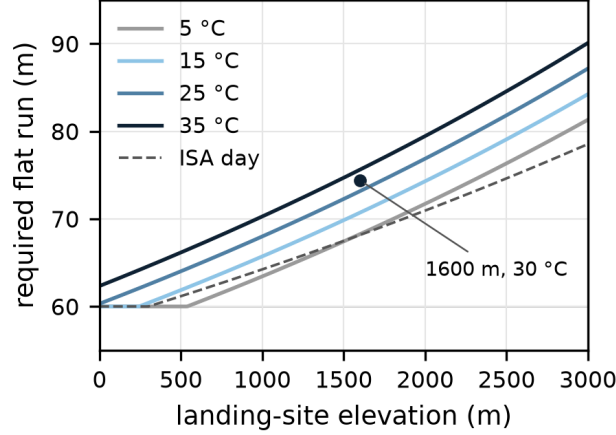
$$S = 0.35 \frac{\min(\ell_r, 150)}{150} + 0.20 \left(1 - \frac{\gamma}{4\%}\right) + 0.25 \frac{1 + \cos(\psi - \psi_w)}{2} + 0.20 \left(1 - \frac{d_s}{2500 \text{ m}}\right), \quad (36)$$

with run ℓ_r in metres, slope γ , the angle $\psi - \psi_w$ between landing heading and wind, and distance d_s from the survey area. A tie-break of at most 0.02 favours the middle of a field. Sites are accepted greedily, at least 300 m apart, up to eight, and the launch field is always added as site H.

People near the sites. Three layers keep people clear. The building and road buffers keep sites away from where people usually are. Before takeoff the planner drafts a notice, with a map link and a time window (90 min by default), to the named contacts of the predicted site and the two runners-up, by push or e-mail. Nothing is sent until the operator types “SEND”. In flight, the thermal check rejects a site with a warm body on it.

Table 31: Landing-site criteria (planner defaults).

Test	Criterion
Slope, roughness	$\leq 4\%$ on a smoothed surface and end to end; RMS roughness ≤ 0.5 m
Land cover	WorldCover grassland, cropland or bare ground
Buffers	≥ 100 m from buildings, ≥ 30 m from roads, ≥ 15 m from water
Run length	$\geq L_{\min}(z, T)$ of Equation (38): 60 m at sea level
Final approach	obstacles (surface +15 m trees, +8 m built-up, OSM buildings, +25 m power lines) below the 5:1 final in a ± 15 m corridor; path ≥ 150 m from power lines
Loiter	the 60 m approach circle ≥ 15 m below the 40 m approach height



z (m)	T (°C)	σ	$L_{\min}$ (m)
0	15	1.000	60.0
0	35	0.935	62.3
1000	30	0.843	69.1
1600	30	0.784	74.4
2000	35	0.734	79.5
2500	25	0.712	81.8

Figure 30: Required flat run against site elevation and temperature, from the planner’s own functions (Equation (38)); the dashed line is a standard day.

10.3 Density-altitude correction of the landing run

The 60 m minimum run holds at sea level. The autopilot flies fixed *indicated* approach and flare speeds, so the true airspeed is $V_{\text{IAS}}/\sqrt{\sigma}$ for density ratio σ , and the flare float and belly slide, which scale with $V_{\text{TAS}}^2/(2a)$, grow as $1/\sigma$. The planner takes the site elevation z as pressure altitude, with the forecast temperature T (`--temp`), in the standard troposphere [24]:

$$p = p_0 \left(1 - \frac{\Lambda z}{T_0}\right)^{g/(R\Lambda)} = 101\,325 \text{ Pa} (1 - 2.25577 \times 10^{-5} z)^{5.25588}, \quad \sigma = \frac{p}{R(T + 273.15 \text{ K}) \rho_0}, \quad (37)$$

with $\Lambda = 6.5 \text{ K/km}$, $T_0 = 288.15 \text{ K}$, $R = 287.05 \text{ J/(kg K)}$ and $\rho_0 = 1.225 \text{ kg/m}^3$. The 60 m already has margin for a mild, low site, which we credit up to a reference density altitude of 300 m ($\sigma_{\text{ref}} = 0.9715$ on a standard day):

$$L_{\min}(z, T) = 60 \text{ m} \times \max\left(1, \frac{\sigma_{\text{ref}}}{\sigma(z, T)}\right). \quad (38)$$

The R12 case (2000 m, 35 °C) needs 79.5 m, a third more than at sea level (Figure 30). Unit tests pin 74.4 m at 1600 m and 30 °C and check that a hot, high day can only remove sites.

10.4 Reachability

The route is sampled every 50 m. A site is within glide reach of a point if

$$d \leq (h - h_{\text{site}} - 30 \text{ m}) \frac{L}{D} \frac{V - V_{\text{hw}}}{V}, \quad (39)$$

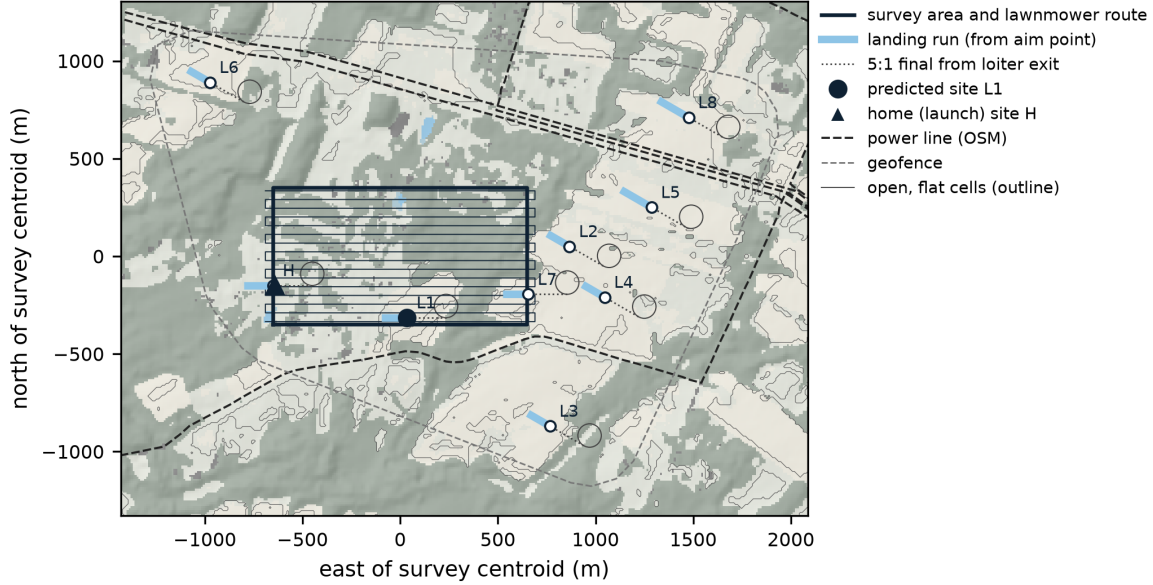


Figure 31: Planner output for the 91 ha New Jersey example (current camera, 5:1 approach). Land cover is WorldCover (dark: trees; pale: grass and crops; blue: water). Each site shows its run from the aim point, its final and its approach loiter. All data came from the repo’s cache; nothing was downloaded.

keeping 30 m for the final turn and approach, with V_{hw} the headwind toward the site (tailwind is not credited) and $L/D = 10$, below the aerodynamic best of 12.0. It is within energy reach if the energy used so far plus the divert energy Pd/V_g fits in the usable 103.7 Wh. A point is covered only if one site passes both tests (the climb-out is exempt from the glide test), and the whole route must fit in 80 % of the usable energy. A plan with a gap is rejected and the gap is drawn on the map.

10.5 Mission compilation

The planner writes the mission, fence and rally files and `mission.json` for the phone, and uploads them over MAVLink. The mission is: home; NAV_TAKEOFF to 40 m (15° pitch for a hand or bungee launch, 12° on the dolly); DO_CHANGE_SPEED 14 m/s; DO_SET_RELAY (relay 0 on, the camera rail); the survey waypoints; then the landing sequences, predicted site first so that a normal mission end lands there. Each is DO_LAND_START → NAV_LOITER_TO_ALT → NAV_LAND; the clockwise 60 m loiter at 40 m, centred 60 m right of the centreline and 200 m before the aim point, exits onto a straight 5:1 final. The fence is the hull of everything, buffered by 200 m, and the parameter snippet routes the link-loss, low-battery and fence failsafes to RTL.

10.6 Worked example

We re-planned the repo’s New Jersey farmland demonstration with the current camera, offline from cached data (Figure 31). The 91 ha rectangle, with wind from 300° at 5 m/s, becomes 16 lines, 22.1 km of imaging and 751 photos. The route to the predicted site is 22.8 km and 28.6 min, using 36.3 Wh of the 83.0 Wh budget. Eight field sites plus H cover the route with no gaps; of 716 candidates tried, 374 failed on power lines, 255 on obstacles under the final and 79 on the loiter. The predicted site L1 is a 150 m cropland run at 1.4 % slope. The launch field’s approach is flagged for trees, the kind of warning an operator must see before committing.

Table 32: Transmitters on BLUEBIRD.

Transmitter	Authorisation	Operating rule
SiK V3, 915 MHz, 100 mW	FCC ID 2AP22-HT03-V3	EIRP 0.16 W vs 4 W (§15.247)
RP3 V2 ELRS, 2.4 GHz	FCC version [pending: ID from label]	FCC-domain firmware, ≤ 100 mW
Ground RC transmitter	FCC-ID'd model	FCC-domain firmware
Dronetag BS Remote ID	FAA-accepted [pending: FCC ID]	as supplied
Phone	FCC-certified	cellular off in flight (preflight)

10.7 Test coverage

We ran the ground suite for this paper: 45 tests pass, covering the landing criteria (13, including density altitude), mission files, the relay item and the upload protocol (10), reachability (6), survey geometry (5), notifications (4), the phone checklist (4) and the server (3). The Android core library's 95 JVM tests passed in the last recorded run (we did not re-run Gradle); they include a MAVLink codec byte-identical to pymavlink (10), the divert gate (10), the thermal verifier (12), payload power (7), preflight with the airplane-mode check (7) and the thermal governor (7). The four SITL tests are in Table 34. These tests show that the logic does what the design says, not that the app runs on a phone: it has been built but never launched on one, and its USB, dual-camera and background-service paths still need a bench run.

11 Regulatory compliance (United States)

11.1 FCC: radios and the airborne phone

This is an engineering checklist, not legal advice; the rules changed twice in the past year, so each purchase is re-checked against the FCC ID database. The FCC regulates the transmitters: the SiK radio, the ELRS receiver (it sends telemetry), the Remote ID module and the phone; motors, batteries, the airframe and the FC are not authorised as parts. The aircraft itself needs no authorisation: under 47 CFR 15.23 [60], home-built devices that are not marketed, not built from a kit and built in quantities of five or fewer for personal use are exempt, provided the builder meets the technical limits. We therefore fly only transmitters with their own FCC ID, operated inside their grant (Table 32). Publishing the open design is not marketing a device, but selling BLUEBIRD as a product or kit would end the exemption.

The SiK's fin dipole (≤ 2.15 dBi) gives $P_t G = 0.1 \text{ W} \times 10^{0.215} = 0.16 \text{ W}$ EIRP, 13.9 dB inside the §15.247 limit; a grant normally assumes the module's own antenna, so the supplied one remains the cleanest option.

The phone in the air. Under 47 CFR 22.925 [61], a cellular telephone aboard any aircraft must not be operated while airborne; the FCC has applied this to balloons, and it reads onto a drone. So the flight phone flies in airplane mode, and USB (cameras and FC link) keeps working. The app's preflight has a *blocking* item that fails if the cellular radio is on, and a failed preflight withholds the interlock value that an FC arming script can require. The rest of the design followed: in-flight events go down the SiK link, notices go out before takeoff, and thumbnails and logs sync after landing. A live in-flight LTE link would need an airborne-approved data plan and an FCC waiver.

11.2 The December 2025 Covered List

On 22 December 2025 the FCC added foreign-produced UAS and UAS critical components to its Covered List [62, 63], including communications, flight controllers, ground stations, navigation, sensors and cameras, batteries and motors. The listing blocks *new* FCC equipment authorisations for these items; it does not prohibit importing, selling or using models already authorised. A January 2026 update exempts Blue UAS Cleared List items and Buy American end products until 1 January 2027 [64]. For us the consequence is to buy transmitters whose FCC IDs predate 22 December 2025, and every transmitter on the BOM qualifies. The non-transmitting parts are not authorised as parts, so the rule does not reach a home-built aircraft through them; it would if BLUEBIRD were sold. If the FCC revokes earlier grants, Table 32 must be redone.

11.3 FAA

- **Registration and pilot.** Over 250 g, so registered under Part 107 [65] with the Remote ID serial; research use is non-recreational, so the pilot holds a Part 107 certificate.
- **Remote ID.** Home-built aircraft are excepted from the *production* rules but must still broadcast [66]. BLUEBIRD carries a Dronetag BS module (5 g, rated to 85 °C); a battery beacon rated to 40 °C was rejected, because the bays soak to 60–73 °C.
- **Line of sight.** Part 107, and a broadcast module, limit flights to visual line of sight with an observer. This binds the mission: the 62.5–64.9 km range is usable only under a waiver. The proposed BVLOS rule, Part 108 [67], has been under OIRA review since 10 July 2026.
- **Airspace and height.** LAANC in controlled airspace; the 100 m survey height is 328 ft, under 400 ft, subject to the line-level margin of Section 10.

11.4 Federal funding and operation abroad

The American Security Drone Act [68] bars federal funds, including grants, from buying or operating UAS with critical components from covered foreign entities, and much of this BOM (FC, radios, cameras, motor) is Chinese-made. Rutgers-internal funding such as Aresty is not federal. A federal grant would need an NDAA-compliant variant with a Blue UAS Framework FC and radios and a US-made thermal core such as the Boson, plus its own CAD pass. Our review covers only the United States; any other country, including the East African sites of R12, needs its own spectrum and aviation review before flight.

12 Verification status and test plan

Nothing has flown, so every result here is a prediction. We state what was verified, by which method, and which test on the first airframe will confirm or overturn each prediction.

12.1 Verification matrix

Table 33 lists each major claim. The status column separates results that are fully simulated from those that are analysed but need a physical test.

12.2 Software-in-the-loop flight tests

The landing-site architecture (R9) is the claim that matters most for safety, so it is tested end to end. We run the ArduPlane 4.7.1 firmware in SITL [6] with the aircraft’s own parameter file, and

Table 33: Verification matrix. *Sim*: verified by simulation or automated tests. *Analysis*: predicted by two or more independent methods. *Bench/Flight*: the physical test that closes the item.

Claim	How it was checked	Status	Closing test
Cruise drag and L/D	AeroBuildup + VLM, VSPAERO, SU2 RANS sections (Section 4)	Analysis	Power at cruise from flight logs
Stall speed and $C_{L,max}$	NeuralFoil with SU2 knock-down (Section 4.2)	Analysis	Slow-flight test at altitude
Static and dynamic stability	Neutral point by two methods; linearised modes (Section 6)	Analysis	FBWA trim, then AUTOTUNE
Crow airbrake authority	Controls model; hinge moments vs servo torque (Section 6)	Analysis	Eleven twist bench test; glide-slope test
Landing-site failsafes	ArduPlane SITL, four end-to-end tests (Table 34)	Sim	First autonomous landing
Ground planner	Unit tests, including density-altitude scaling (Section 10)	Sim	Field survey of a planned site
Phone app pipeline	Unit tests of the core library; the APK builds (Section 9)	Sim	Bench run with both cameras
Structure	Beam models, CalculiX FEA, landing impact (Section 8)	Analysis	Coupon Gt test; bench twist; first-airframe drop test
Dolly launch	3-D dolly simulation over rough strips (Section 6)	Sim	Spring rates, sag, release trip on the bench
Propulsion	BEMT vs UIUC data (Section 5)	Analysis	Thrust stand: static current ≤ 40 A
Cameras	GSD and blur geometry (Section 9)	Analysis	Readout time; lens calibration
Manufacture and service	Automated DFM and service audits of the CAD	Analysis	[pending: re-run on the final export]

pytest drives it over MAVLink.

Every test uploads the same mission:

- a takeoff;
- six survey waypoints at 80 m;
- two landing sites, each compiled as DO_LAND_START $\rightarrow$ NAV_LOITER_TO_ALT (40 m, 300 m out) $\rightarrow$ NAV_LAND.

Site A is listed first in the mission on purpose, so the tests show that the autopilot picks the *nearest* site and not the first one. Table 34 gives the results.

The third test is why the phone app follows every go-around with an explicit mission change. On its own, ArduPlane returns after a go-around to the nearest landing sequence, which is usually the site it has just rejected.

12.3 Design audits of the CAD

Two automated audits read the exported CAD solids rather than the build scripts.

Manufacturability (DFM) audit. It checks print orientation, support, bridge spans, wall and hole minima, bed fit and assembly paths for every part. The first run was against the earlier printed

Table 34: ArduPlane SITL end-to-end tests (all pass, 148 s at 10× speed-up).

Test	What it proves	Result
RTL lands at nearest sequence	A failsafe (RTL with RTL_AUTOLAND=2) jumps to the nearest landing sequence with no phone involvement. The crow airbrake opens on the landing leg.	22 m from B; crow split 320 μ s
Companion divert	The phone can move the aircraft to another vetted site during the approach.	23 m from A
Go-around on final, then divert	Rejecting a site on final (the camera check fails): climb out, then land at the other site.	abort 36 m, peak 48 m; 24 m from A
Plain RTL needs AUTO	Documents a firmware behaviour the phone must handle: in plain RTL a mission change does nothing until the mode changes to AUTO.	24 m from A

outer wing. It found seven blockers, which are local geometry errors: a dowel bored on the wrong axis, servo trays wider than their pockets, a fin tongue on the hinge-pin line, a missing lattice, a thin insert wall, missing dowel holes and elevon trailing edges too thin to print. With fixes, the build is about 86 h of printing and 1.35 kg of filament. Changing to foam outer wings removed the wing-segment items. The rest are being fixed in the final CAD export.

Service audit. It tries to remove each of the 35 line-replaceable units along a collision-free path with real tool access. On the same export:

- 10 units came out cleanly;
 - 14 came out only with help;
 - 11 could not come out at all, including the battery, whose seat lip overhung the pack by 5.5 mm.
- Each has a specific geometric fix. Both audits must pass on the final export before printing starts [pending: final export re-run].

These audits matter because a design that cannot be assembled or serviced fails R10 and R14 however good its aerodynamics are. Running them on the solids caught errors that no drawing review had found.

12.4 Test plan for the first airframe

The first airframe is tested in the following order. Each step de-risks the next, and each has a pass criterion taken from the analysis in this paper.

1. Component bench tests.

- Motor and 11×7 prop on a fresh 4S pack: static current ≤ 40 A. Otherwise cap TKOFF_THR_MAX at 90 %.
- Both cameras streaming through the hub at full resolution; measure the rolling-shutter read-out time.
- Phone thermal soak in the closed bay.
- Weigh every module whose mass is an estimate.

2. Coupons and structure.

- Weigh the first covered foam panel and re-run the foam-wing model.
- Torsion coupon to $Gt \geq 54$ N/mm.

- Elevon twist under a 50 g tip load (4.4° predicted).
 - Sun coupon for the skin: $r \leq 0.16$ on the wing tops.
 - Drop test of the first airframe on its skid sled.
3. **Systems on the bench, prop off.**
 - Surface directions in FBWA.
 - The crow clamshell: the upper halves open up and the lower halves down.
 - The phone’s preflight checks, including airplane mode, and the payload power relay.
 4. **Dolly.**
 - Spring rates and static sag (20 mm).
 - Release trip at an average of 15 mm across the two feelers.
 5. **Flight tests**, under Part 107 in visual line of sight:
 - (a) Hand-launched trim flights in FBWA.
 - (b) AUTOTUNE for roll and pitch, then write `TRIM_THROTTLE` back to the parameter file.
 - (c) A slow-flight stall check at altitude before trusting anything below about 10 m/s.
 - (d) Crow-airbrake glide-slope runs below the 15 m/s crow limit.
 - (e) Autonomous landings at a surveyed site.
 - (f) The SITL divert scenarios, repeated in the air.

13 Concept of operations for detections (planned)

The survey itself needs no radio link for data. Each flight writes about 2 400 full-resolution photos, or 7–8 GB, to the phone, and they come home with the aircraft (Section 9). Conservation users also want to act on what the aircraft sees while it is still in the air: a person on a reserve at dawn, or an animal outside its owner’s fence. This section describes how BLUEBIRD will handle such detections. It is planned work: the flight-control pieces it relies on exist and are tested in SITL, but the detector and the alert path do not exist yet.

13.1 Alert, evidence and human confirmation

A detection on the phone becomes two messages on the existing SiK telemetry link:

- a short alert: class, confidence, geolocated position, time and an identifier;
- an evidence crop: a 128×128 cut-out from the RGB and thermal frames, about 4 kB to 5 kB each. It is sent with MAVLink’s encapsulated-data messages and arrives in a few seconds at the link’s spare capacity.

A full photo would take minutes; a crop is enough for a person to judge the detection.

The operator sees each detection as a card and chooses one of three actions: *confirm*, *dismiss* or *closer look*. Nothing happens on the aircraft until a person decides. This extends the architecture’s rule: the flight controller flies, the phone proposes, and a person decides.

13.2 The closer look: a pylon turn

For a closer look, the aircraft leaves its survey line, circles the detection for a bounded number of turns, and then returns to the survey with the same divert commands verified in Section 12.2.

The camera is fixed and points straight down through the belly. In a coordinated turn the aircraft banks by ϕ , which tilts the camera toward the centre of the circle by the same angle (Figure 32). The camera stays aimed at the centre when

$$\tan \phi = \frac{r}{h}, \quad (40)$$

and a coordinated level turn at true airspeed V requires

$$\tan \phi = \frac{V^2}{g r}. \quad (41)$$

Eliminating ϕ gives the orbit radius that holds the target at the image centre:

$$r^* = V \sqrt{\frac{h}{g}}, \quad \phi^* = \arctan \frac{r^*}{h}. \quad (42)$$

This is the pylon-turn geometry used by side-firing gunships, adapted to a camera that looks down the body axis instead of along the wing.

At the survey condition, $V = 14 \text{ m/s}$ and $h = 100 \text{ m}$:

- the orbit radius is $r^* = 44.7 \text{ m}$ and the bank is $\phi^* = 24.1^\circ$;
- the slant range is 110 m , so the ground sample distance only coarsens from 3.9 to about 4.3 cm/px ;
- one lap takes about 20 s .

The load factor in the bank is $n = 1/\cos \phi^* = 1.10$. The stall speed rises by $\sqrt{n}$: the trimmed stall of about 9.5 m/s at 2.52 kg becomes about 9.9 m/s , which leaves a 1.4 margin at 14 m/s .

For an identification pass on animals at 35 m , the same equations give $r^* \approx 26 \text{ m}$ and $\phi^* \approx 37^\circ$ at 14 m/s . At 12 m/s they give 23 m and 33° .

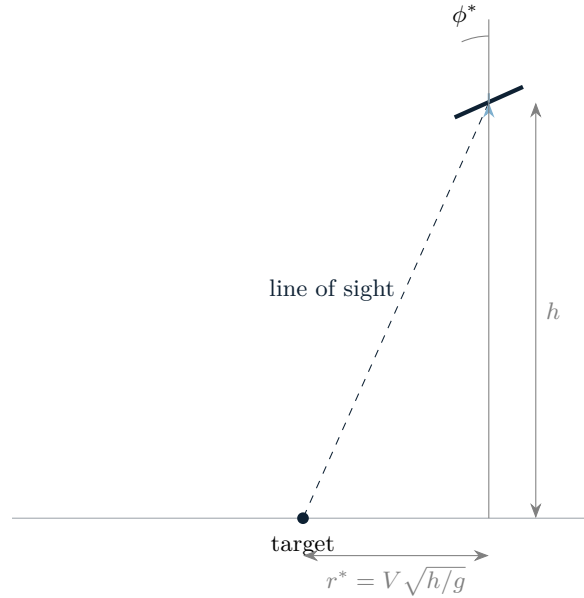


Figure 32: Pylon turn for a closer look. The camera points down the body axis, so banking by ϕ^* aims it along the line of sight to the orbit centre. The aircraft never passes over the target.

Rules for the closer look.

- **People.** The aircraft reports people; it does not follow them. At about 2.5 kg , BLUEBIRD does not qualify for operations over people under 14 CFR 107.39. The orbit therefore keeps it $r^* \approx 45 \text{ m}$ to the side, and the phone enforces a 30 m keep-out ring around the target. If the tracking error in wind would breach it, the phone widens the circle.
- **Animals.** The orbit stays at 30 m above ground or higher and avoids breeding sites.

- **Always.**
 - The circle is centred on a fixed point, lasts at most a few laps, and stays inside the geofence and in visual line of sight.
 - The phone refuses the look if it would break the landing-site energy reserve.
 - Two laps cost about 40 s, or roughly 1 Wh of the 104 Wh usable.

Wind. In a 5 m/s wind, the groundspeed around the orbit varies between about 9 and 19 m/s. Holding a circle over the ground then needs the bank to vary between roughly 10° and 40°. The aim point swings by about $\pm 15^\circ$, but that is within the camera’s half field of view of about 30°, so the target stays in frame.

13.3 Individuals and property boundaries

Counting is straightforward at survey height: a cow is about 56×18 pixels at 3.9 cm/px. Checking whether an animal is outside its owner’s land then only needs the property polygon, drawn once in the planner.

Knowing *which* animal it is needs more than pixels. An ear tag of about 6 cm is one or two pixels at 100 m. There are two routes to individual identity:

- **Coat-pattern re-identification** on a low identification pass, for species with markings visible from above (for example Holstein cattle, zebra and giraffe).
- **Electronic tags.** Many ranches already fit LoRaWAN GPS tags and collars. A small receive-only LoRa gateway on the aircraft could read every tag it overflies. LoRa suits this job: it carries a few bytes a long way, where imagery needs far more bandwidth.

14 Limitations and future work

Limitations.

- Every performance, stability and strength number in this paper is a prediction.
- The largest aerodynamic uncertainty is the low-Reynolds-number stall of the outer wing, where the models disagree by about 25 % (Section 4.2).
- The largest structural uncertainty is the landing impact. It depends on the ground, the sink rate and the as-printed skid.
- Both are addressed first in the test plan (Section 12).
- Several items are marked [pending:] in this paper because they await the final CAD export or a measurement.

Next steps.

1. Complete the final CAD export and re-run the full check chain: DFM and service audits, mass and CG, aero, structures and SITL.
2. Build the first airframe and run the bench, coupon and flight tests of Section 12.
3. Train and deploy the onboard detector, then the alert, confirmation and closer-look workflow of Section 13.
4. Add the upgrade path: a 640 px thermal core for counting from survey height, and a LoRa tag reader for livestock work.

15 Conclusion

BLUEBIRD shows that a capable survey aircraft can be designed around parts a student can buy, print and repair, and around a phone that the operator already owns.

The design is a set of decisions, each made with numbers and re-checked by the next discipline:

- the blended-wing layout and its reflexed airfoils;
- the CFD-corrected stall speed;
- the taller canted fins;
- the split-elevon airbrake;
- the foam outer wing;
- the deep skid sled that makes a printed airframe survive landing;
- landing sites compiled into the autopilot’s mission before takeoff.

The analysis and simulation are complete to the level a first build needs. The next step is to build the aircraft and find out where the predictions are wrong.

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A Key design parameters

All values are from the shared parameter file (`params.yaml`), which the CAD and every analysis read. Lengths are in millimetres unless stated.

Table 35: Key parameters of BLUEBIRD.

Mission	
Cruise speed; survey height	14 m/s; 100 m AGL
Energy reserve	20 % of the battery, kept for reaching a landing site
Airframe	
Span; centre-body root chord	1800; 520
Centre-body half-width; wing root chord; tip chord	180; 330; 180
Outer-wing LE sweep; washout; dihedral	22°; −3° (tip nose down); 1°
Airfoils	MH60 centre body, scaled to 11.6 % thick; MH45 outer wing
Fins	200 mm tall at the tips, canted 10° outboard
Static-margin target	6–10 % of MAC, trimmed by the battery position
Controls	
Elevons	22 % chord, 25–90 % of the outer-wing span; split into upper and lower halves on one hinge
Deflections	±25° pitch/roll; crow 51° up (upper) and 12° down (lower)
Servos	EMAX ES08MD II, 2.4 kg cm at 6 V, one per half (4 in total)
Crow limit	15 m/s airspeed (servo torque)
Propulsion and energy	
Motor; prop	2814 class (~800 KV); 11×7 folding pusher
Battery	Li-ion 4S2P Molicel P45B (14.4 V nominal, 9 Ah)
Structure	
Limit load factors	−3.5 / +5.6 (gust-critical at $V_C = 22$ m/s); factor of safety 1.5
Outer spar; joiner; tip spigot	14×12 roll-wrapped carbon to $y = 820$; 12×8 pultruded; 12×10
Spar node	CNC 7075-T6 V-node, M3 8.8 through-bolts, PETG-CF keel pad
Outer wing	EPP 30 kg/m ³ core, two layers of Oracover; about 357 g per side
Landing design load	50 g ultimate at the CG (50–70 g by station) with the deep TPU skid sled
Touchdown	2.0 m/s sink (limit), 3.0 m/s abuse
Payload and avionics	
RGB camera	Arducam B0433, 12 MP, 3840 × 3032, 4.0 mm lens, HFOV 73.3°
Thermal camera	InfiRay P2 Pro class, 256 × 192
Flight controller	Matek H743-WING running ArduPlane
Phone bay	phones up to 170 × 82 × 12, 150–240 g, mounted under the flip-forward hatch
Launch	
Dolly	passive release at 8° seat pitch; 3.5 Hz torsion-bar suspension on 4-inch wheels; 719 g
Strip	nothing over 15 mm in the first 20 m; crosswind component ≤ 3 m/s

B Toolchain

Every tool used is free. Table 36 lists the versions used for the results in this paper.

Table 36: Software used for design, analysis and simulation.

Tool	Version	Used for
FreeCAD	1.1.3	Parametric CAD, STEP export, automated DFM and service audits
AeroSandbox / NeuralFoil	4.2.10 / 0.3.3	Section polars, 3-D aero build-up, vortex lattice, trim and performance
OpenVSP + VSPAERO	3.53	Independent 3-D panel and VLM cross-check
SU2	8.5	RANS CFD of the MH45 and MH60 sections
gmsh	4.15.2	Meshing for FEA
CalculiX	2.23	Finite-element analysis of printed and machined parts
ArduPlane (SITL)	4.7.1	Flight-control parameters; software-in-the-loop mission tests
pymavlink / MAVProxy	2.4.50 / 1.8.75	Mission upload and test harness
Python, NumPy, SciPy, Matplotlib	3.12, 2.5, 1.18, 3.11	Analysis scripts and figures
WireViz	0.4.1	Wiring harness drawings
Android SDK (Kotlin)	platform 35	Onboard phone application
Tectonic (XeLaTeX)	0.17	This document